

Higher Education Process Flows — CAUDIT HERM Reference Model

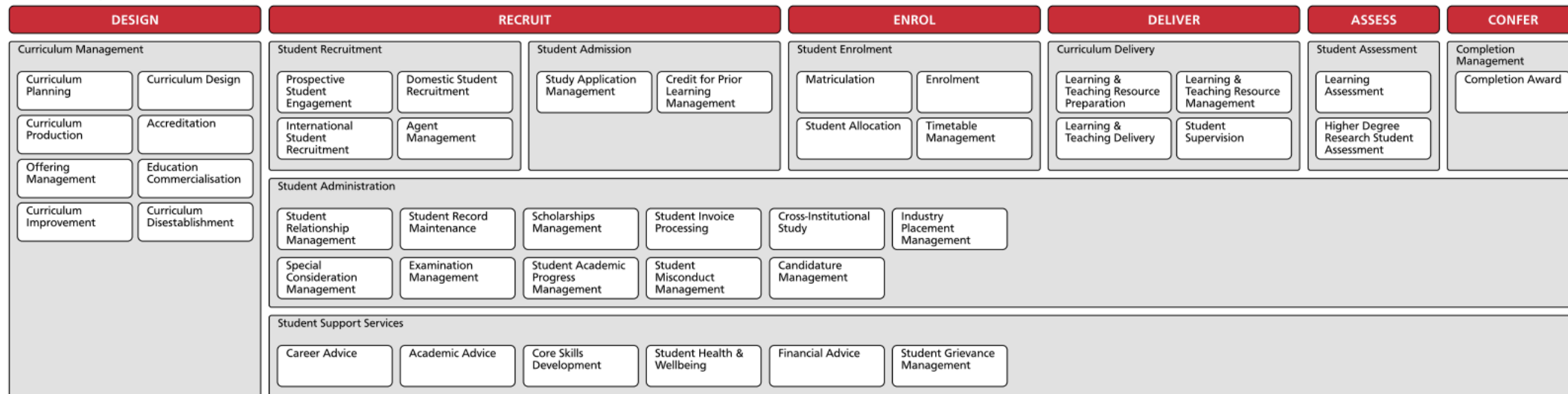
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CAUDIT HERM Business Capability Model Reference

The CAUDIT Higher Education Reference Model (HERM) **Business Reference Model v3.2** defines the standard business capabilities for higher education institutions. The table below lists all Level 2 capabilities grouped by Level 1 domain. Each process flow in this document is mapped to its relevant HERM capability codes.

CORE CAPABILITIES AND VALUE CHAINS

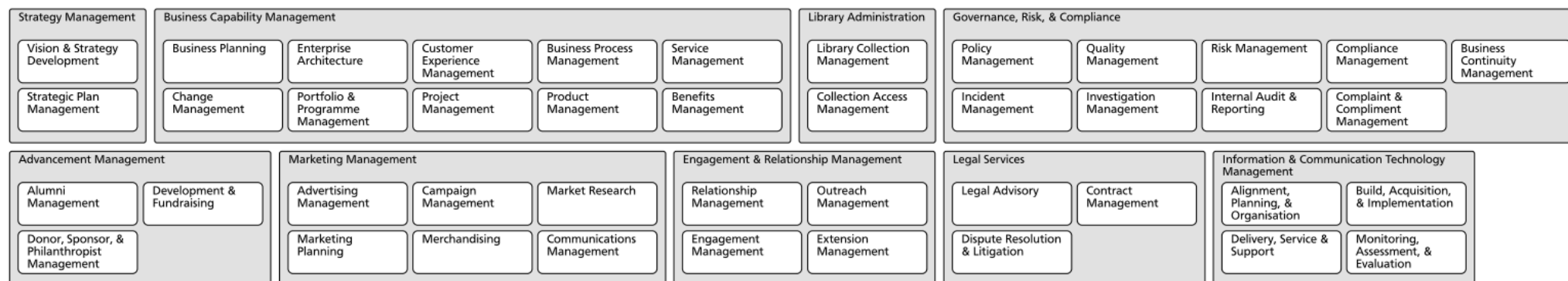
LEARNING AND TEACHING

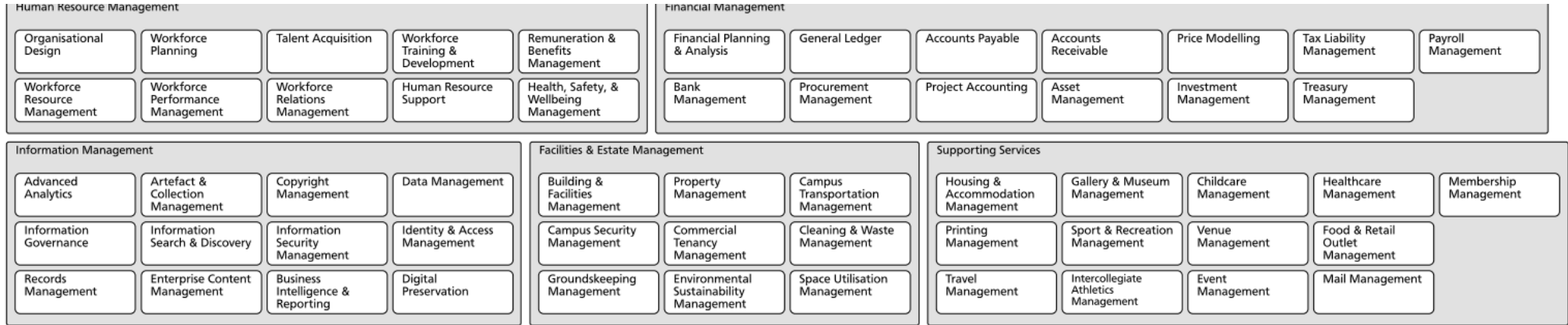


RESEARCH



ENABLING CAPABILITIES





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Learning & Teaching

Capability	Code	Description
Curriculum Management	BC001	Design, develop, and maintain academic programs, courses, and curricula
Curriculum Delivery	BC023	Deliver teaching and learning experiences across modalities
Student Assessment	BC028	Assess, grade, and certify student learning outcomes

Research & Innovation

Capability	Code	Description
Research Opportunities & Planning	BC065	Identify, evaluate, and plan research opportunities and strategy
Research Funding	BC071	Secure, administer, and report on research funding and grants
Research Activity	BC074	Conduct and manage research projects and investigations
Research Dissemination	BC086	Publish, disseminate, and commercialise research outputs
Research Management	BC093	Manage research portfolios, performance, and reporting
Research Assurance	BC245	Ensure research integrity, ethics, and regulatory compliance

Student Administration

Capability	Code	Description
Student Recruitment	BC008	Attract, engage, and recruit prospective students
Student Admission	BC014	Process applications, make admission decisions, and manage offers
Student Enrolment	BC019	Enrol, register, and orient students into programs
Student Management	BC044	Maintain student records, progression, and academic history
Student Support	BC052	Provide wellbeing, academic support, and student services
Completion Management	BC032	Manage graduation, certification, and program completion

Corporate Services

Capability	Code	Description
Financial Management	BC184	Manage budgeting, accounting, procurement, and financial control
Human Resource Management	BC171	Manage workforce planning, recruitment, payroll, and development
Legal Services	BC155	Provide legal counsel, contracts, and compliance advice
Governance Risk & Compliance	BC160	Oversee governance, risk management, audit, and regulatory compliance
Strategy Management	BC147	Develop institutional strategy, planning, and performance monitoring
Business Capability Management	BC206	Manage enterprise architecture and business capability evolution

Engagement & Advancement

Capability	Code	Description
Marketing Management	BC107	Manage brand, communications, and market engagement
Advancement Management	BC232	Manage alumni relations, fundraising, and donor stewardship
Engagement & Relationship Management	BC238	Manage stakeholder engagement, partnerships, and community relations

Infrastructure & Operations

Capability	Code	Description
Facilities & Estate Management	BC125	Manage campus, buildings, space, and physical infrastructure
Supporting Services	BC114	Provide ancillary services (catering, printing, transport, etc.)

Information & Library

Capability	Code	Description
Library Administration	BC133	Manage library collections, access, and information services
Information Management	BC135	Manage information assets, records, and data governance
Publishing Management	BC250	Manage institutional publishing and university press operations
Archive Management	BC255	Preserve and provide access to permanently valuable records

Technology

Capability	Code	Description
ICT Management	BC201	Plan, deliver, and support information and communication technology services

Table of Contents

- [Student Administration - BC008, BC014, BC019, BC044, BC052, BC032](#)
 - [1.1 Student Onboarding \(Prospect → Enrollment\)](#)
 - [1.2 Enrollment & Registration](#)
- [Student Administration - BC008, BC014, BC019, BC044, BC052, BC032](#)
 - [1.3 Academic Records Management](#)
 - [1.4 Graduation & Certification](#)
- [Learning & Teaching - BC001, BC023, BC028](#)
 - [2.1 Curriculum Management](#)
 - [2.2 Course Delivery & Assessment](#)
 - [2.3 Timetable & Scheduling](#)
- [HR & Faculty Management - BC171](#)
 - [3.1 Faculty Recruitment](#)
 - [3.2 Payroll & Benefits](#)
 - [3.3 Faculty Performance & Development](#)
- [Finance & Resource Management - BC184](#)
 - [4.1 Fee Assessment & Billing](#)
 - [4.2 Financial Aid Management](#)
 - [4.3 Procurement & Vendor Management](#)
- [Research Management - BC065, BC071, BC074, BC086, BC093, BC245](#)
 - [5.1 Grant Lifecycle Management](#)
 - [5.2 Research Outputs & Publishing](#)
- [Institutional Engagement - BC107, BC232, BC238](#)
 - [6.1 Marketing & Student Recruitment](#)
 - [6.2 Alumni Relations & Fundraising](#)
- [Library & Learning Support - BC133, BC135](#)
 - [7.1 Library Resource Management](#)
- [Enterprise Data & Analytics - BC201, BC135](#)
 - [8.1 Data Integration & Warehousing](#)
 - [8.2 Institutional Reporting & BI](#)

Test Data

The following test data is used consistently across all sequence diagrams:

Entity	Field	Value
Student	id	STU-2026-001
Student	name	John Doe
Student	email	john.doe@university.edu
Student	program	B.Sc. Computer Science
Student	semester	Fall 2026
Student	gpa	3.75
Prospect	id	PRO-2026-042
Prospect	name	Jane Smith
Prospect	email	jane.smith@prospect.com
Prospect	program	M.Sc. Data Science
Prospect	source	Open Day
Course	id	CS501
Course	title	Artificial Intelligence & ML
Course	credits	4
Course	instructor	Dr. Sarah Smith
Course	capacity	120
Course	enrolled	98
Faculty	id	FAC-101
Faculty	name	Dr. Sarah Smith
Faculty	dept	Computer Science
Faculty	email	sarah.smith@university.edu
Faculty	hire_date	2024-08-15
Faculty	salary	95000
Finance	fee_amount	12500.0
Finance	scholarship	2500.0
Finance	grant_amount	5000.0
Finance	invoice_no	INV-2026-8901
Finance	payment_ref	PAY-7X3K2
Research	grant_id	GR-2026-045
Research	title	AI-Driven Personalized Learning
Research	amount	250000.0
Research	sponsor	NSF
Research	pi	Dr. Sarah Smith
Research	status	Active
Alumni	id	ALUM-2019-088
Alumni	name	Emily Johnson
Alumni	grad_year	2019
Alumni	degree	B.Sc. Computer Science

Entity	Field	Value
Alumni	email	emily.j@corp.com

Table of Contents

- [Student Administration - BC008, BC014, BC019, BC044, BC052, BC032](#)
 - [1.1 Student Onboarding \(Prospect → Enrollment\)](#)
 - [1.2 Enrollment & Registration](#)
 - [1.3 Academic Records Management](#)
 - [1.4 Graduation & Certification](#)
- [Learning & Teaching - BC001, BC023, BC028](#)
 - [2.1 Curriculum Management](#)
 - [2.2 Course Delivery & Assessment](#)
 - [2.3 Timetable & Scheduling](#)
 - [2.4 Academic Advising & Progression Management](#)
- [HR & Faculty Management - BC171](#)
 - [3.1 Faculty Recruitment](#)
 - [3.2 Payroll & Benefits](#)
 - [3.3 Faculty Performance & Development](#)
- [Finance & Resource Management - BC184](#)
 - [4.1 Fee Assessment & Billing](#)
 - [4.2 Financial Aid Management](#)
 - [4.3 Procurement & Vendor Management](#)
- [Research Management - BC065, BC071, BC074, BC086, BC093, BC245](#)
 - [5.1 Grant Lifecycle Management](#)
 - [5.2 Research Outputs & Publishing](#)
 - [5.3 Research Compliance & Ethics](#)
- [Institutional Engagement - BC107, BC232, BC238](#)
 - [6.1 Marketing & Student Recruitment](#)
 - [6.2 Alumni Relations & Fundraising](#)
 - [6.3 Community & Industry Partnerships](#)
- [Library & Learning Support - BC133, BC135](#)
 - [7.1 Library Resource Management](#)
 - [7.2 Digital Content Delivery & Course Reserves](#)
- [Enterprise Data & Analytics - BC201, BC135](#)
 - [8.1 Data Integration & Warehousing](#)
 - [8.2 Institutional Reporting & BI](#)

- [8.3 Data Governance & Quality Management](#)
- [Student Services & Wellbeing - BC052, BC019](#)
 - [9.1 Student Support & Counselling Services](#)
 - [9.2 Career Services & Placement](#)
- [Facilities & Campus Operations - BC125, BC114](#)
 - [10.1 Facility Management & Space Allocation](#)
 - [10.2 Security & Access Control](#)
 - [10.3 Ancillary Services Management](#)
- [Governance, Risk & Compliance - BC160, BC155](#)
 - [11.1 Policy Management & Accreditation](#)
 - [11.2 Risk Management & Audit](#)

Test Data

The following test data is used consistently across all sequence diagrams:

Entity	Field	Value
Admission	admission_id	ADM-2026-042
	decision	Accepted
	decision_date	2026-04-01
	conditions	Submit final transcripts
	application_id	APP-2026-088
Alumni	id	ALUM-2019-088
	name	Emily Johnson
	grad_year	2019
	degree	B.Sc. Computer Science
	email	emily.j@corp.com
Application	application_id	APP-2026-088
	program	M.Sc. Data Science
	submitted_date	2026-03-15
	status	Admitted
	prospect_id	PRO-2026-042
Assessment	assessment_id	ASMT-2026-001
	type	Midterm
	score	42
	max_score	50
	student_id	STU-2026-001
	course_id	CSS01
Building	building_id	BLD-001
	name	Adams Hall
	location	Main Campus - East Quad
Campaign	campaign_id	CMP-2026-001
	name	Fall 2026 Open Day
	type	Recruitment
	target	12,000 prospects
	status	Active
CareerListing	listing_id	CL-2026-088
	company	Google
	role	Software Engineer Intern
	deadline	2026-09-30
	status	Open
Compliance	compliance_id	COMP-2026-001
	type	IRB Approval
	status	Approved
	approval_date	2026-04-15

Entity	Field	Value
Contract	dept	Computer Science
	contract_id	CON-2026-001
	vendor_id	VEN-2026-001
	amount	120000.0
	start_date	2026-02-01
	end_date	2028-01-31
Course	id	CS501
	title	Artificial Intelligence & ML
	credits	4
	instructor	Dr. Sarah Smith
	capacity	120
	enrolled	98
Dataset	dataset_id	DS-2026-001
	name	Student Enrollment Fact
	source	SIS (Ellucian Banner)
	frequency	Daily
Department	dept_id	DEPT-CS
	name	Computer Science
	budget	5000000.0
Donation	donation_id	DON-2026-042
	donor_id	ALUM-2019-088
	amount	5000.0
	fund	CS Scholarship
	date	2026-06-01
Enrollment	enrollment_id	ENR-2026-001
	semester	Fall 2026
	status	Active
	credits	15
	student_id	STU-2026-001
Facility	facility_id	FAC-BLD-001
	name	Adams Hall Building
	type	Academic Building
	location	Main Campus
Faculty	id	FAC-101
	name	Dr. Sarah Smith
	dept	Computer Science
	email	sarah.smith@university.edu
	hire_date	2024-08-15

Entity	Field	Value
Finance	salary	95000
	fee_amount	12500.0
	scholarship	2500.0
	grant_amount	5000.0
Grant	invoice_no	INV-2026-8901
	payment_ref	PAY-7X3K2
	grant_id	GR-2026-045
	title	AI-Driven Personalized Learning
	amount	250000.0
	sponsor	NSF
Invoice	pi	Dr. Sarah Smith
	status	Active
	invoice_id	INV-2026-8901
	amount	12500.0
	due_date	2026-08-15
	status	Paid
LibraryResource	student_id	STU-2026-001
	resource_id	LR-2026-001
	title	Deep Learning - Goodfellow et al
	type	eBook
	status	Available
Payment	copies	5
	payment_id	PAY-7X3K2
	amount	12500.0
	method	Credit Card
	ref	CC-4242
Placement	invoice_id	INV-2026-8901
	placement_id	PL-2026-001
	student_id	STU-2026-001
	company	Google
	role	SWE Intern
Policy	status	Confirmed
	policy_id	POL-2026-001
	name	Student Data Privacy Policy
	category	Data Governance
	effective_date	2026-01-01
Program	program_id	PG-BSC-CS
	name	B.Sc. Computer Science

Entity	Field	Value
Prospect	level	Undergraduate
	dept	Computer Science
	id	PRO-2026-042
	name	Jane Smith
	email	jane.smith@prospect.com
	program	M.Sc. Data Science
Publication	source	Open Day
	publication_id	PUB-2026-001
	title	Deep Learning in Higher Education
	doi	10.1234/edu.2026.005
	journal	Journal of Educational AI
	date	2026-06-15
PurchaseOrder	grant_id	GR-2026-045
	po_id	PO-2026-301
	amount	6000.0
	date	2026-05-15
	status	Delivered
	vendor_id	VEN-2026-001
Report	report_id	RPT-2026-001
	name	Institutional KPI Dashboard
	type	Executive Dashboard
	frequency	Real-time
Research	grant_id	GR-2026-045
	title	AI-Driven Personalized Learning
	amount	250000.0
	sponsor	NSF
	pi	Dr. Sarah Smith
	status	Active
RiskAssessment	risk_id	RISK-2026-001
	area	Data Privacy
	score	Medium
	mitigation	Encryption and access controls implemented
Room	room_id	RM-HALLB-204
	building	Adams Hall
	capacity	60
	type	Lecture Theatre
Student	id	STU-2026-001
	name	John Doe

Entity	Field	Value
	email	john.doe@university.edu
	program	B.Sc. Computer Science
	semester	Fall 2026
	gpa	3.75
SupportCase	case_id	SC-2026-042
	type	Academic Counselling
	status	Open
	student_id	STU-2026-001
	priority	Medium
Syllabus	syllabus_id	SYL-2026-014
	course_id	CS501
	learning_outcomes	AI/ML fundamentals, neural networks, ethics
Timetable	timetable_id	TT-2026-FALL
	semester	Fall 2026
	status	Published
Vendor	vendor_id	VEN-2026-001
	name	LabTech Supplies Inc
	category	Laboratory Equipment
	status	Approved

Student Administration - BC008 BC014 BC019 BC044 BC052 BC032

End-to-end management of the student lifecycle from prospect engagement through graduation.

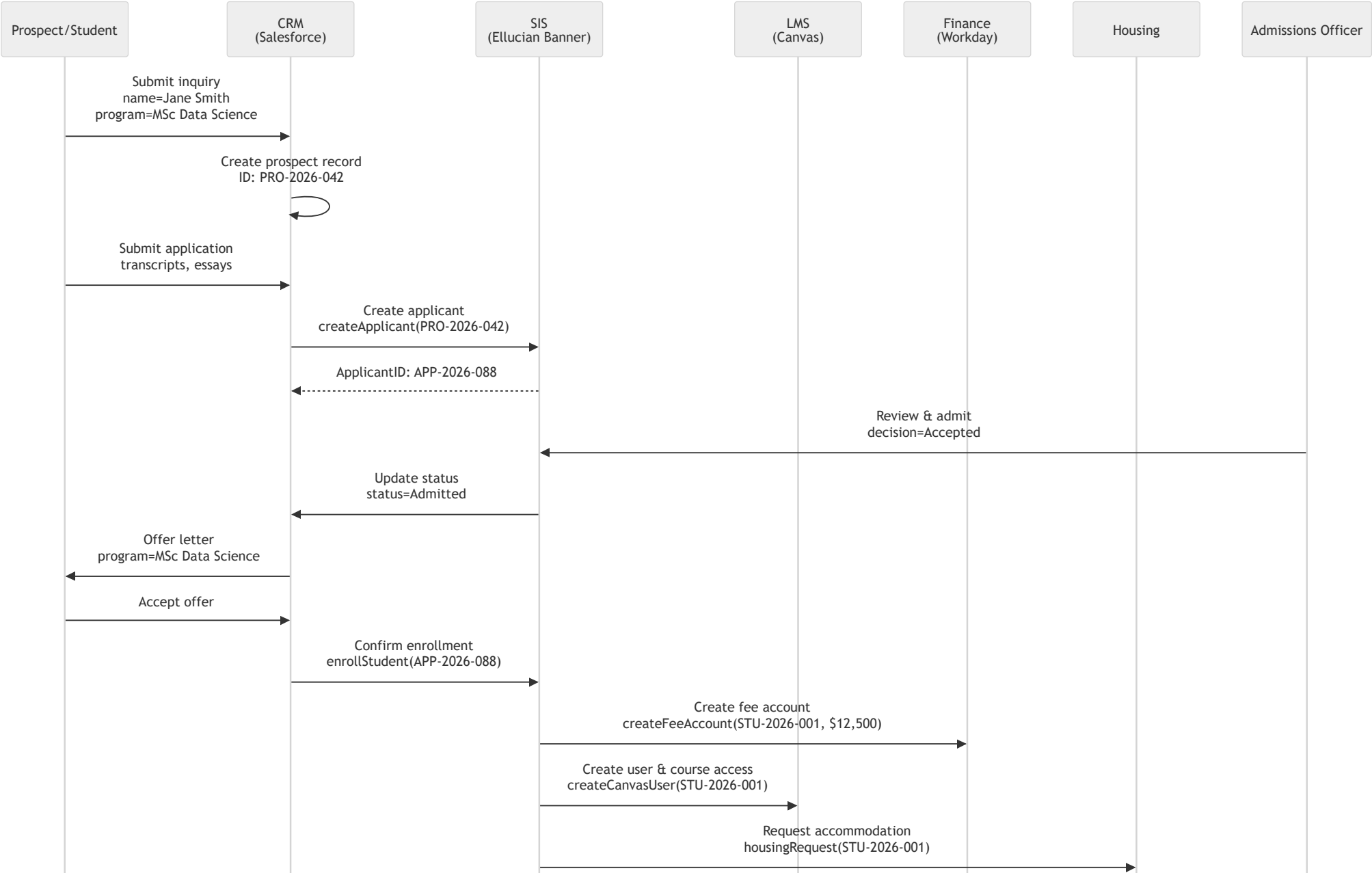
1.1 Student Onboarding (Prospect → Enrollment)

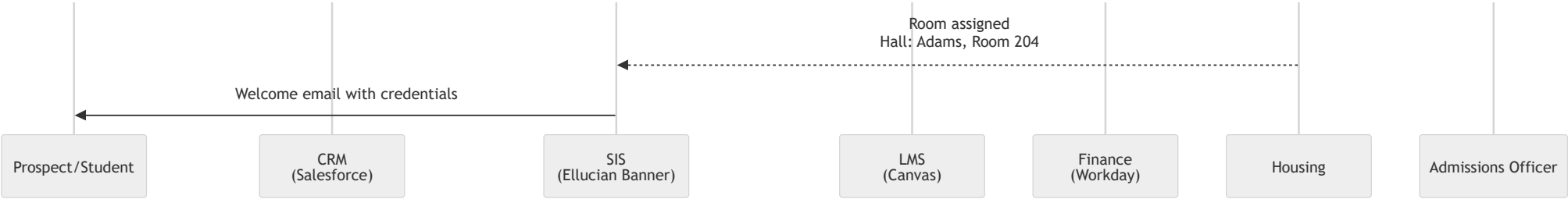
Description: A prospective student expresses interest, submits an application, is admitted, accepts the offer, enrolls, and registers for courses.

Actors: Prospect/Student, Admissions Officer

Systems: CRM, SIS, LMS, Finance, Housing

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Student Onboarding	Prospect	CreateProspect	CRM (Salesforce Education Cloud)	SIS (Ellucian Banner)	REST API (OAuth2)	REST API (API Key)	API-led (Real-time)	Medium
Student Onboarding	Application	SubmitApplication	CRM (Salesforce Education Cloud)	SIS (Ellucian Banner)	REST API (OAuth2)	REST API (API Key)	API-led (Real-time)	Medium
Student Onboarding	Student	EnrollStudent	SIS (Ellucian Banner)	LMS (Canvas)	Banner API (Basic Auth)	Canvas API (OAuth2)	Event-driven	Simple
Student Onboarding	Fee Account	CreateFeeAccount	SIS (Ellucian Banner)	Finance ERP (Workday Financials)	Banner Workflow (SOAP)	Workday API (WS-Security)	API-led (Real-time)	Medium
Student Onboarding	Transcript	SubmitTranscript	CRM (Salesforce Education Cloud)	Credential Evaluation Service	Salesforce REST API (OAuth2)	Credential API (REST)	API-led (Real-time)	Medium
Student Onboarding	Visa	InitiateVisaProcess	SIS (Ellucian Banner)	International Office Portal	Banner API (API Key)	Portal REST API	Event-driven	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.1-01	Data integrity: Prospect data transferred from CRM (Salesforce Education Cloud) to SIS (Ellucian Banner) via CreateProspect must be complete and consistent	CreateProspect	Prospect records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-02	Data integrity: Application data transferred from CRM (Salesforce Education Cloud) to SIS (Ellucian Banner) via SubmitApplication must be complete and consistent	SubmitApplication	Application records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-03	Data integrity: Student data transferred from SIS (Ellucian Banner) to LMS (Canvas) via EnrollStudent must be complete and consistent	EnrollStudent	Student records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.1-04	Data integrity: Fee Account data transferred from SIS (Ellucian Banner) to Finance ERP (Workday Financials) via CreateFeeAccount must be complete and consistent	CreateFeeAccount	Fee Account records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-05	Data integrity: Transcript data transferred from CRM (Salesforce Education Cloud) to Credential Evaluation Service via SubmitTranscript must be complete and consistent	SubmitTranscript	Transcript records in Credential Evaluation Service match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-06	Data integrity: Visa data transferred from SIS (Ellucian Banner) to International Office Portal via InitiateVisaProcess must be complete and consistent	InitiateVisaProcess	Visa records in International Office Portal match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-07	Error handling: Failed transactions between CRM (Salesforce Education Cloud) and SIS (Ellucian Banner) are logged with error context and trigger automatic retry or alert	CreateProspect	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-1.1-08	Security: All endpoints enforce authentication (Salesforce REST API (OAuth2), REST API (OAuth2), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

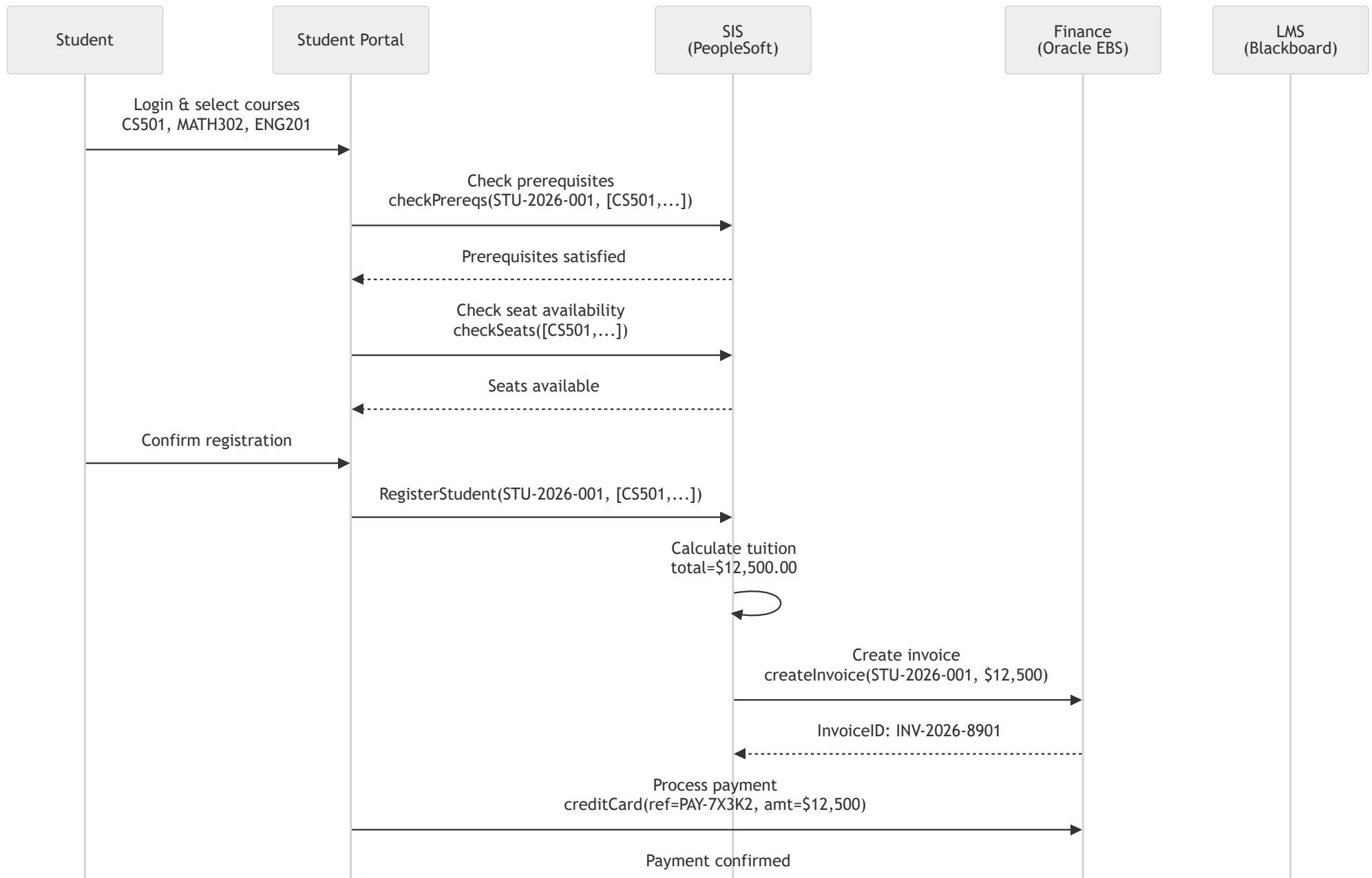
1.2 Enrollment & Registration

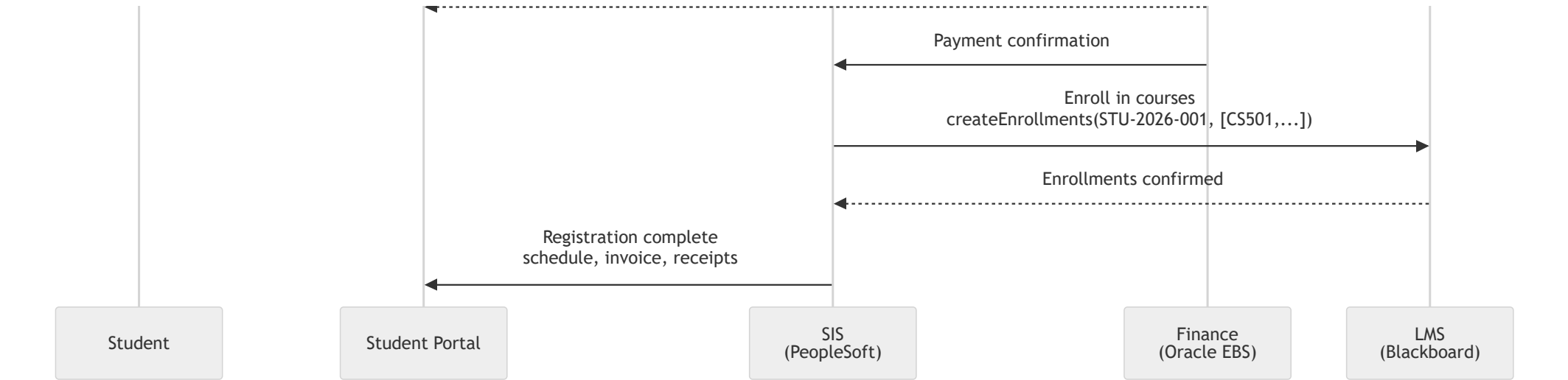
Description: Existing students register for courses each semester. System validates prerequisites, checks seat availability, processes fees, and confirms enrollment.

Actors: Student, Registrar

Systems: Student Portal, SIS, LMS, Finance

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Enrollment & Registration	Course Enrollment	RegisterCourse	Student Portal	SIS (PeopleSoft Campus Solutions)	Portal REST API	PeopleSoft REST API (JWT)	API-led (Real-time)	Simple
Enrollment & Registration	Invoice	CreateInvoice	SIS (PeopleSoft Campus Solutions)	Finance ERP (Oracle EBS)	PeopleSoft Integration Broker	Oracle EBS API (SOAP)	API-led (Real-time)	Medium
Enrollment & Registration	Payment	ProcessPayment	Payment Gateway (TouchNet)	Finance ERP (Oracle EBS)	Payment Gateway API	Oracle EBS REST API	Event-driven	High
Enrollment & Registration	Enrollment	SyncEnrollmentToLMS	SIS (PeopleSoft Campus Solutions)	LMS (Blackboard Learn)	PeopleSoft REST API (JWT)	Blackboard REST API (OAuth2)	Batch (Scheduled)	Simple
Enrollment & Registration	Schedule	BuildStudentSchedule	SIS (PeopleSoft Campus Solutions)	Student Portal	PeopleSoft REST API (JWT)	Portal REST API	API-led (Real-time)	Simple
Enrollment & Registration	Prerequisite	ValidatePrerequisite	Student Portal	SIS (PeopleSoft Campus Solutions)	Portal REST API	PeopleSoft REST API (JWT)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.2-01	Data integrity: Course Enrollment data transferred from Student Portal to SIS (PeopleSoft Campus Solutions) via RegisterCourse must be complete and consistent	RegisterCourse	Course Enrollment records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.2-02	Data integrity: Invoice data transferred from SIS (PeopleSoft Campus Solutions) to Finance ERP (Oracle EBS) via CreateInvoice must be complete and consistent	CreateInvoice	Invoice records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.2-03	Data integrity: Payment data transferred from Payment Gateway (TouchNet) to Finance ERP (Oracle EBS) via ProcessPayment must be complete and consistent	ProcessPayment	Payment records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-1.2-04	Data integrity: Enrollment data transferred from SIS (PeopleSoft Campus Solutions) to LMS (Blackboard Learn) via SyncEnrollmentToLMS must be complete and consistent	SyncEnrollmentToLMS	Enrollment records in LMS (Blackboard Learn) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.2-05	Data integrity: Schedule data transferred from SIS (PeopleSoft Campus Solutions) to Student Portal via BuildStudentSchedule must be complete and consistent	BuildStudentSchedule	Schedule records in Student Portal match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.2-06	Data integrity: Prerequisite data transferred from Student Portal to SIS (PeopleSoft Campus Solutions) via ValidatePrerequisite must be complete and consistent	ValidatePrerequisite	Prerequisite records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.2-07	Error handling: Failed transactions between Student Portal and SIS (PeopleSoft Campus Solutions) are logged with error context and trigger automatic retry or alert	RegisterCourse	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-1.2-08	Security: All endpoints enforce authentication (Payment Gateway API, PeopleSoft REST API (JWT), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

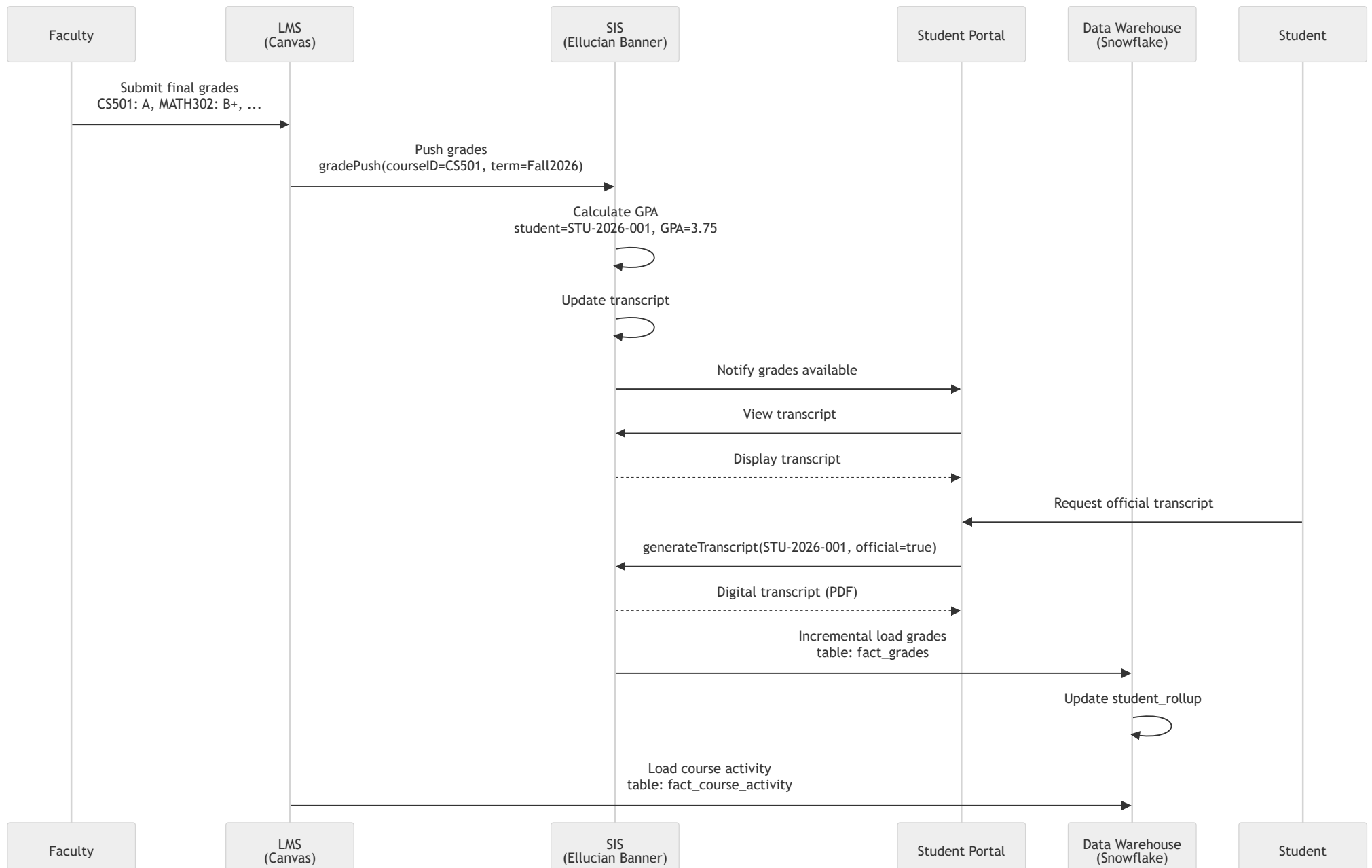
1.3 Academic Records Management

Description: Faculty submit grades. SIS calculates GPA, updates transcripts, notifies students. Transcript requests fulfilled digitally.

Actors: Faculty, Student, Registrar

Systems: LMS, SIS, Student Portal, Data Warehouse

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Academic Records	Grade	GradePush	LMS (Canvas)	SIS (Ellucian Banner)	Canvas API (OAuth2)	Banner API (API Key)	Batch (Scheduled)	Simple
Academic Records	Transcript	GenerateTranscript	SIS (Ellucian Banner)	Student Portal	Banner REST API	Portal REST API	API-led (Real-time)	Simple
Academic Records	Grade	GradeToDW	SIS (Ellucian Banner)	Data Warehouse (Snowflake)	Banner ODBC / CDC	Snowflake JDBC	Batch (ETL)	Medium
Academic Records	Enrollment	EnrollmentStatusToDW	SIS (Ellucian Banner)	Data Warehouse (Snowflake)	Banner ODBC	Snowflake JDBC	Batch (ETL)	Simple
Academic Records	Notification	NotifyGradeAvailable	SIS (Ellucian Banner)	Notification Service	Banner REST API	SMTP / Push API	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.3-01	Data integrity: Grade data transferred from LMS (Canvas) to SIS (Ellucian Banner) via GradePush must be complete and consistent	GradePush	Grade records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.3-02	Data integrity: Transcript data transferred from SIS (Ellucian Banner) to Student Portal via GenerateTranscript must be complete and consistent	GenerateTranscript	Transcript records in Student Portal match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.3-03	Data integrity: Grade data transferred from SIS (Ellucian Banner) to Data Warehouse (Snowflake) via GradeToDW must be complete and consistent	GradeToDW	Grade records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.3-04	Data integrity: Enrollment data transferred from SIS (Ellucian Banner) to Data Warehouse (Snowflake) via EnrollmentStatusToDW must be complete and consistent	EnrollmentStatusToDW	Enrollment records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.3-05	Data integrity: Notification data transferred from SIS (Ellucian Banner) to Notification Service via NotifyGradeAvailable must be complete and consistent	NotifyGradeAvailable	Notification records in Notification Service match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.3-06	Error handling: Failed transactions between LMS (Canvas) and SIS (Ellucian Banner) are logged with error context and trigger automatic retry or alert	GradePush	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-1.3-07	Security: All endpoints enforce authentication (Banner REST API, Banner ODBC / CDC, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

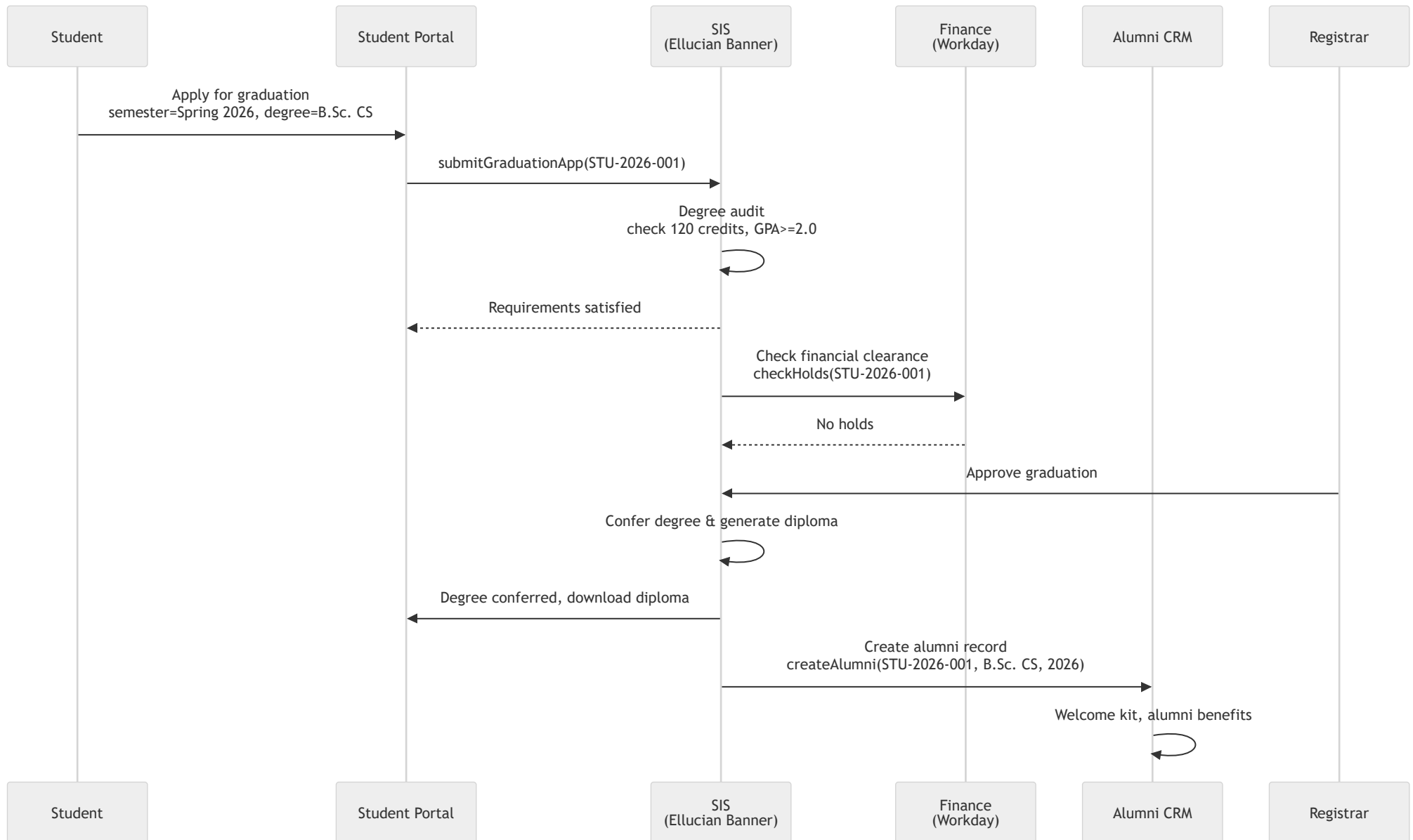
1.4 Graduation & Certification

Description: Students apply to graduate. System audits degree requirements, clears financial holds, confers degree, issues digital diploma.

Actors: Student, Registrar

Systems: Student Portal, SIS, Finance, Alumni CRM

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Graduation & Certification	Graduation Application	SubmitGradApp	Student Portal	SIS (Ellucian Banner)	Portal REST API	Banner REST API (API Key)	API-led (Real-time)	Simple
Graduation & Certification	Financial Clearance	CheckFinancialHolds	SIS (Ellucian Banner)	Finance ERP (Workday Financials)	Banner Workflow (SOAP)	Workday API (WS-Security)	API-led (Real-time)	Medium
Graduation & Certification	Alumni Record	CreateAlumniRecord	SIS (Ellucian Banner)	Alumni CRM (Blackbaud)	Banner REST API	Blackbaud API (OAuth2)	Event-driven	Simple
Graduation & Certification	Degree	GenerateDiploma	SIS (Ellucian Banner)	Diploma Printing Service	Banner REST API	Print Service API (SFTP)	Batch (Scheduled)	Medium
Graduation & Certification	Survey	SendGraduateSurvey	SIS (Ellucian Banner)	Alumni CRM (Blackbaud)	Banner REST API	Blackbaud API (OAuth2)	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.4-01	Data integrity: Graduation Application data transferred from Student Portal to SIS (Ellucian Banner) via SubmitGradApp must be complete and consistent	SubmitGradApp	Graduation Application records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.4-02	Data integrity: Financial Clearance data transferred from SIS (Ellucian Banner) to Finance ERP (Workday Financials) via CheckFinancialHolds must be complete and consistent	CheckFinancialHolds	Financial Clearance records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-1.4-03	Data integrity: Alumni Record data transferred from SIS (Ellucian Banner) to Alumni CRM (Blackbaud) via CreateAlumniRecord must be complete and consistent	CreateAlumniRecord	Alumni Record records in Alumni CRM (Blackbaud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.4-04	Data integrity: Degree data transferred from SIS (Ellucian Banner) to Diploma Printing Service via GenerateDiploma must be complete and consistent	GenerateDiploma	Degree records in Diploma Printing Service match source; no data loss; reconciliation report confirms accuracy	High
AC-1.4-05	Data integrity: Survey data transferred from SIS (Ellucian Banner) to Alumni CRM (Blackbaud) via SendGraduateSurvey must be complete and consistent	SendGraduateSurvey	Survey records in Alumni CRM (Blackbaud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-1.4-06	Error handling: Failed transactions between Student Portal and SIS (Ellucian Banner) are logged with error context and trigger automatic retry or alert	SubmitGradApp	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-1.4-07	Security: All endpoints enforce authentication (Banner REST API, Banner Workflow (SOAP), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	C

AC ID	Description	Related Info Flow	Expected Outcome	Priority
				ritical

Learning & Teaching - BC001 BC023 BC028

Design, develop, deliver, and assess academic programs and courses. Spans curriculum planning, timetable scheduling, delivery, and assessment.

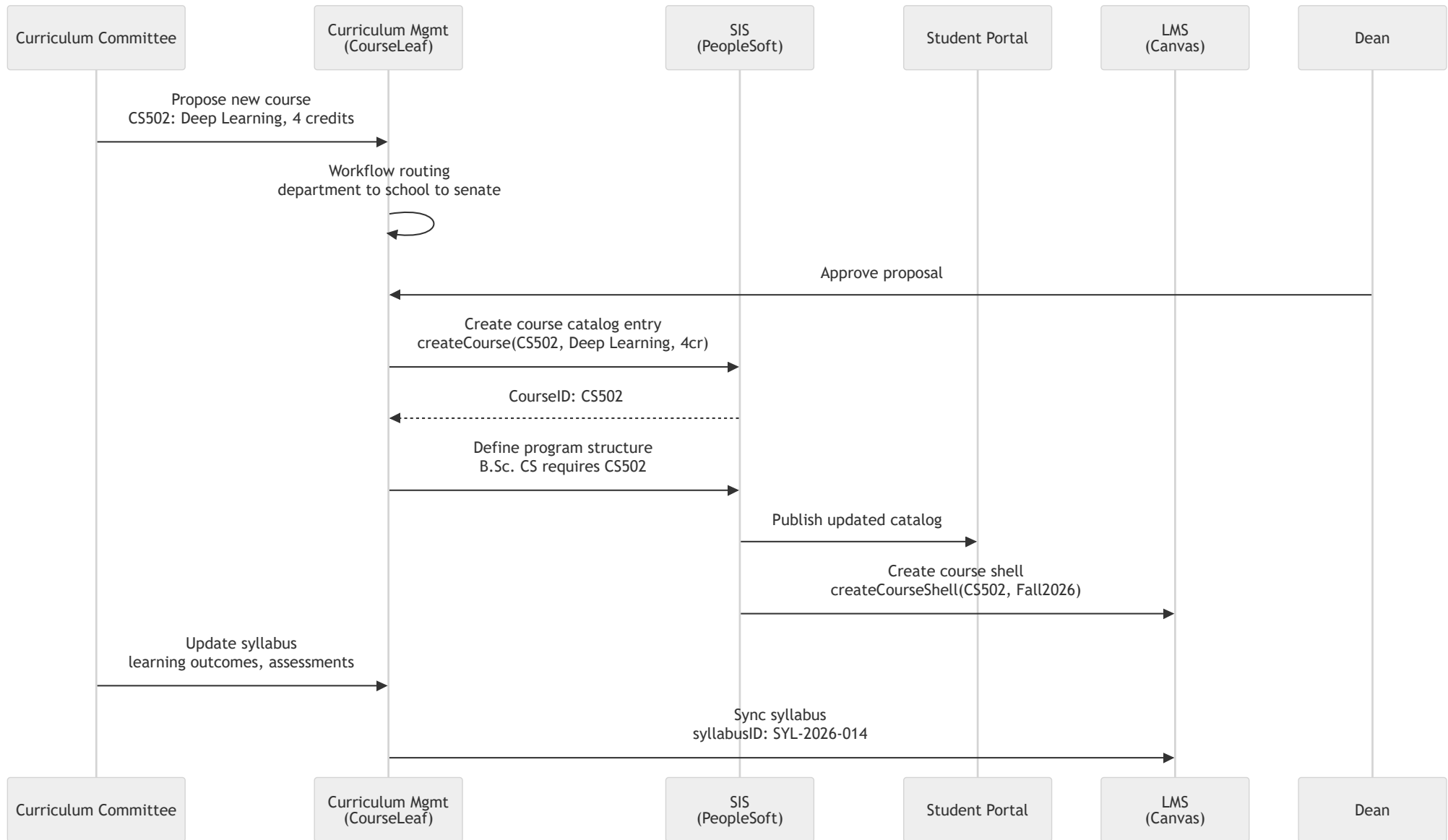
2.1 Curriculum Management

Description: Academic committees design new programs/courses. Approved curricula published to SIS catalog and LMS.

Actors: Curriculum Committee, Dean, Registrar

Systems: Curriculum Mgmt System, SIS, Student Portal, LMS

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Curriculum Management	Course	CreateCourse	Curriculum Mgmt (CourseLeaf)	SIS (PeopleSoft Campus Solutions)	CourseLeaf CIM API	PeopleSoft REST API (JWT)	API-led (Real-time)	Medium
Curriculum Management	Course Shell	CreateCourseShell	SIS (PeopleSoft Campus Solutions)	LMS (Canvas)	PeopleSoft REST API (JWT)	Canvas API (OAuth2)	Event-driven	Simple
Curriculum Management	Syllabus	SyncSyllabus	Curriculum Mgmt (CourseLeaf)	LMS (Canvas)	CourseLeaf REST API	Canvas API (OAuth2)	Batch (Scheduled)	Simple
Curriculum Management	Program	DefineProgramStructure	Curriculum Mgmt (CourseLeaf)	SIS (PeopleSoft Campus Solutions)	CourseLeaf CIM API	PeopleSoft REST API (JWT)	API-led (Real-time)	Medium
Curriculum Management	Catalog	PublishCatalog	SIS (PeopleSoft Campus Solutions)	Student Portal	PeopleSoft REST API (JWT)	Portal REST API	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.1-01	Data integrity: Course data transferred from Curriculum Mgmt (CourseLeaf) to SIS (PeopleSoft Campus Solutions) via CreateCourse must be complete and consistent	CreateCourse	Course records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.1-02	Data integrity: Course Shell data transferred from SIS (PeopleSoft Campus Solutions) to LMS (Canvas) via CreateCourseShell must be complete and consistent	CreateCourseShell	Course Shell records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.1-03	Data integrity: Syllabus data transferred from Curriculum Mgmt (CourseLeaf) to LMS (Canvas) via SyncSyllabus must be complete and consistent	SyncSyllabus	Syllabus records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.1-04	Data integrity: Program data transferred from Curriculum Mgmt (CourseLeaf) to SIS (PeopleSoft Campus Solutions) via DefineProgramStructure must be complete and consistent	DefineProgramStructure	Program records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.1-05	Data integrity: Catalog data transferred from SIS (PeopleSoft Campus Solutions) to Student Portal via PublishCatalog must be complete and consistent	PublishCatalog	Catalog records in Student Portal match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.1-06	Error handling: Failed transactions between Curriculum Mgmt (CourseLeaf) and SIS (PeopleSoft Campus Solutions) are logged with error context and trigger automatic retry or alert	CreateCourse	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-2.1-07	Security: All endpoints enforce authentication (PeopleSoft REST API (JWT), CourseLeaf REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

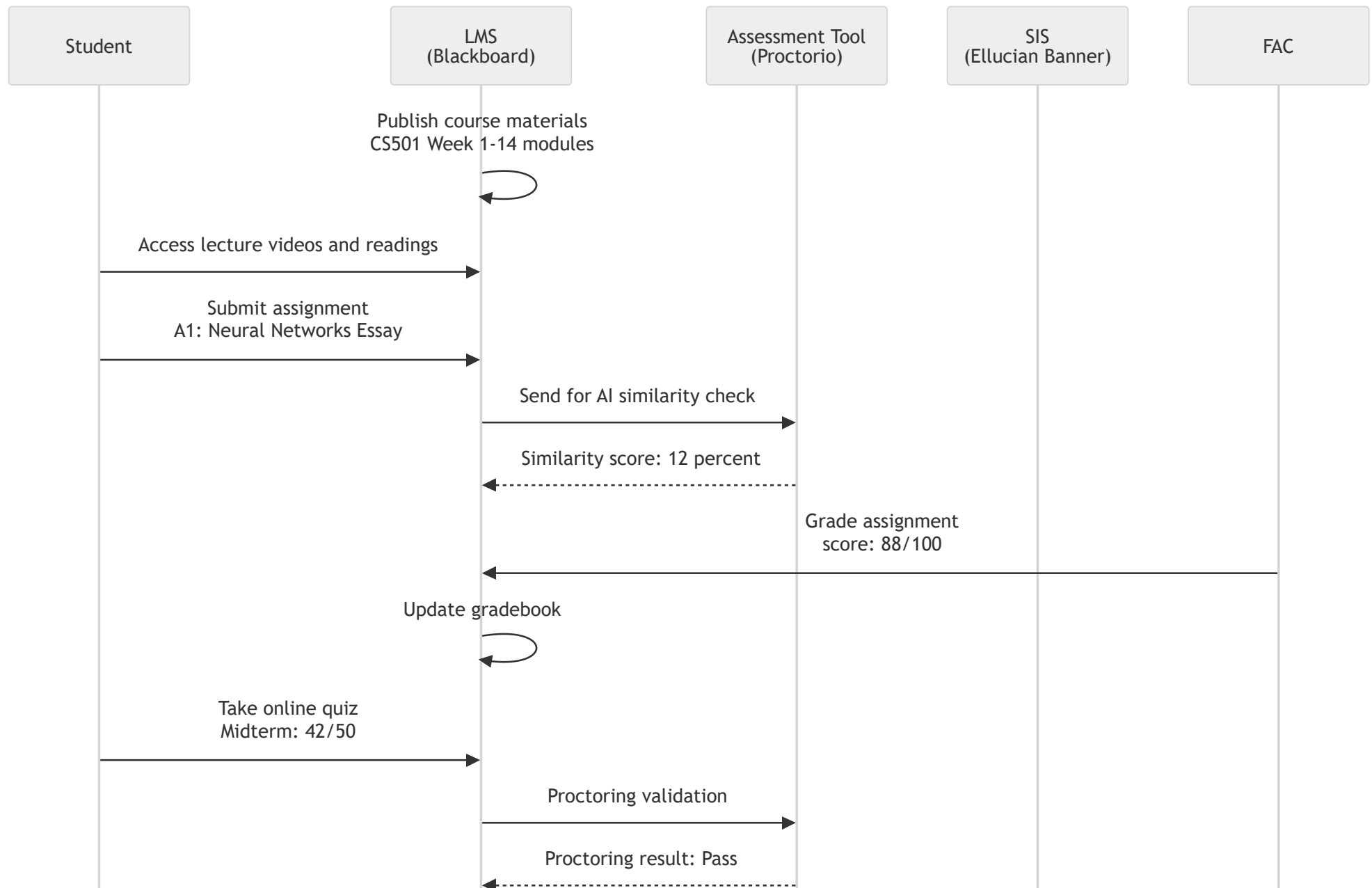
2.2 Course Delivery & Assessment

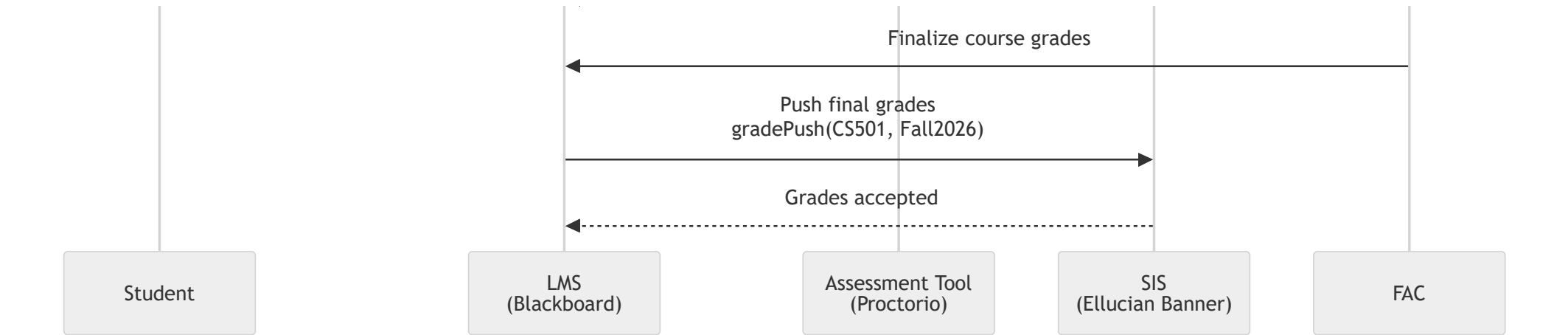
Description: Courses delivered via LMS. Students submit assignments, take quizzes. Faculty grade and final grades push to SIS.

Actors: Student, Faculty

Systems: LMS, Assessment Tool, SIS

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Course Delivery & Assessment	Assignment Submission	SubmitAssignment	LMS (Blackboard Learn)	Assessment Tool (Proctorio)	Blackboard REST API (OAuth2)	Proctorio API (API Key)	API-led (Real-time)	Medium
Course Delivery & Assessment	Final Grade	GradePushToSIS	LMS (Blackboard Learn)	SIS (Ellucian Banner)	Blackboard REST API (OAuth2)	Banner API (API Key)	Batch (Scheduled)	Simple
Course Delivery & Assessment	Content	PublishCourseContent	LMS (Blackboard Learn)	LMS (Blackboard Learn)	Internal	Internal	API-led (Real-time)	Simple
Course Delivery & Assessment	Gradebook	SyncGradebook	LMS (Blackboard Learn)	Assessment Tool (Proctorio)	Blackboard REST API (OAuth2)	Proctorio API (API Key)	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.2-01	Data integrity: Assignment Submission data transferred from LMS (Blackboard Learn) to Assessment Tool (Proctorio) via SubmitAssignment must be complete and consistent	SubmitAssignment	Assignment Submission records in Assessment Tool (Proctorio) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.2-02	Data integrity: Final Grade data transferred from LMS (Blackboard Learn) to SIS (Ellucian Banner) via GradePushToSIS must be complete and consistent	GradePushToSIS	Final Grade records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.2-03	Data integrity: Content data transferred from LMS (Blackboard Learn) to LMS (Blackboard Learn) via PublishCourseContent must be complete and consistent	PublishCourseContent	Content records in LMS (Blackboard Learn) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.2-04	Data integrity: Gradebook data transferred from LMS (Blackboard Learn) to Assessment Tool (Proctorio) via SyncGradebook must be complete and consistent	SyncGradebook	Gradebook records in Assessment Tool (Proctorio) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.2-05	Error handling: Failed transactions between LMS (Blackboard Learn) and Assessment Tool (Proctorio) are logged with error context and trigger automatic retry or alert	SubmitAssignment	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-2.2-06	Security: All endpoints enforce authentication (Blackboard REST API (OAuth2), Internal, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

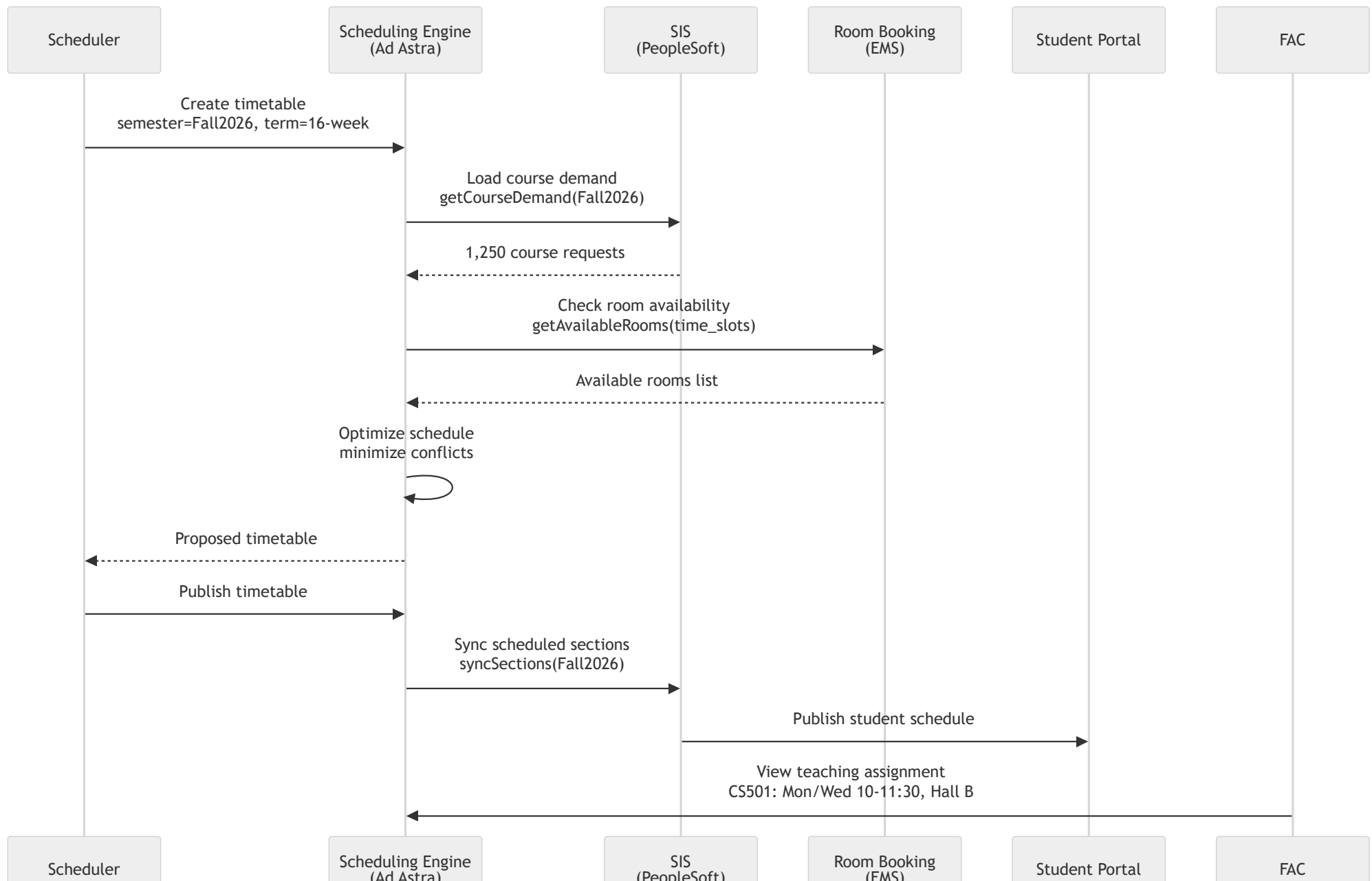
2.3 Timetable & Scheduling

Description: Schedulers create master timetable, assign rooms, check conflicts, publish schedules.

Actors: Scheduler, Faculty, Student

Systems: Scheduling Engine, SIS, Room Booking, Portal

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Timetable & Scheduling	Course Demand	LoadCourseDemand	SIS (PeopleSoft Campus Solutions)	Scheduling Engine (Ad Astra)	PeopleSoft REST API (JWT)	Ad Astra API (API Key)	Batch (Scheduled)	Simple
Timetable & Scheduling	Room Availability	CheckRoomAvailability	Scheduling Engine (Ad Astra)	Room Booking (EMS)	Ad Astra REST API	EMS Web API (Basic Auth)	API-led (Real-time)	Medium
Timetable & Scheduling	Scheduled Section	SyncSections	Scheduling Engine (Ad Astra)	SIS (PeopleSoft Campus Solutions)	Ad Astra REST API	PeopleSoft REST API (JWT)	Batch (Scheduled)	Simple
Timetable & Scheduling	Conflict	DetectScheduleConflict	Scheduling Engine (Ad Astra)	Scheduling Engine (Ad Astra)	Internal	Internal	Batch (Scheduled)	Medium
Timetable & Scheduling	Faculty Assignment	AssignFaculty	Scheduling Engine (Ad Astra)	SIS (PeopleSoft Campus Solutions)	Ad Astra REST API	PeopleSoft REST API (JWT)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.3-01	Data integrity: Course Demand data transferred from SIS (PeopleSoft Campus Solutions) to Scheduling Engine (Ad Astra) via LoadCourseDemand must be complete and consistent	LoadCourseDemand	Course Demand records in Scheduling Engine (Ad Astra) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.3-02	Data integrity: Room Availability data transferred from Scheduling Engine (Ad Astra) to Room Booking (EMS) via CheckRoomAvailability must be complete and consistent	CheckRoomAvailability	Room Availability records in Room Booking (EMS) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.3-03	Data integrity: Scheduled Section data transferred from Scheduling Engine (Ad Astra) to SIS (PeopleSoft Campus Solutions) via SyncSections must be complete and consistent	SyncSections	Scheduled Section records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.3-04	Data integrity: Conflict data transferred from Scheduling Engine (Ad Astra) to Scheduling Engine (Ad Astra) via DetectScheduleConflict must be complete and consistent	DetectScheduleConflict	Conflict records in Scheduling Engine (Ad Astra) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.3-05	Data integrity: Faculty Assignment data transferred from Scheduling Engine (Ad Astra) to SIS (PeopleSoft Campus Solutions) via AssignFaculty must be complete and consistent	AssignFaculty	Faculty Assignment records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.3-06	Error handling: Failed transactions between SIS (PeopleSoft Campus Solutions) and Scheduling Engine (Ad Astra) are logged with error context and trigger automatic retry or alert	LoadCourseDemand	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-2.3-07	Security: All endpoints enforce authentication (PeopleSoft REST API (JWT), Ad Astra REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

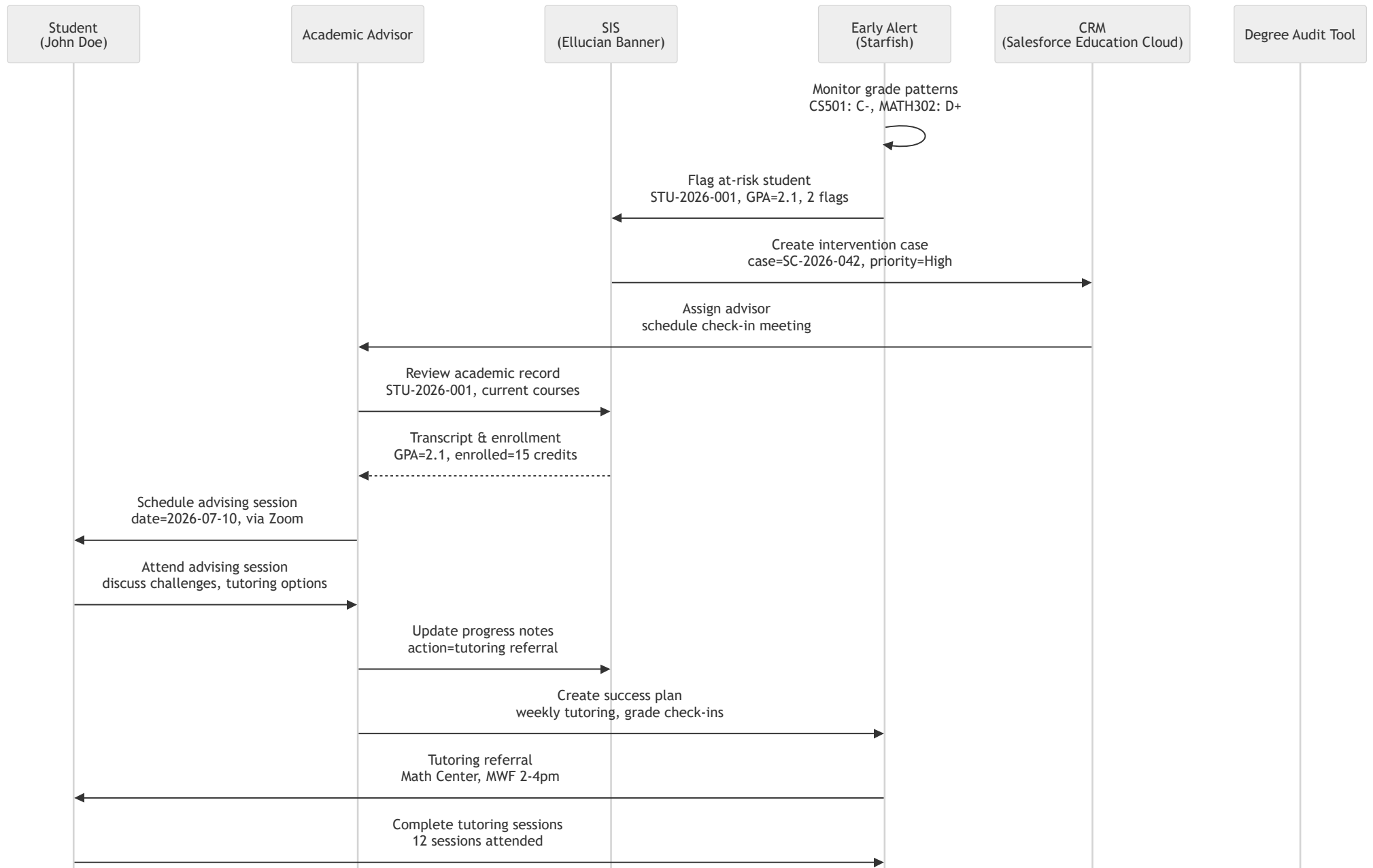
2.4 Academic Advising & Progression Management

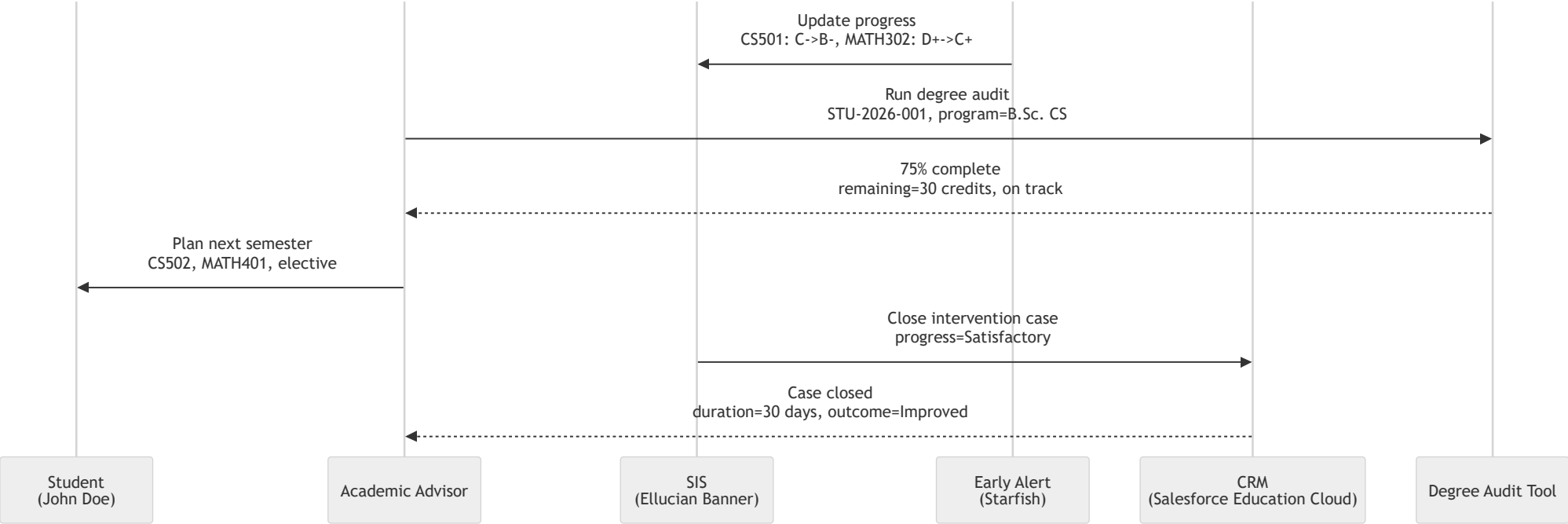
Description: Academic advisors monitor student progress, review grades and course completion, intervene for at-risk students, and plan degree pathways.

Actors: Student, Academic Advisor, Registrar

Systems: SIS (Ellucian Banner), CRM (Salesforce Education Cloud), Early Alert System (Starfish), LMS (Canvas), Degree Audit Tool

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Academic Advising	Flag	FlagAtRiskStudent	Early Alert (Starfish)	SIS (Ellucian Banner)	Starfish API (API Key)	Banner API (API Key)	Event-driven	Medium
Academic Advising	Case	CreateInterventionCase	SIS (Ellucian Banner)	CRM (Salesforce Education Cloud)	Banner API (API Key)	Salesforce REST API (OAuth2)	Event-driven	Medium
Academic Advising	Advising	ScheduleAdvisingSession	CRM (Salesforce Education Cloud)	CRM (Salesforce Education Cloud)	Internal	Internal	API-led (Real-time)	Simple
Academic Advising	Success Plan	CreateSuccessPlan	Early Alert (Starfish)	Early Alert (Starfish)	Internal	Internal	API-led (Real-time)	Simple
Academic Advising	Degree Audit	RunDegreeAudit	Degree Audit Tool	SIS (Ellucian Banner)	Degree Audit API	Banner API (API Key)	API-led (Real-time)	Medium
Academic Advising	Progress	UpdateAcademicProgress	SIS (Ellucian Banner)	CRM (Salesforce Education Cloud)	Banner API (API Key)	Salesforce REST API (OAuth2)	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.4-01	Data integrity: Flag data transferred from Early Alert (Starfish) to SIS (Ellucian Banner) via FlagAtRiskStudent must be complete and consistent	FlagAtRiskStudent	Flag records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.4-02	Data integrity: Case data transferred from SIS (Ellucian Banner) to CRM (Salesforce Education Cloud) via CreateInterventionCase must be complete and consistent	CreateInterventionCase	Case records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.4-03	Data integrity: Advising data transferred from CRM (Salesforce Education Cloud) to CRM (Salesforce Education Cloud) via ScheduleAdvisingSession must be complete and consistent	ScheduleAdvisingSession	Advising records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.4-04	Data integrity: Success Plan data transferred from Early Alert (Starfish) to Early Alert (Starfish) via CreateSuccessPlan must be complete and consistent	CreateSuccessPlan	Success Plan records in Early Alert (Starfish) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.4-05	Data integrity: Degree Audit data transferred from Degree Audit Tool to SIS (Ellucian Banner) via RunDegreeAudit must be complete and consistent	RunDegreeAudit	Degree Audit records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-2.4-06	Data integrity: Progress data transferred from SIS (Ellucian Banner) to CRM (Salesforce Education Cloud) via UpdateAcademicProgress must be complete and consistent	UpdateAcademicProgress	Progress records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-2.4-07	Error handling: Failed transactions between Early Alert (Starfish) and SIS (Ellucian Banner) are logged with error context and trigger automatic retry or alert	FlagAtRiskStudent	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-2.4-08	Security: All endpoints enforce authentication (Degree Audit API, Banner API (API Key), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; inv	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			alid/expired tokens logged	

HR & Faculty Management - BC171

Full lifecycle of faculty and staff: recruitment, onboarding, payroll, performance management, professional development.

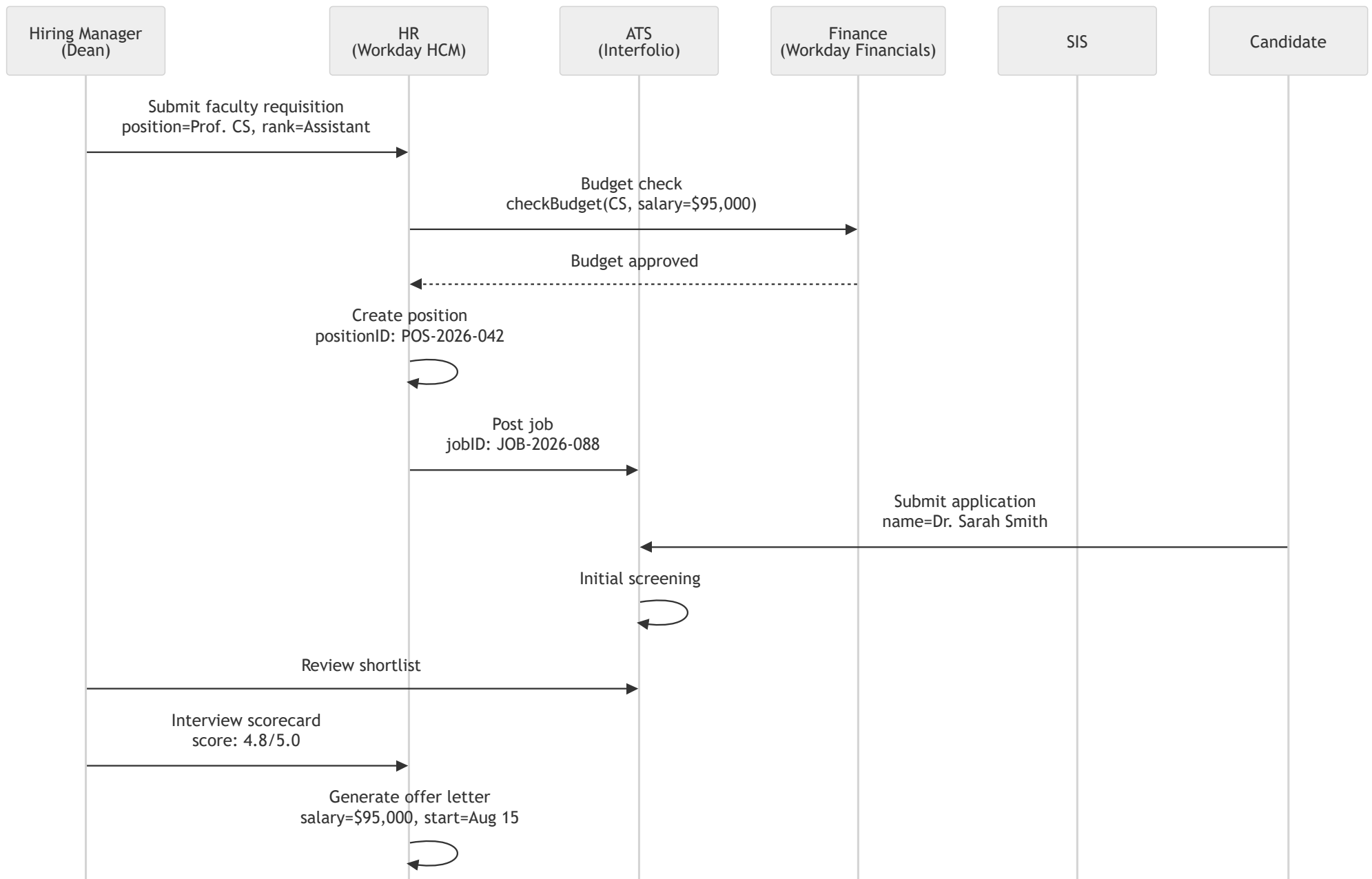
3.1 Faculty Recruitment

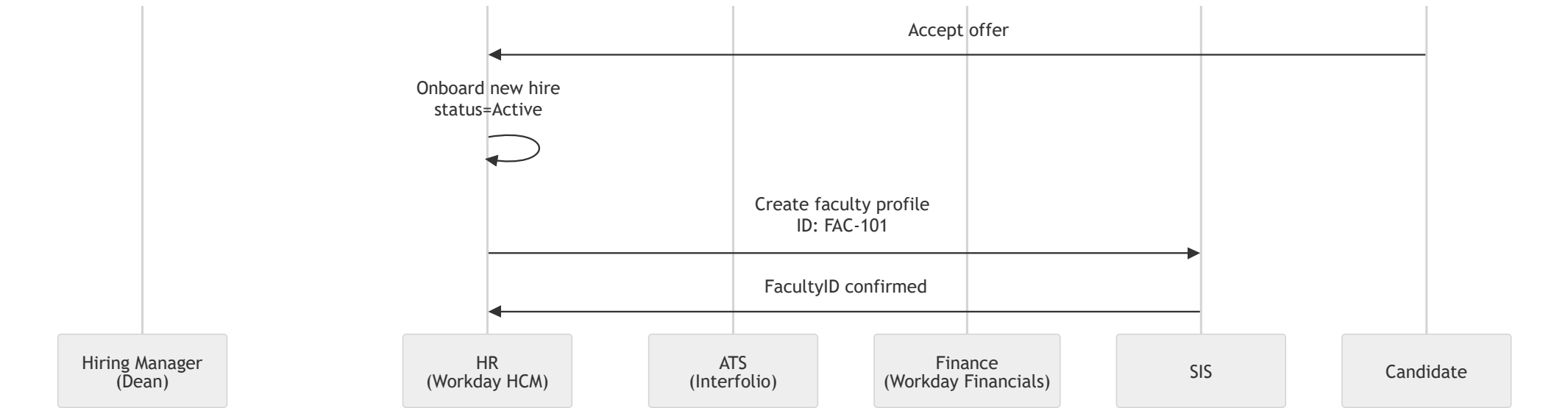
Description: Departments request new positions. HR posts job, screens applicants, coordinates interviews, manages offer/acceptance.

Actors: Hiring Manager, HR, Candidate

Systems: HRIS, ATS, Finance, SIS

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Faculty Recruitment	Position Requisition	SubmitRequisition	HRIS (Workday HCM)	Finance ERP (Workday Financials)	Workday API (WS-Security)	Workday API (WS-Security)	API-led (Real-time)	Simple
Faculty Recruitment	Job Posting	PostJob	HRIS (Workday HCM)	ATS (Interfolio)	Workday Recruiting API	Interfolio API (OAuth2)	API-led (Real-time)	Medium
Faculty Recruitment	Faculty Profile	CreateFacultyProfile	HRIS (Workday HCM)	SIS (Ellucian Banner)	Workday API (WS-Security)	Banner API (API Key)	Event-driven	Simple
Faculty Recruitment	Background	RunBackgroundCheck	HRIS (Workday HCM)	Background Check Provider	Workday API (WS-Security)	Provider API (SOAP)	API-led (Real-time)	High
Faculty Recruitment	Orientation	ScheduleOrientation	HRIS (Workday HCM)	LMS (Canvas)	Workday API (WS-Security)	Canvas API (OAuth2)	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-3.1-01	Data integrity: Position Requisition data transferred from HRIS (Workday HCM) to Finance ERP (Workday Financials) via SubmitRequisition must be complete and consistent	SubmitRequisition	Position Requisition records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.1-02	Data integrity: Job Posting data transferred from HRIS (Workday HCM) to ATS (Interfolio) via PostJob must be complete and consistent	PostJob	Job Posting records in ATS (Interfolio) match source; no data loss; reconciliation report confirms accuracy	High
AC-3.1-03	Data integrity: Faculty Profile data transferred from HRIS (Workday HCM) to SIS (Ellucian Banner) via CreateFacultyProfile must be complete and consistent	CreateFacultyProfile	Faculty Profile records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.1-04	Data integrity: Background data transferred from HRIS (Workday HCM) to Background Check Provider via RunBackgroundCheck must be complete and consistent	RunBackgroundCheck	Background records in Background Check Provider match source; no data loss; reconciliation report confirms accuracy	Critical
AC-3.1-05	Data integrity: Orientation data transferred from HRIS (Workday HCM) to LMS (Canvas) via ScheduleOrientation must be complete and consistent	ScheduleOrientation	Orientation records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.1-06	Error handling: Failed transactions between HRIS (Workday HCM) and Finance ERP (Workday Financials) are logged with error context and trigger automatic retry or alert	SubmitRequisition	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-3.1-07	Security: All endpoints enforce authentication (Workday API (WS-Security), Workday Recruiting API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

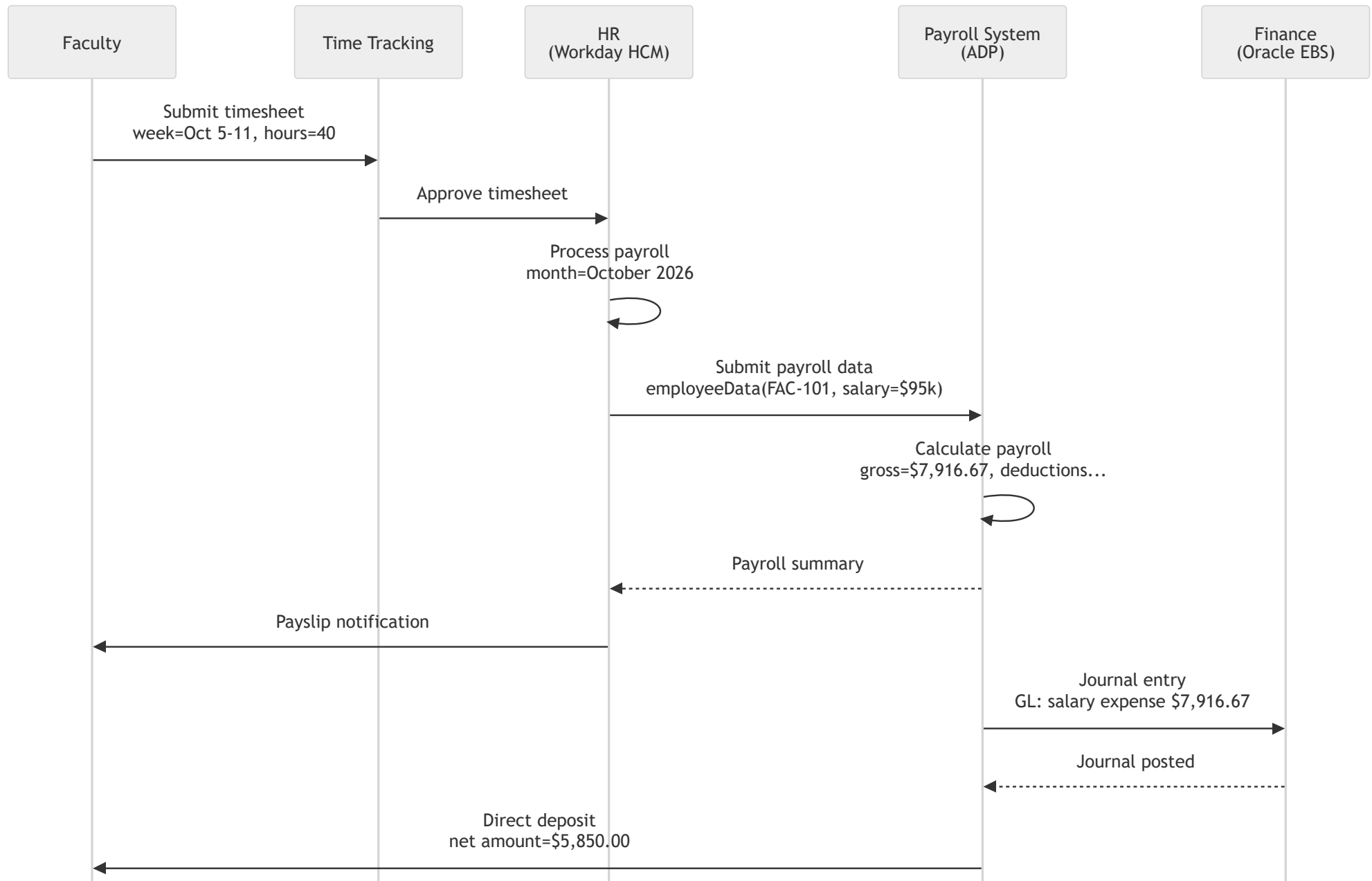
3.2 Payroll & Benefits

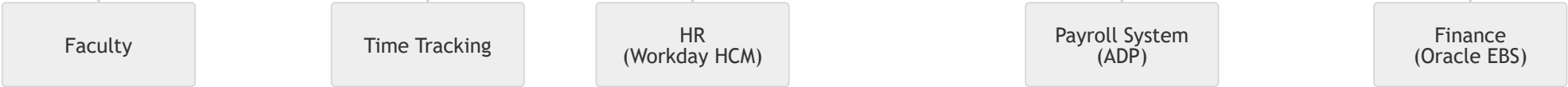
Description: Monthly payroll: time/attendance to payroll, calculations, payslips, GL journal entries.

Actors: Faculty, HR, Finance

Systems: HRIS, Payroll, Finance, Time Tracking

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Payroll & Benefits	Timesheet	SubmitTimesheet	Time Tracking System	HRIS (Workday HCM)	Time API (REST)	Workday API (WS-Security)	API-led (Real-time)	Simple
Payroll & Benefits	Payroll Data	SubmitPayroll	HRIS (Workday HCM)	Payroll System (ADP)	Workday Payroll Interface	ADP API (SFTP)	Batch (Scheduled)	High
Payroll & Benefits	Journal Entry	PostGLJournal	Payroll System (ADP)	Finance ERP (Oracle EBS)	ADP GL Interface (SFTP)	Oracle EBS REST API (SOAP)	Batch (Scheduled)	Medium
Payroll & Benefits	Benefits	EnrollBenefits	HRIS (Workday HCM)	Benefits Provider	Workday API (WS-Security)	Provider API (REST)	API-led (Real-time)	Medium
Payroll & Benefits	Tax	CalculateTaxWithholding	Payroll System (ADP)	Tax Authority Portal	ADP API (SOAP)	HMRC RTI API (SOAP)	Batch (Scheduled)	High

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-3.2-01	Data integrity: Timesheet data transferred from Time Tracking System to HRIS (Workday HCM) via SubmitTimesheet must be complete and consistent	SubmitTimesheet	Timesheet records in HRIS (Workday HCM) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.2-02	Data integrity: Payroll Data data transferred from HRIS (Workday HCM) to Payroll System (ADP) via SubmitPayroll must be complete and consistent	SubmitPayroll	Payroll Data records in Payroll System (ADP) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-3.2-03	Data integrity: Journal Entry data transferred from Payroll System (ADP) to Finance ERP (Oracle EBS) via PostGLJournal must be complete and consistent	PostGLJournal	Journal Entry records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	High
AC-3.2-04	Data integrity: Benefits data transferred from HRIS (Workday HCM) to Benefits Provider via EnrollBenefits must be complete and consistent	EnrollBenefits	Benefits records in Benefits Provider match source; no data loss; reconciliation report confirms accuracy	High
AC-3.2-05	Data integrity: Tax data transferred from Payroll System (ADP) to Tax Authority Portal via CalculateTaxWithholding must be complete and consistent	CalculateTaxWithholding	Tax records in Tax Authority Portal match source; no data loss; reconciliation report confirms accuracy	Critical
AC-3.2-06	Error handling: Failed transactions between Time Tracking System and HRIS (Workday HCM) are logged with error context and trigger automatic retry or alert	SubmitTimesheet	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-3.2-07	Security: All endpoints enforce authentication (ADP GL Interface (SFTP), ADP API (SOAP), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

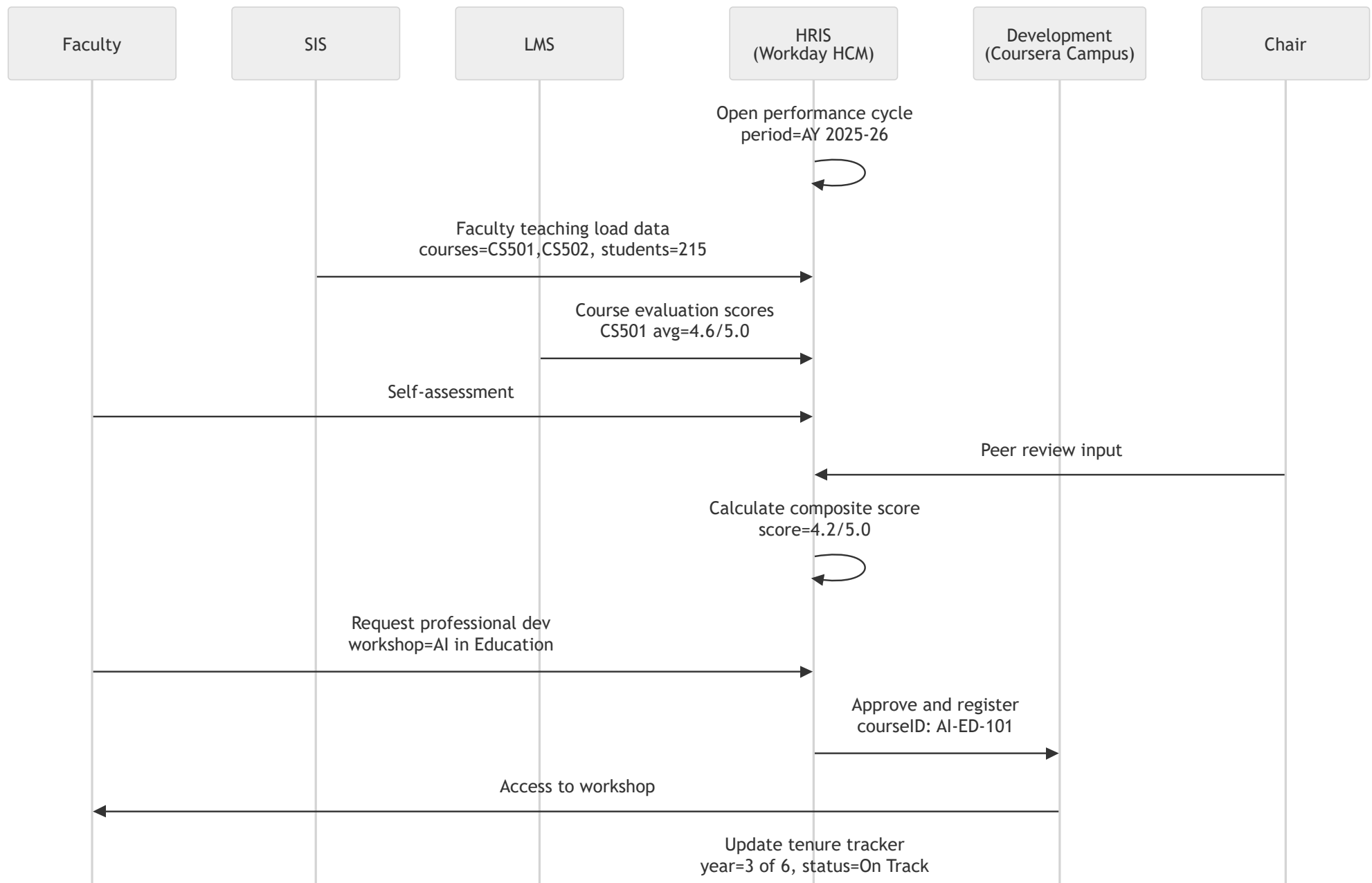
3.3 Faculty Performance & Development

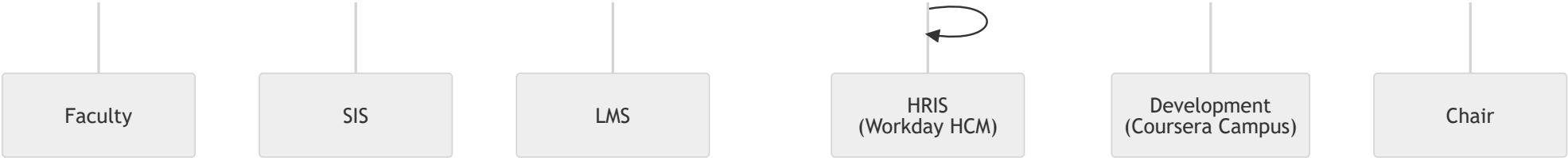
Description: Performance reviews, professional development, tenure tracking with inputs from SIS and LMS.

Actors: Faculty, Department Chair, HR

Systems: HRIS, LMS, SIS, Development Platform

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Faculty Performance & Development	Teaching Load	GetTeachingLoad	SIS (Ellucian Banner)	HRIS (Workday HCM)	Banner REST API	Workday API (WS-Security)	Batch (Scheduled)	Simple
Faculty Performance & Development	Course Evaluation	SyncEvalScores	LMS (Canvas)	HRIS (Workday HCM)	Canvas API (OAuth2)	Workday API (WS-Security)	Batch (Scheduled)	Medium
Faculty Performance & Development	Development Request	ApproveTraining	HRIS (Workday HCM)	Development Platform (Coursera Campus)	Workday API (WS-Security)	Coursera API (OAuth2)	API-led (Real-time)	Simple
Faculty Performance & Development	Goal	SetPerformanceGoal	HRIS (Workday HCM)	HRIS (Workday HCM)	Internal	Internal	API-led (Real-time)	Simple
Faculty Performance & Development	Review	SubmitPerformanceReview	HRIS (Workday HCM)	HRIS (Workday HCM)	Internal	Internal	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-3.3-01	Data integrity: Teaching Load data transferred from SIS (Ellucian Banner) to HRIS (Workday HCM) via GetTeachingLoad must be complete and consistent	GetTeachingLoad	Teaching Load records in HRIS (Workday HCM) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.3-02	Data integrity: Course Evaluation data transferred from LMS (Canvas) to HRIS (Workday HCM) via SyncEvalScores must be complete and consistent	SyncEvalScores	Course Evaluation records in HRIS (Workday HCM) match source; no data loss; reconciliation report confirms accuracy	High
AC-3.3-03	Data integrity: Development Request data transferred from HRIS (Workday HCM) to Development Platform (Coursera Campus) via ApproveTraining must be complete and consistent	ApproveTraining	Development Request records in Development Platform (Coursera Campus) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.3-04	Data integrity: Goal data transferred from HRIS (Workday HCM) to HRIS (Workday HCM) via SetPerformanceGoal must be complete and consistent	SetPerformanceGoal	Goal records in HRIS (Workday HCM) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-3.3-05	Data integrity: Review data transferred from HRIS (Workday HCM) to HRIS (Workday HCM) via SubmitPerformanceReview must be complete and consistent	SubmitPerformanceReview	Review records in HRIS (Workday HCM) match source; no data loss; reconciliation report confirms accuracy	High
AC-3.3-06	Error handling: Failed transactions between SIS (Ellucian Banner) and HRIS (Workday HCM) are logged with error context and trigger automatic retry or alert	GetTeachingLoad	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-3.3-07	Security: All endpoints enforce authentication (Banner REST API, Workday API (WS-Security), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected wit	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			h 401; tokens validated per call; invalid/expired tokens logged	

Finance & Resource Management - BC184

Manage financial resources: student fees, financial aid, procurement, budgeting, general ledger.

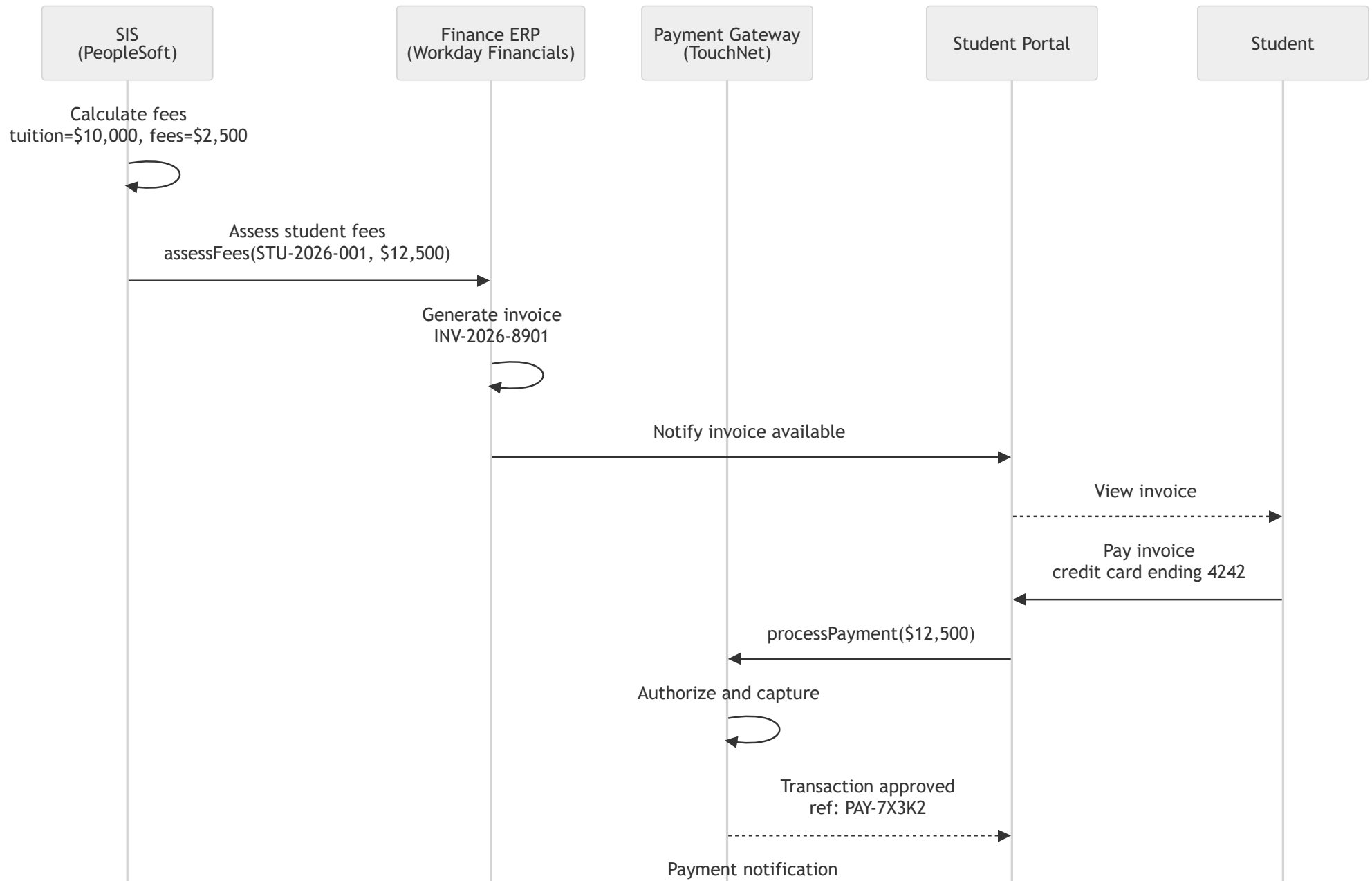
4.1 Fee Assessment & Billing

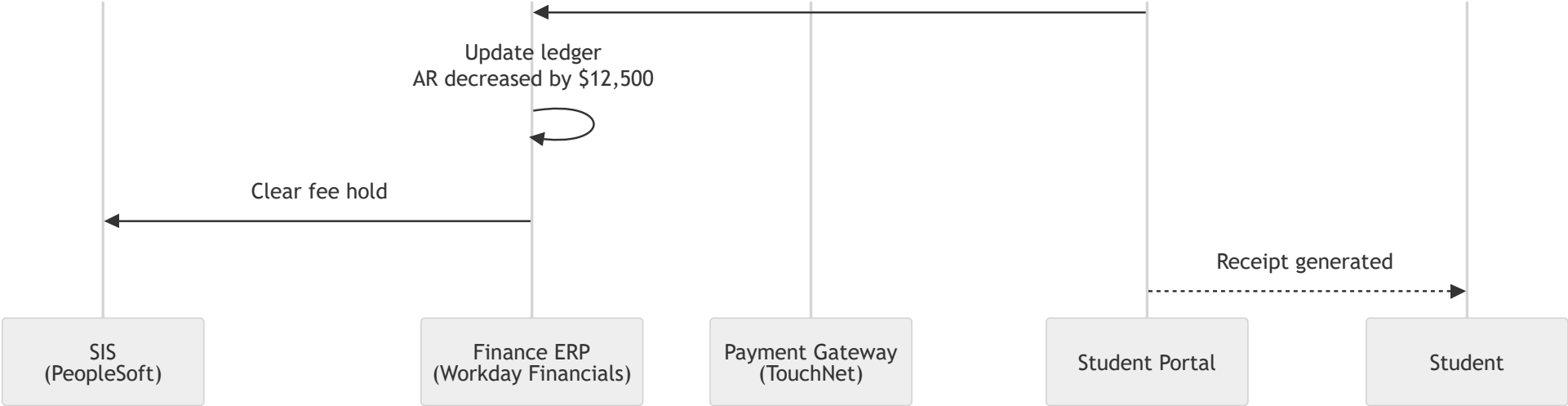
Description: Student fees assessed per term. Invoices generated, payments processed, holds cleared.

Actors: Student, Finance, Bursar

Systems: SIS, Finance ERP, Payment Gateway, Student Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Fee Assessment & Billing	Student Fee	AssessFees	SIS (PeopleSoft Campus Solutions)	Finance ERP (Workday Financials)	PeopleSoft Integration Broker	Workday API (WS-Security)	API-led (Real-time)	Medium
Fee Assessment & Billing	Payment	ProcessPayment	Payment Gateway (TouchNet)	Finance ERP (Workday Financials)	TouchNet API	Workday API (WS-Security)	Event-driven	High
Fee Assessment & Billing	Fee Hold	ClearFeeHold	Finance ERP (Workday Financials)	SIS (PeopleSoft Campus Solutions)	Workday API (WS-Security)	PeopleSoft REST API (JWT)	Event-driven	Medium
Fee Assessment & Billing	Scholarship	ApplyScholarshipCredit	Financial Aid (PowerFAIDS)	Finance ERP (Workday Financials)	PowerFAIDS REST API	Workday API (WS-Security)	Event-driven	Medium
Fee Assessment & Billing	Receipt	GenerateReceipt	Finance ERP (Workday Financials)	Student Portal	Workday API (WS-Security)	Portal REST API	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.1-01	Data integrity: Student Fee data transferred from SIS (PeopleSoft Campus Solutions) to Finance ERP (Workday Financials) via AssessFees must be complete and consistent	AssessFees	Student Fee records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-4.1-02	Data integrity: Payment data transferred from Payment Gateway (TouchNet) to Finance ERP (Workday Financials) via ProcessPayment must be complete and consistent	ProcessPayment	Payment records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-4.1-03	Data integrity: Fee Hold data transferred from Finance ERP (Workday Financials) to SIS (PeopleSoft Campus Solutions) via ClearFeeHold must be complete and consistent	ClearFeeHold	Fee Hold records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	High
AC-4.1-04	Data integrity: Scholarship data transferred from Financial Aid (PowerFAIDS) to Finance ERP (Workday Financials) via ApplyScholarshipCredit must be complete and consistent	ApplyScholarshipCredit	Scholarship records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-4.1-05	Data integrity: Receipt data transferred from Finance ERP (Workday Financials) to Student Portal via GenerateReceipt must be complete and consistent	GenerateReceipt	Receipt records in Student Portal match source; no data loss; reconciliation report confirms accuracy	Medium
AC-4.1-06	Error handling: Failed transactions between SIS (PeopleSoft Campus Solutions) and Finance ERP (Workday Financials) are logged with error context and trigger automatic retry or alert	AssessFees	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-4.1-07	Security: All endpoints enforce authentication (TouchNet API, Workday API (WS-Security), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

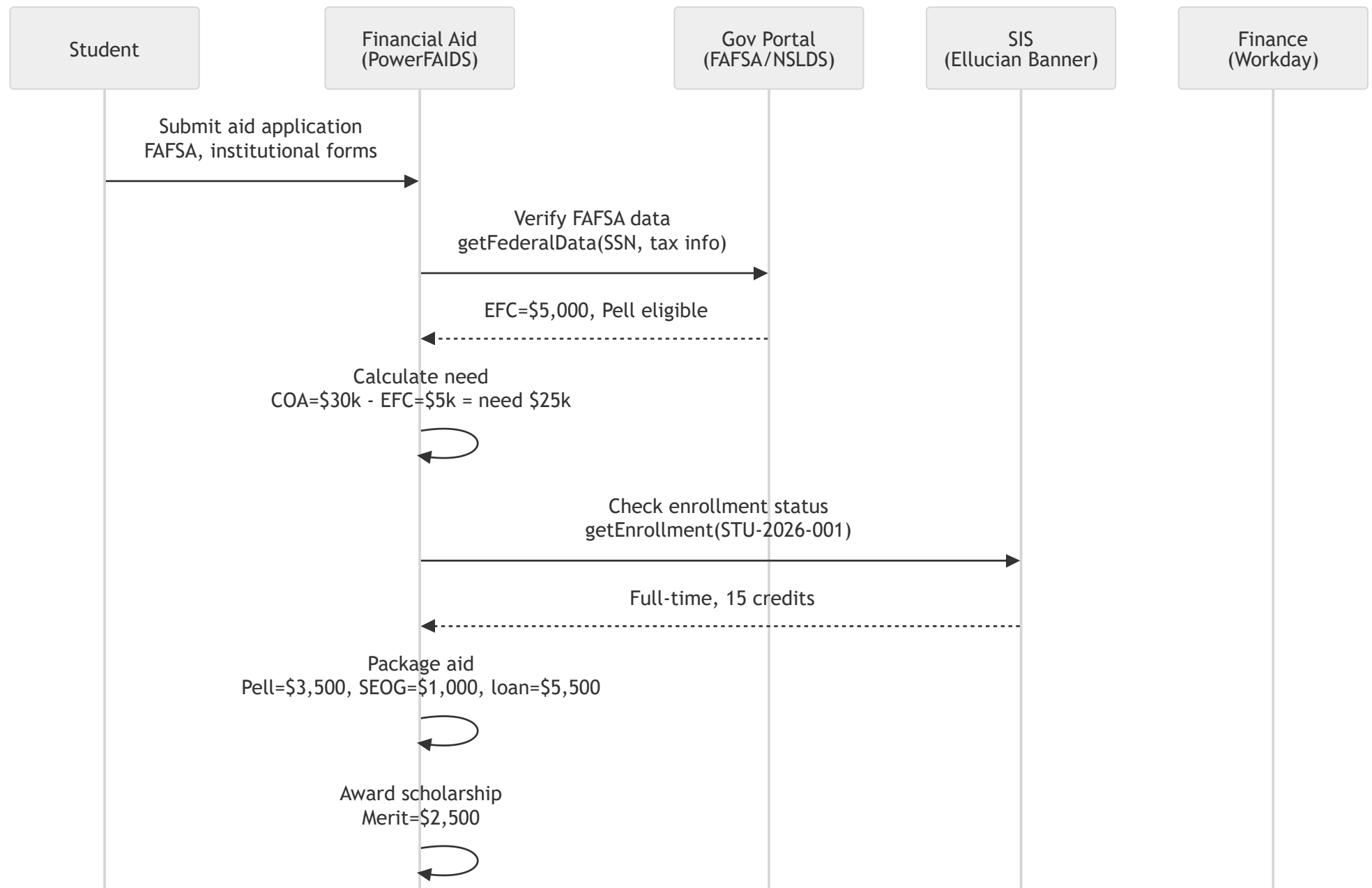
4.2 Financial Aid Management

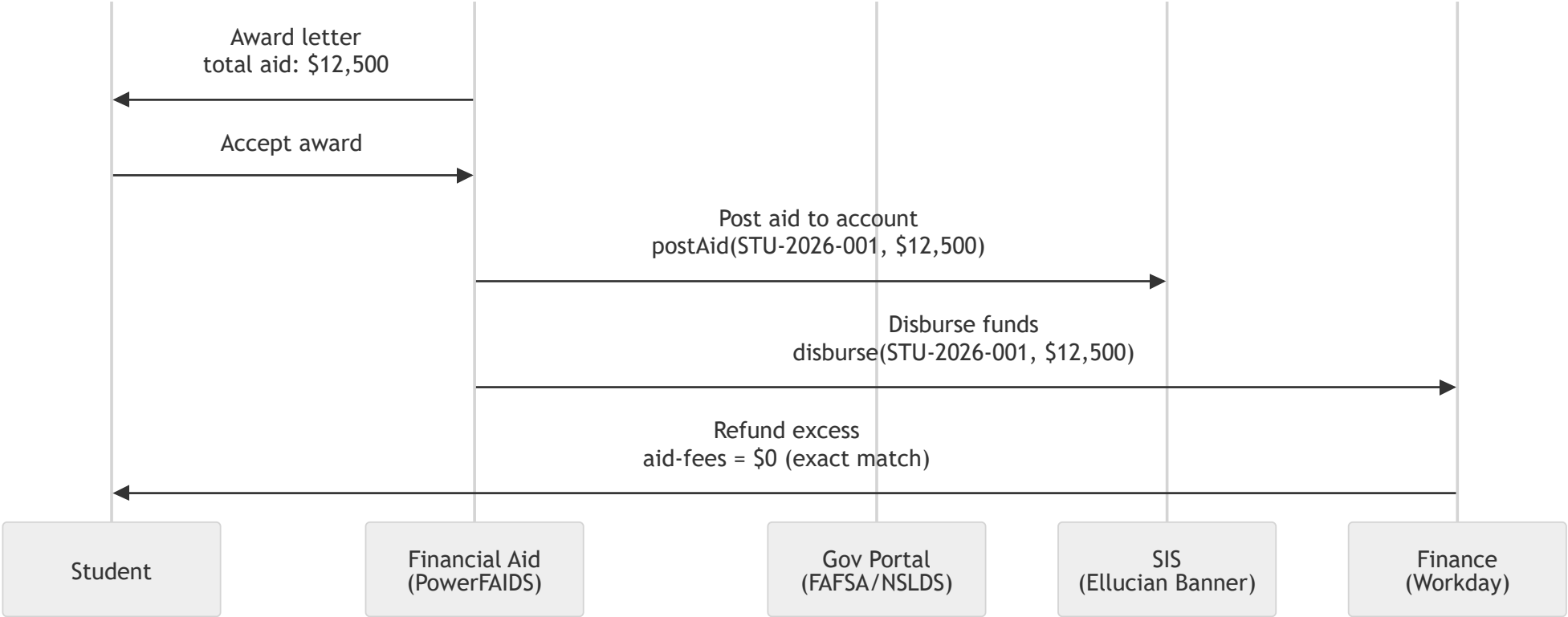
Description: Students apply for aid. System determines eligibility, packages aid, disburses funds.

Actors: Student, Financial Aid Officer, Finance

Systems: Financial Aid System, SIS, Finance ERP, Government Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Financial Aid Management	FAFSA Data	VerifyFAFSA	Financial Aid System (PowerFAIDS)	Government Portal (FAFSA/NSLDS)	PowerFAIDS API	NSLDS Web Services (SOAP)	API-led (Real-time)	High
Financial Aid Management	Enrollment Status	CheckEnrollment	Financial Aid System (PowerFAIDS)	SIS (Ellucian Banner)	PowerFAIDS REST API	Banner API (API Key)	API-led (Real-time)	Simple
Financial Aid Management	Aid Disbursement	DisburseAid	Financial Aid System (PowerFAIDS)	Finance ERP (Workday Financials)	PowerFAIDS REST API	Workday API (WS-Security)	Event-driven	Medium
Financial Aid Management	Cost	CalculateCOA	Financial Aid (PowerFAIDS)	Financial Aid (PowerFAIDS)	Internal	Internal	Batch (Scheduled)	Medium
Financial Aid Management	Verification	VerifyTaxData	Financial Aid (PowerFAIDS)	Government Portal (FAFSA/NSLDS)	PowerFAIDS API	NSLDS Web Services (SOAP)	API-led (Real-time)	High

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.2-01	Data integrity: FAFSA Data data transferred from Financial Aid System (PowerFAIDS) to Government Portal (FAFSA/NSLDS) via VerifyFAFSA must be complete and consistent	VerifyFAFSA	FAFSA Data records in Government Portal (FAFSA/NSLDS) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-4.2-02	Data integrity: Enrollment Status data transferred from Financial Aid System (PowerFAIDS) to SIS (Ellucian Banner) via CheckEnrollment must be complete and consistent	CheckEnrollment	Enrollment Status records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-4.2-03	Data integrity: Aid Disbursement data transferred from Financial Aid System (PowerFAIDS) to Finance ERP (Workday Financials) via DisburseAid must be complete and consistent	DisburseAid	Aid Disbursement records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-4.2-04	Data integrity: Cost data transferred from Financial Aid (PowerFAIDS) to Financial Aid (PowerFAIDS) via CalculateCOA must be complete and consistent	CalculateCOA	Cost records in Financial Aid (PowerFAIDS) match source; no data loss; reconciliation report confirms accuracy	High
AC-4.2-05	Data integrity: Verification data transferred from Financial Aid (PowerFAIDS) to Government Portal (FAFSA/NSLDS) via VerifyTaxData must be complete and consistent	VerifyTaxData	Verification records in Government Portal (FAFSA/NSLDS) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-4.2-06	Error handling: Failed transactions between Financial Aid System (PowerFAIDS) and Government Portal (FAFSA/NSLDS) are logged with error context and trigger automatic retry or alert	VerifyFAFSA	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-4.2-07	Security: All endpoints enforce authentication (PowerFAIDS API, PowerFAIDS REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

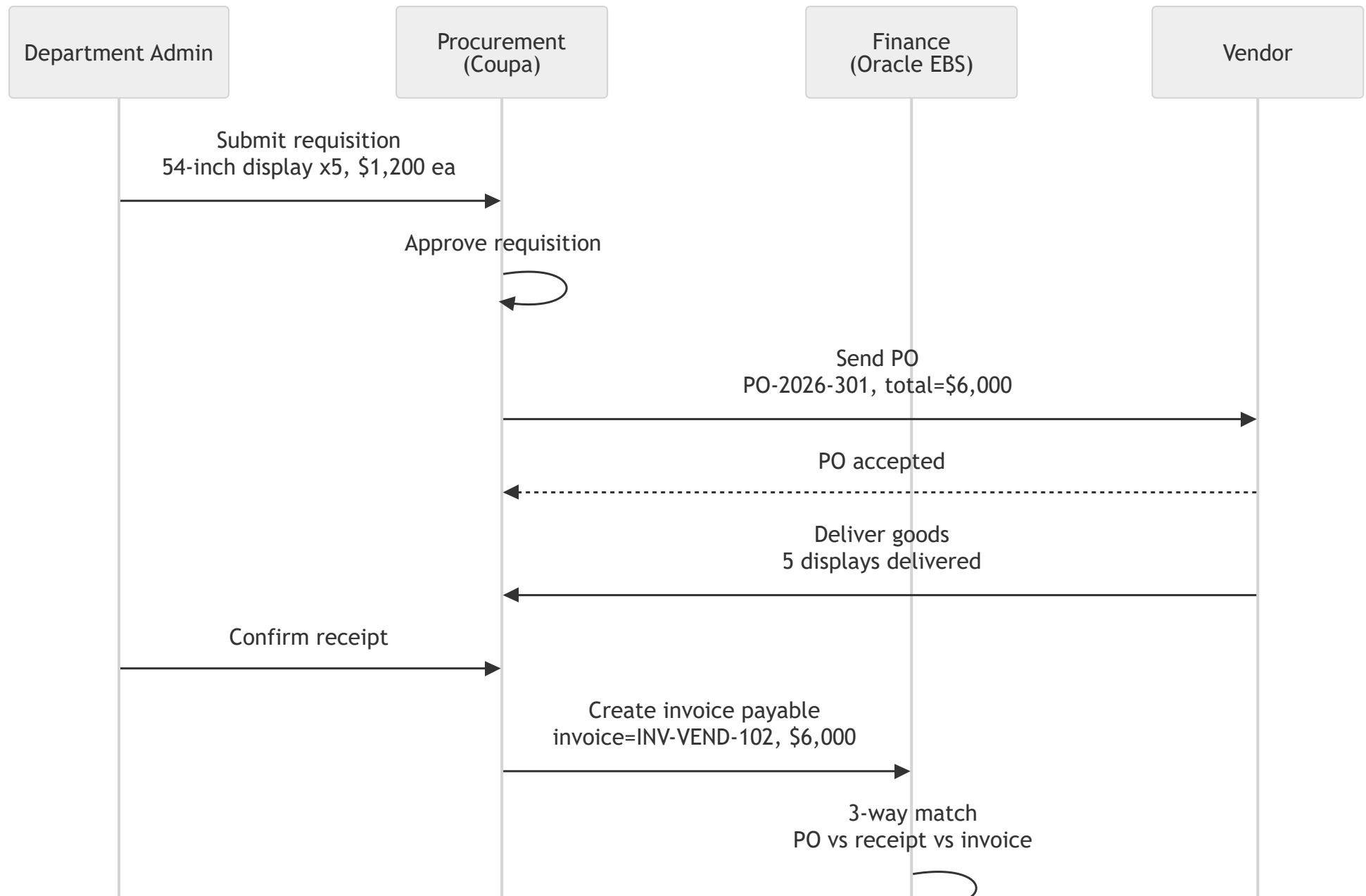
4.3 Procurement & Vendor Management

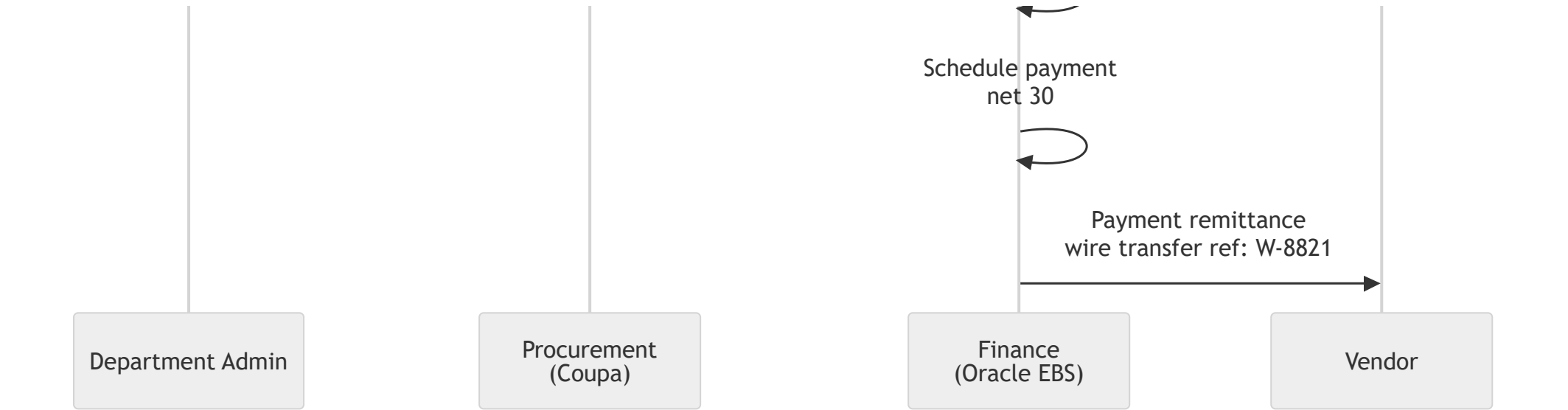
Description: Departments submit requisitions. Procurement processes POs, confirms receipt, matches invoices.

Actors: Department Admin, Procurement, Finance, Vendor

Systems: Procurement System, Finance ERP, Vendor Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Procurement & Vendor Management	Purchase Order	SendPO	Procurement System (Coupa)	Vendor Portal	Coupa API (OAuth2)	Vendor cXML / EDI	API-led (Real-time)	High
Procurement & Vendor Management	Invoice	SyncInvoice	Procurement System (Coupa)	Finance ERP (Oracle EBS)	Coupa API (OAuth2)	Oracle EBS REST API (SOAP)	Batch (Scheduled)	Medium
Procurement & Vendor Management	Requisition	ApproveRequisition	Procurement (Coupa)	Procurement (Coupa)	Internal	Internal	API-led (Real-time)	Simple
Procurement & Vendor Management	Receipt	ConfirmGoodsReceipt	Procurement (Coupa)	Finance ERP (Oracle EBS)	Coupa API (OAuth2)	Oracle EBS REST API (SOAP)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.3-01	Data integrity: Purchase Order data transferred from Procurement System (Coupa) to Vendor Portal via SendPO must be complete and consistent	SendPO	Purchase Order records in Vendor Portal match source; no data loss; reconciliation report confirms accuracy	Critical
AC-4.3-02	Data integrity: Invoice data transferred from Procurement System (Coupa) to Finance ERP (Oracle EBS) via SyncInvoice must be complete and consistent	SyncInvoice	Invoice records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	High
AC-4.3-03	Data integrity: Requisition data transferred from Procurement (Coupa) to Procurement (Coupa) via ApproveRequisition must be complete and consistent	ApproveRequisition	Requisition records in Procurement (Coupa) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-4.3-04	Data integrity: Receipt data transferred from Procurement (Coupa) to Finance ERP (Oracle EBS) via ConfirmGoodsReceipt must be complete and consistent	ConfirmGoodsReceipt	Receipt records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-4.3-05	Error handling: Failed transactions between Procurement System (Coupa) and Vendor Portal are logged with error context and trigger automatic retry or alert	SendPO	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.3-06	Security: All endpoints enforce authentication (Coupa API (OAuth2), Internal, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

Research Management - BC065 BC071 BC074 BC086 BC093 BC245

Full research lifecycle: opportunity discovery, grant applications, award management, project execution, publication, compliance.

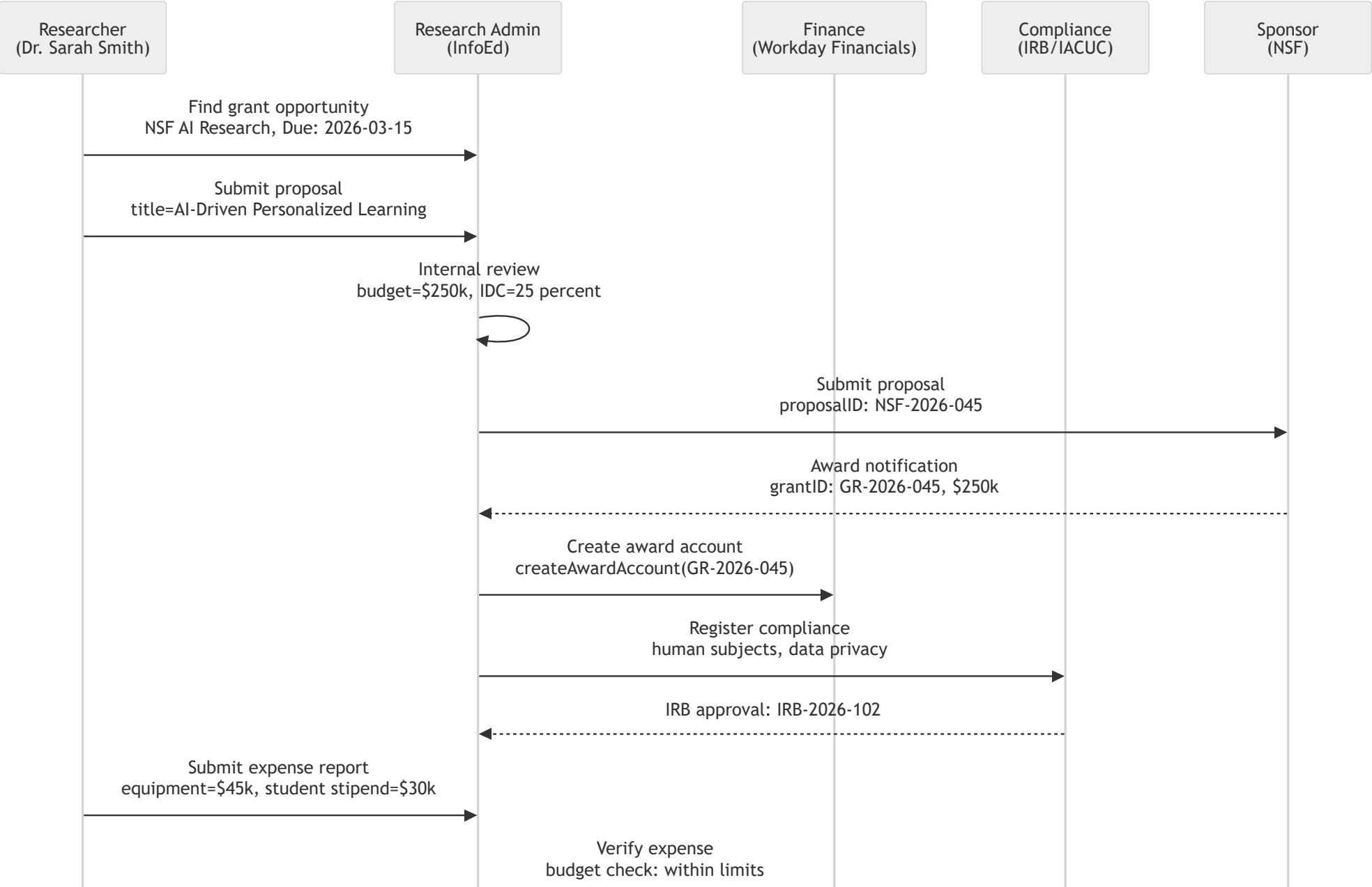
5.1 Grant Lifecycle Management

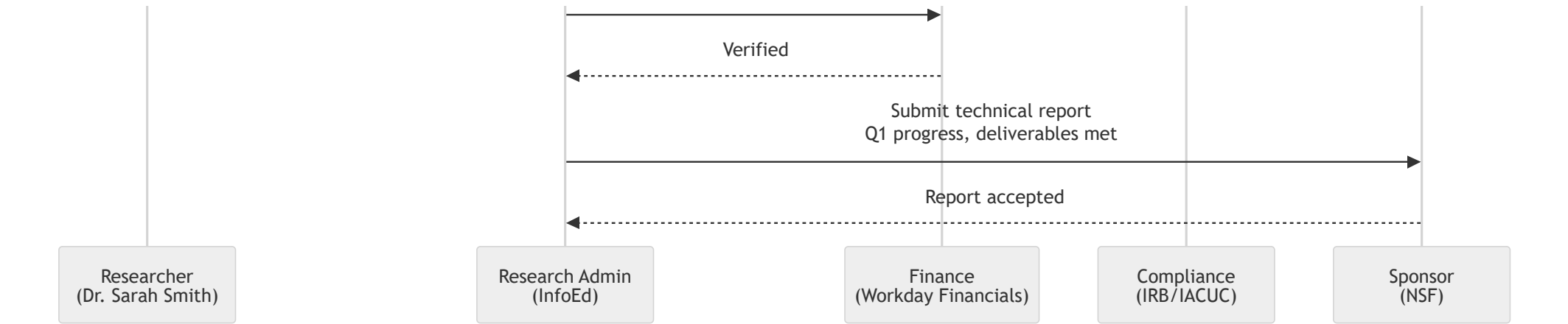
Description: Researchers identify funding, submit proposals, manage awarded grants, track expenditures, report to sponsors.

Actors: Researcher, Research Admin, Finance, Sponsor

Systems: Research Admin, Finance ERP, Compliance, Sponsor Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Grant Lifecycle Management	Proposal	SubmitProposal	Research Admin (InfoEd)	Sponsor Portal (NSF FastLane)	InfoEd API (OAuth2)	NSF FastLane API (Research.gov SOAP)	API-led (Real-time)	High
Grant Lifecycle Management	Award Account	CreateAwardAccount	Research Admin (InfoEd)	Finance ERP (Workday Financials)	InfoEd API (OAuth2)	Workday API (WS-Security)	Event-driven	Medium
Grant Lifecycle Management	Compliance	RegisterCompliance	Research Admin (InfoEd)	Compliance System (IRB Manager)	InfoEd REST API	IRB Manager API (API Key)	API-led (Real-time)	Simple
Grant Lifecycle Management	Expense	VerifyExpense	Research Admin (InfoEd)	Finance ERP (Workday Financials)	InfoEd API (OAuth2)	Workday API (WS-Security)	API-led (Real-time)	Medium
Grant Lifecycle Management	Budget	ValidateGrantBudget	Research Admin (InfoEd)	Finance ERP (Workday Financials)	InfoEd API (OAuth2)	Workday API (WS-Security)	API-led (Real-time)	Medium
Grant Lifecycle Management	Report	SubmitProgressReport	Research Admin (InfoEd)	Sponsor Portal (NSF FastLane)	InfoEd API (OAuth2)	NSF FastLane API (Research.gov SOAP)	API-led (Real-time)	High

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-5.1-01	Data integrity: Proposal data transferred from Research Admin (InfoEd) to Sponsor Portal (NSF FastLane) via SubmitProposal must be complete and consistent	SubmitProposal	Proposal records in Sponsor Portal (NSF FastLane) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-5.1-02	Data integrity: Award Account data transferred from Research Admin (InfoEd) to Finance ERP (Workday Financials) via CreateAwardAccount must be complete and consistent	CreateAwardAccount	Award Account records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-5.1-03	Data integrity: Compliance data transferred from Research Admin (InfoEd) to Compliance System (IRB Manager) via RegisterCompliance must be complete and consistent	RegisterCompliance	Compliance records in Compliance System (IRB Manager) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.1-04	Data integrity: Expense data transferred from Research Admin (InfoEd) to Finance ERP (Workday Financials) via VerifyExpense must be complete and consistent	VerifyExpense	Expense records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-5.1-05	Data integrity: Budget data transferred from Research Admin (InfoEd) to Finance ERP (Workday Financials) via ValidateGrantBudget must be complete and consistent	ValidateGrantBudget	Budget records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-5.1-06	Data integrity: Report data transferred from Research Admin (InfoEd) to Sponsor Portal (NSF FastLane) via SubmitProgressReport must be complete and consistent	SubmitProgressReport	Report records in Sponsor Portal (NSF FastLane) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-5.1-07	Error handling: Failed transactions between Research Admin (InfoEd) and Sponsor Portal (NSF FastLane) are logged with error context and trigger automatic retry or alert	SubmitProposal	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-5.1-08	Security: All endpoints enforce authentication (InfoEd API (OAuth2), InfoEd REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

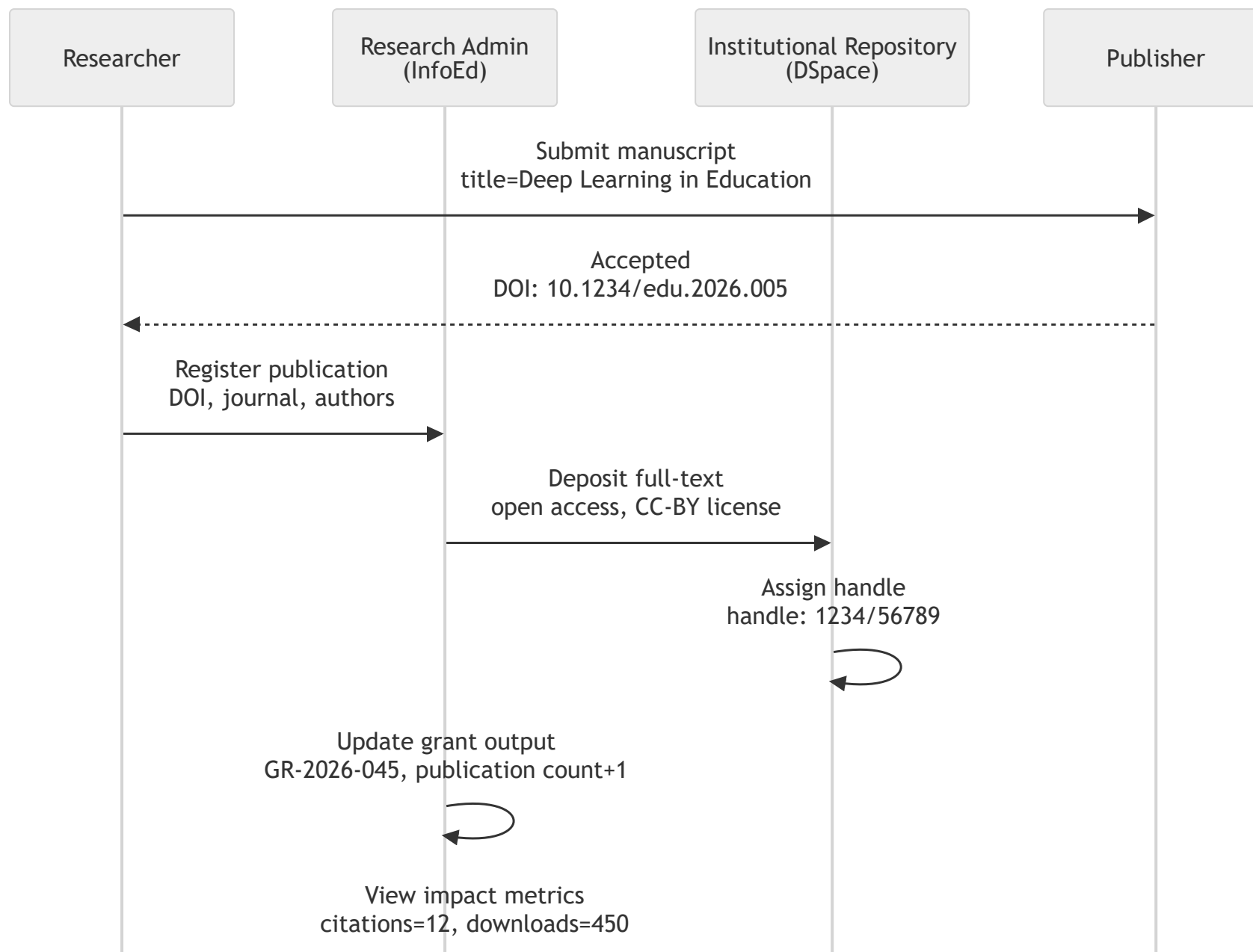
5.2 Research Outputs & Publishing

Description: Researchers publish findings. Publications tracked in institutional repository and linked to grants.

Actors: Researcher, Library, Publisher

Systems: Research Admin, Institutional Repository, Publisher Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Research Outputs & Publishing	Publication	RegisterPublication	Research Admin (InfoEd)	Institutional Repository (DSpace)	InfoEd API (OAuth2)	DSpace REST API (API Key)	API-led (Real-time)	Simple
Research Outputs & Publishing	Grant Output	UpdateGrantOutput	Research Admin (InfoEd)	Research Admin (InfoEd)	Internal	Internal	API-led (Real-time)	Simple
Research Outputs & Publishing	Citation	TrackCitationMetrics	Institutional Repository (DSpace)	Research Admin (InfoEd)	DSpace REST API (API Key)	InfoEd API (OAuth2)	Batch (Scheduled)	Simple
Research Outputs & Publishing	Altmetric	SyncAltmetricScore	Institutional Repository (DSpace)	Altmetric Service	DSpace REST API (API Key)	Altmetric API (REST)	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-5.2-01	Data integrity: Publication data transferred from Research Admin (InfoEd) to Institutional Repository (DSpace) via RegisterPublication must be complete and consistent	RegisterPublication	Publication records in Institutional Repository (DSpace) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.2-02	Data integrity: Grant Output data transferred from Research Admin (InfoEd) to Research Admin (InfoEd) via UpdateGrantOutput must be complete and consistent	UpdateGrantOutput	Grant Output records in Research Admin (InfoEd) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.2-03	Data integrity: Citation data transferred from Institutional Repository (DSpace) to Research Admin (InfoEd) via TrackCitationMetrics must be complete and consistent	TrackCitationMetrics	Citation records in Research Admin (InfoEd) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.2-04	Data integrity: Altmetric data transferred from Institutional Repository (DSpace) to Altmetric Service via SyncAltmetricScore must be complete and consistent	SyncAltmetricScore	Altmetric records in Altmetric Service match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.2-05	Error handling: Failed transactions between Research Admin (InfoEd) and Institutional Repository (DSpace) are logged with error context and trigger automatic retry or alert	RegisterPublication	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-5.2-06	Security: All endpoints enforce authentication (DSpace REST API (API Key), InfoEd API (OAuth2), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

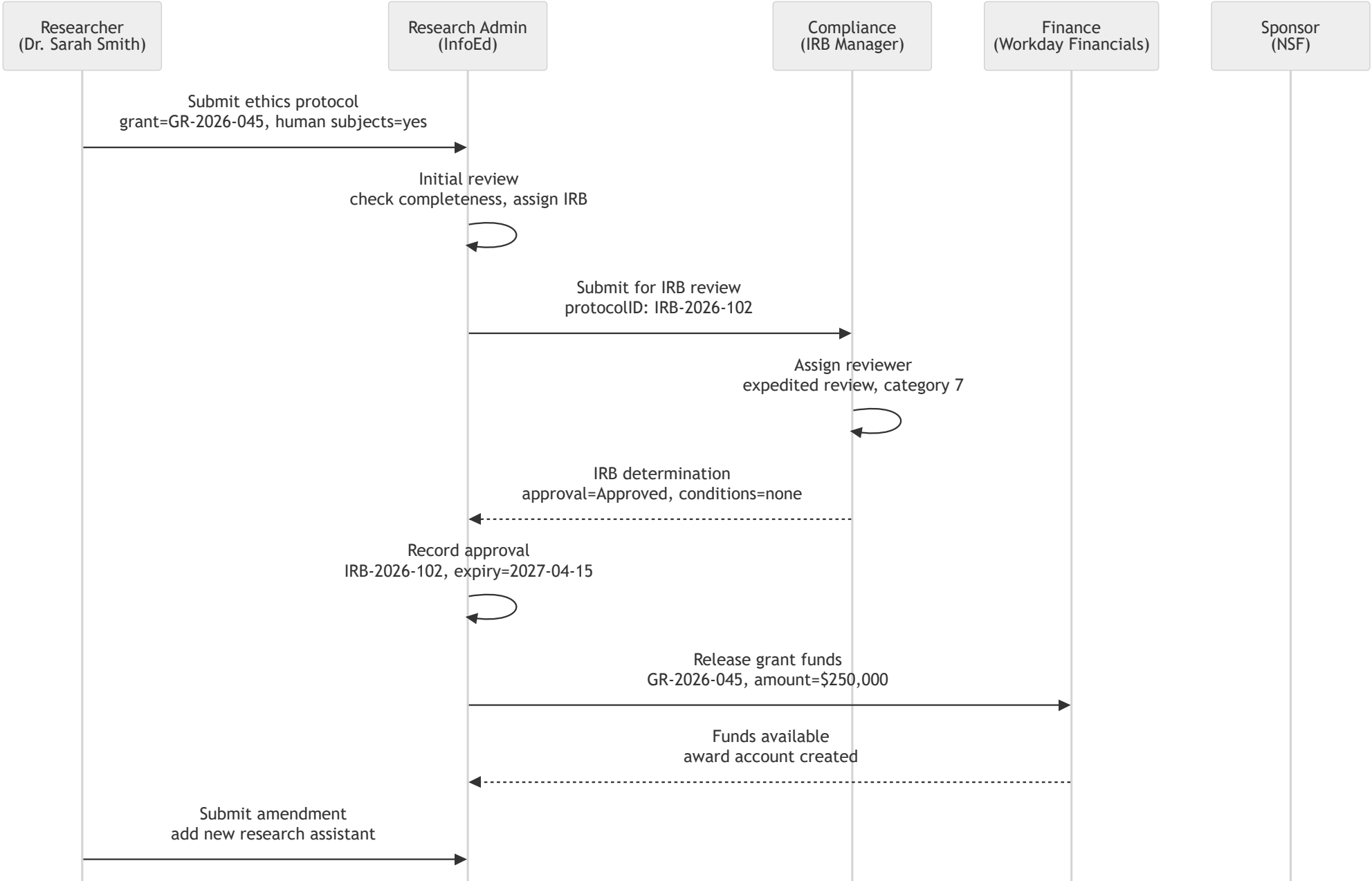
5.3 Research Compliance & Ethics

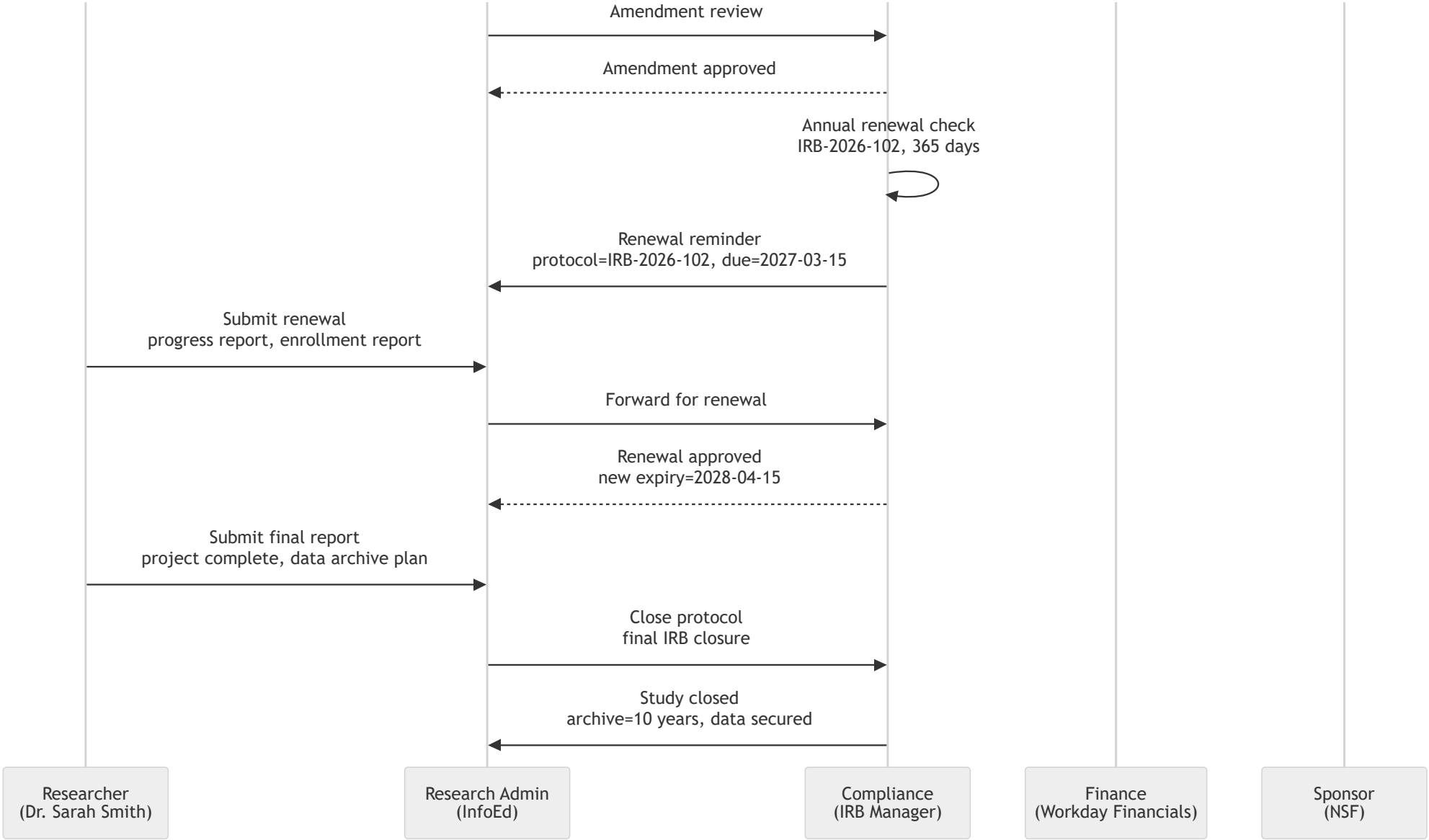
Description: Researchers submit protocols for ethics/IRB review. Compliance system tracks approvals, manages renewals, ensures regulatory adherence throughout the project lifecycle.

Actors: Researcher, IRB Committee, Compliance Officer

Systems: Research Admin (InfoEd), Compliance System (IRB Manager), SIS, Finance ERP (Workday Financials), Grantor Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Research Compliance	Protocol	SubmitEthicsProtocol	Research Admin (InfoEd)	Compliance (IRB Manager)	InfoEd API (OAuth2)	IRB Manager API (API Key)	API-led (Real-time)	Medium
Research Compliance	IRB	ProcessIRBReview	Compliance (IRB Manager)	Compliance (IRB Manager)	Internal	Internal	API-led (Real-time)	Medium
Research Compliance	Funds	ReleaseGrantFunds	Research Admin (InfoEd)	Finance ERP (Workday Financials)	InfoEd API (OAuth2)	Workday API (WS-Security)	Event-driven	Medium
Research Compliance	Renewal	ProcessAnnualRenewal	Compliance (IRB Manager)	Research Admin (InfoEd)	IRB Manager API (API Key)	InfoEd API (OAuth2)	Event-driven	Simple
Research Compliance	Closure	CloseProtocol	Compliance (IRB Manager)	Research Admin (InfoEd)	IRB Manager API (API Key)	InfoEd API (OAuth2)	Event-driven	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-5.3-01	Data integrity: Protocol data transferred from Research Admin (InfoEd) to Compliance (IRB Manager) via SubmitEthicsProtocol must be complete and consistent	SubmitEthicsProtocol	Protocol records in Compliance (IRB Manager) match source; no data loss; reconciliation report confirms accuracy	High
AC-5.3-02	Data integrity: IRB data transferred from Compliance (IRB Manager) to Compliance (IRB Manager) via ProcessIRBReview must be complete and consistent	ProcessIRBReview	IRB records in Compliance (IRB Manager) match source; no data loss; reconciliation report confirms accuracy	High
AC-5.3-03	Data integrity: Funds data transferred from Research Admin (InfoEd) to Finance ERP (Workday Financials) via ReleaseGrantFunds must be complete and consistent	ReleaseGrantFunds	Funds records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-5.3-04	Data integrity: Renewal data transferred from Compliance (IRB Manager) to Research Admin (InfoEd) via ProcessAnnualRenewal must be complete and consistent	ProcessAnnualRenewal	Renewal records in Research Admin (InfoEd) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.3-05	Data integrity: Closure data transferred from Compliance (IRB Manager) to Research Admin (InfoEd) via CloseProtocol must be complete and consistent	CloseProtocol	Closure records in Research Admin (InfoEd) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-5.3-06	Error handling: Failed transactions between Research Admin (InfoEd) and Compliance (IRB Manager) are logged with error context and trigger automatic retry or alert	SubmitEthicsProtocol	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-5.3-07	Security: All endpoints enforce authentication (IRB Manager API (API Key), InfoEd API (OAuth2), ...) and encrypt data in transit via	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	TLS 1.2+			

Institutional Engagement - BC107 BC232 BC238

Marketing, recruitment, alumni relations, fundraising, and community partnerships.

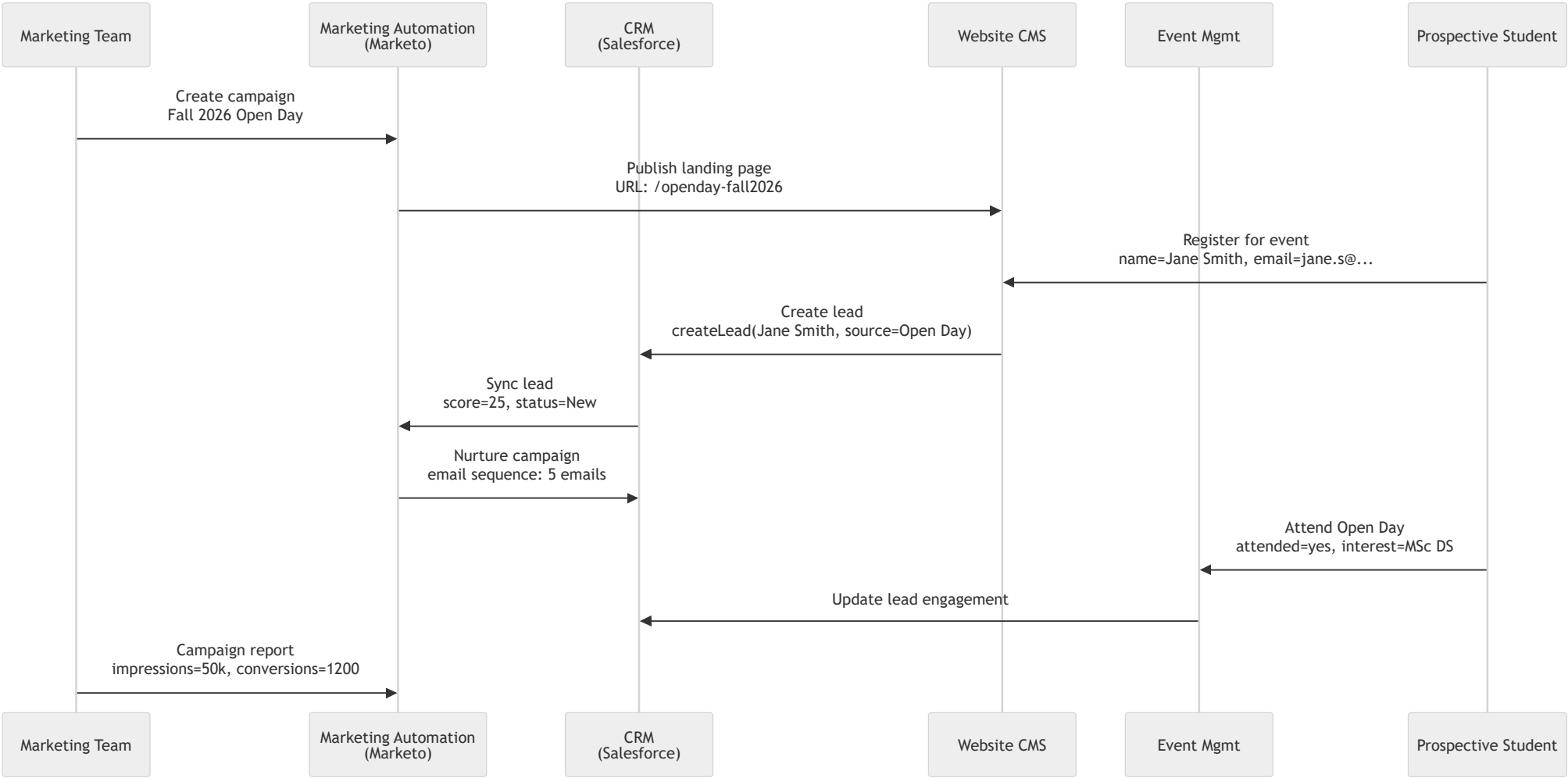
6.1 Marketing & Student Recruitment

Description: Marketing campaigns generate prospects. Inquiries route to CRM for nurturing. Events tracked. Applications originate from recruitment activities.

Actors: Marketing Team, Prospective Student, Admissions

Systems: Marketing Automation, CRM, CMS, Event Mgmt

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Marketing & Recruitment	Lead	CreateLead	Website CMS	CRM (Salesforce Education Cloud)	Webhook / REST API	Salesforce REST API (OAuth2)	Event-driven	Simple
Marketing & Recruitment	Campaign Lead	SyncCampaignLead	CRM (Salesforce Education Cloud)	Marketing Automation (Marketo)	Salesforce REST API (OAuth2)	Marketo REST API (OAuth2)	Batch (Scheduled)	Medium
Marketing & Recruitment	Event Engagement	UpdateEventEngagement	Event Mgmt System	CRM (Salesforce Education Cloud)	Event API (REST)	Salesforce REST API (OAuth2)	API-led (Real-time)	Simple
Marketing & Recruitment	Inquiry	RouteWebInquiry	Website CMS	CRM (Salesforce Education Cloud)	Webhook	Salesforce REST API (OAuth2)	Event-driven	Simple
Marketing & Recruitment	ROI	CalculateCampaignROI	Marketing Automation (Marketo)	CRM (Salesforce Education Cloud)	Marketo REST API (OAuth2)	Salesforce REST API (OAuth2)	Batch (Scheduled)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-6.1-01	Data integrity: Lead data transferred from Website CMS to CRM (Salesforce Education Cloud) via CreateLead must be complete and consistent	CreateLead	Lead records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.1-02	Data integrity: Campaign Lead data transferred from CRM (Salesforce Education Cloud) to Marketing Automation (Marketo) via SyncCampaignLead must be complete and consistent	SyncCampaignLead	Campaign Lead records in Marketing Automation (Marketo) match source; no data loss; reconciliation report confirms accuracy	High
AC-6.1-03	Data integrity: Event Engagement data transferred from Event Mgmt System to CRM (Salesforce Education Cloud) via UpdateEventEngagement must be complete and consistent	UpdateEventEngagement	Event Engagement records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.1-04	Data integrity: Inquiry data transferred from Website CMS to CRM (Salesforce Education Cloud) via RouteWebInquiry must be complete and consistent	RouteWebInquiry	Inquiry records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.1-05	Data integrity: ROI data transferred from Marketing Automation (Marketo) to CRM (Salesforce Education Cloud) via CalculateCampaignROI must be complete and consistent	CalculateCampaignROI	ROI records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	High
AC-6.1-06	Error handling: Failed transactions between Website CMS and CRM (Salesforce Education Cloud) are logged with error context and trigger automatic retry or alert	CreateLead	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-6.1-07	Security: All endpoints enforce authentication (Webhook / REST API, Salesforce REST API (OAuth2), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

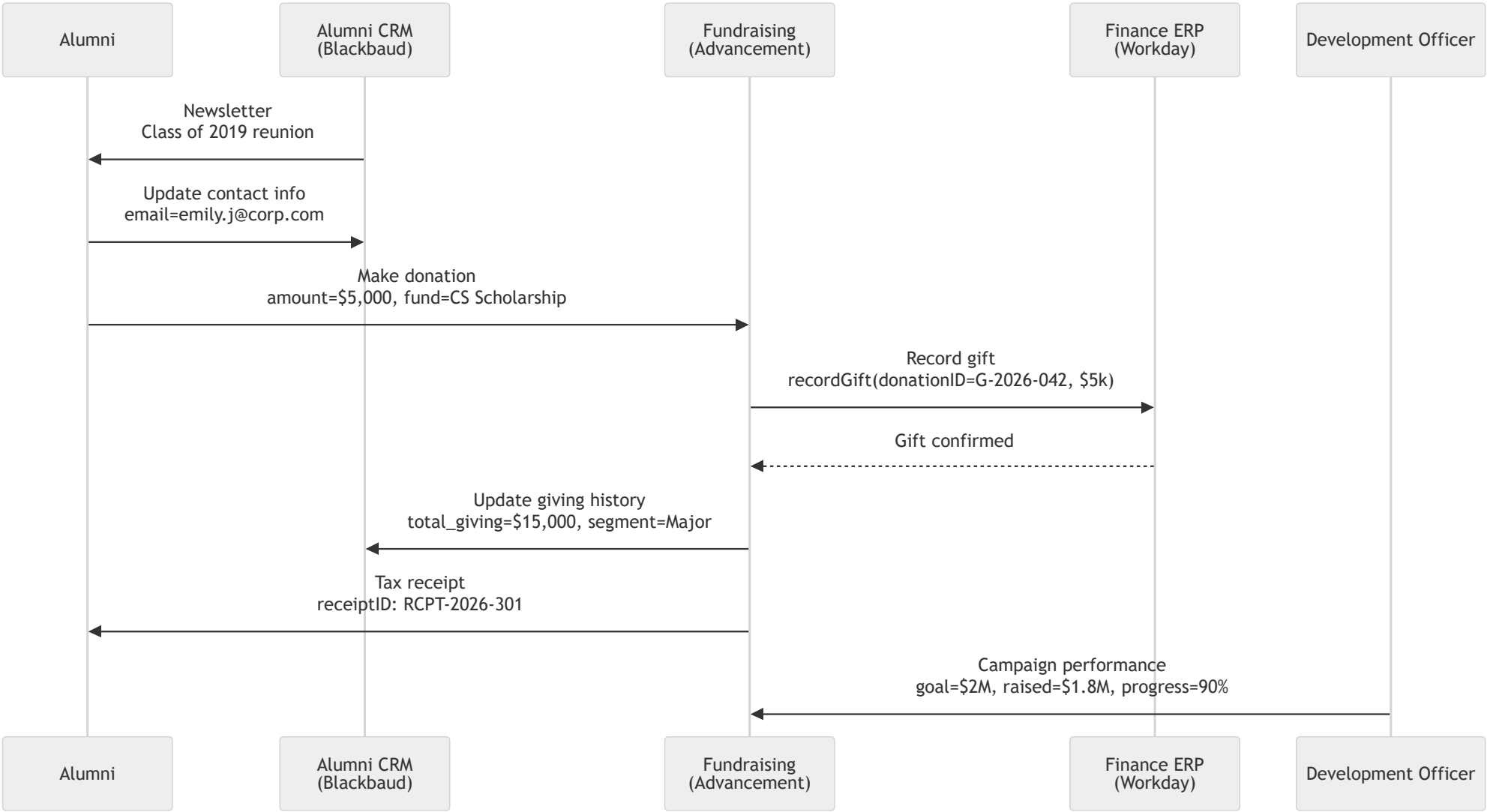
6.2 Alumni Relations & Fundraising

Description: Alumni records maintained. Fundraising campaigns target donors. Donations processed and receipts issued.

Actors: Alumni, Development Officer, Finance

Systems: Alumni CRM, Fundraising Platform, Finance ERP

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Alumni Relations & Fundraising	Donation	RecordGift	Fundraising Platform (Advancement)	Finance ERP (Workday Financials)	Advancement API (OAuth2)	Workday API (WS-Security)	API-led (Real-time)	Medium
Alumni Relations & Fundraising	Giving History	SyncGivingHistory	Fundraising Platform (Advancement)	Alumni CRM (Blackbaud)	Advancement API (OAuth2)	Blackbaud API (OAuth2)	Event-driven	Simple
Alumni Relations & Fundraising	Contact	UpdateContactInfo	Alumni Portal	Alumni CRM (Blackbaud)	Portal REST API	Blackbaud API (OAuth2)	API-led (Real-time)	Simple
Alumni Relations & Fundraising	Campaign	CreateFundraisingCampaign	Fundraising Platform (Advancement)	Alumni CRM (Blackbaud)	Advancement API (OAuth2)	Blackbaud API (OAuth2)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-6.2-01	Data integrity: Donation data transferred from Fundraising Platform (Advancement) to Finance ERP (Workday Financials) via RecordGift must be complete and consistent	RecordGift	Donation records in Finance ERP (Workday Financials) match source; no data loss; reconciliation report confirms accuracy	High
AC-6.2-02	Data integrity: Giving History data transferred from Fundraising Platform (Advancement) to Alumni CRM (Blackbaud) via SyncGivingHistory must be complete and consistent	SyncGivingHistory	Giving History records in Alumni CRM (Blackbaud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.2-03	Data integrity: Contact data transferred from Alumni Portal to Alumni CRM (Blackbaud) via UpdateContactInfo must be complete and consistent	UpdateContactInfo	Contact records in Alumni CRM (Blackbaud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.2-04	Data integrity: Campaign data transferred from Fundraising Platform (Advancement) to Alumni CRM (Blackbaud) via CreateFundraisingCampaign must be complete and consistent	CreateFundraisingCampaign	Campaign records in Alumni CRM (Blackbaud) match source; no data loss; reconciliation report confirms accuracy	High
AC-6.2-05	Error handling: Failed transactions between Fundraising Platform (Advancement) and Finance ERP (Workday Financials) are logged with error context and trigger automatic retry or alert	RecordGift	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-6.2-06	Security: All endpoints enforce authentication (Advancement API (OAuth2), Portal REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

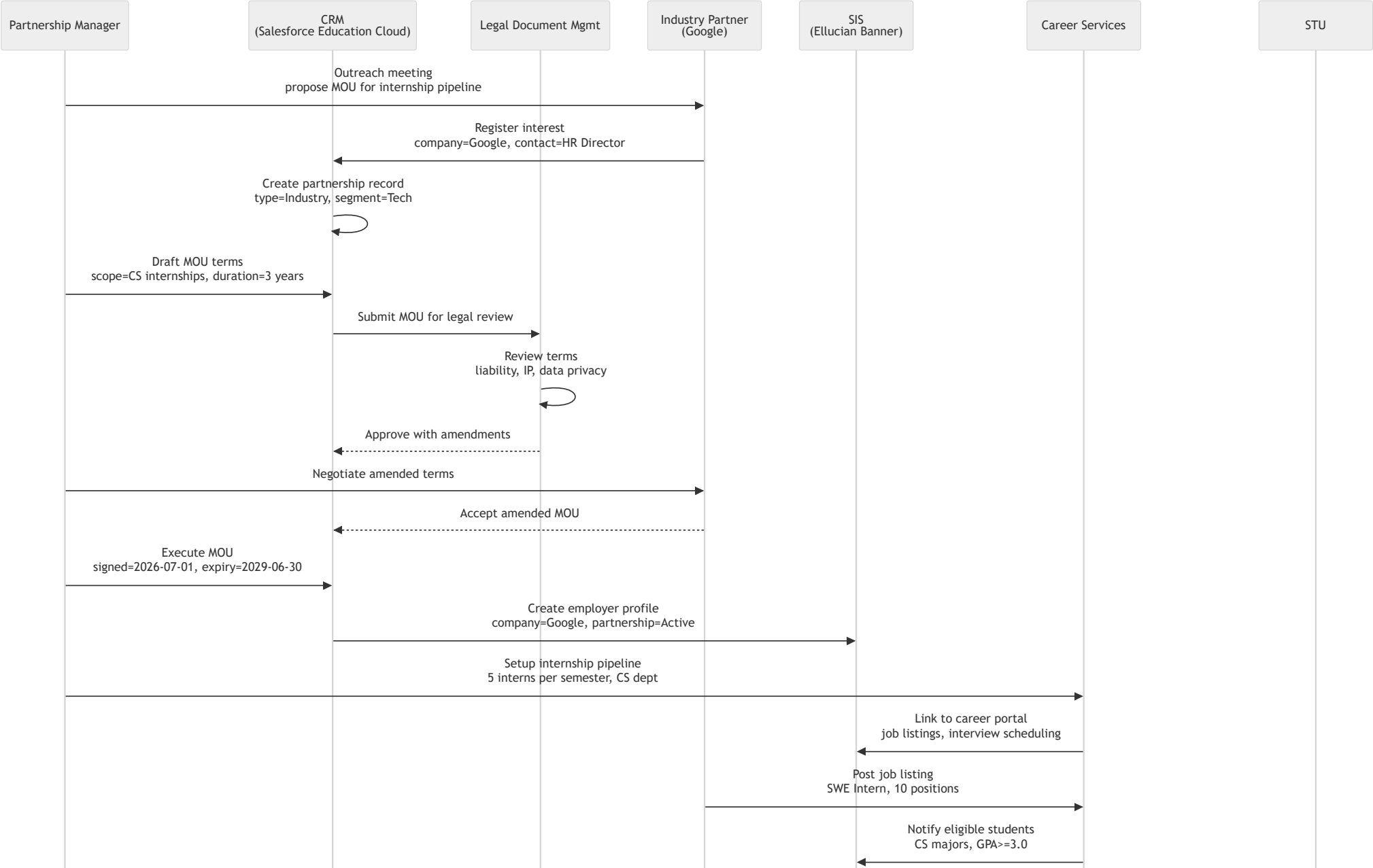
6.3 Community & Industry Partnerships

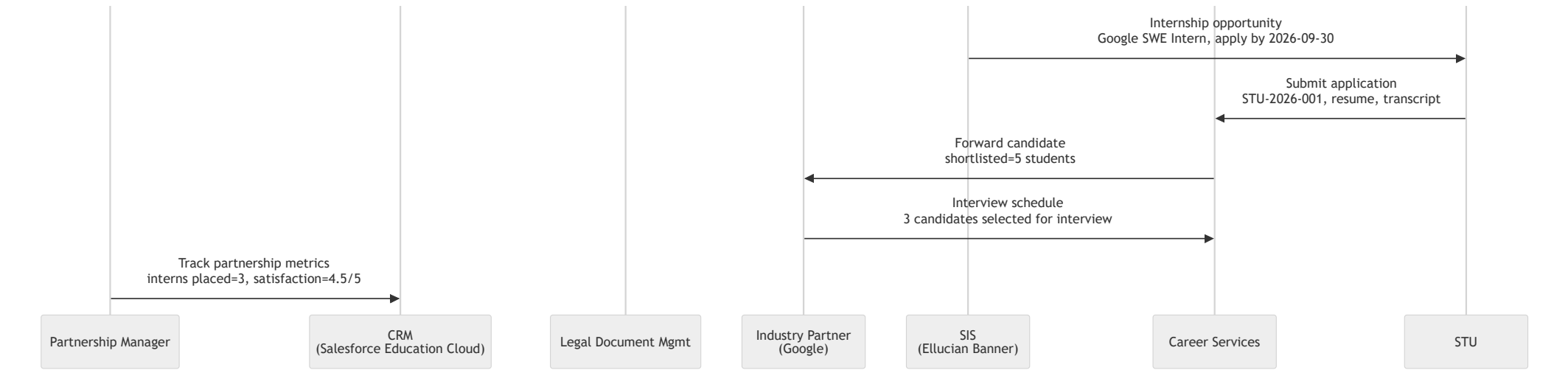
Description: Institution establishes partnerships with industry and community organizations. MOUs signed, internship pipelines created, joint research agreements managed.

Actors: Partnership Manager, Industry Partner, Legal, Dean

Systems: CRM (Salesforce Education Cloud), Legal Document Mgmt, Partner Portal, SIS, Career Services Platform

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Industry Partnership	Registration	RegisterPartnership	CRM (Salesforce Education Cloud)	CRM (Salesforce Education Cloud)	Internal	Internal	API-led (Real-time)	Simple
Industry Partnership	MOU	SubmitMOUForReview	CRM (Salesforce Education Cloud)	Legal Document Mgmt	Salesforce REST API (OAuth2)	DocMgmt API (REST)	API-led (Real-time)	Medium
Industry Partnership	Employer	CreateEmployerProfile	CRM (Salesforce Education Cloud)	SIS (Ellucian Banner)	Salesforce REST API (OAuth2)	Banner API (API Key)	Event-driven	Simple
Industry Partnership	Internship	SetupInternshipPipeline	Career Services Platform	SIS (Ellucian Banner)	Career API (REST)	Banner API (API Key)	API-led (Real-time)	Medium
Industry Partnership	Placement	ForwardCandidate	Career Services Platform	Partner Portal	Career API (REST)	Partner API (OAuth2)	API-led (Real-time)	Medium
Industry Partnership	Metrics	TrackPartnershipKPIs	CRM (Salesforce Education Cloud)	BI Platform (Tableau)	Salesforce REST API (OAuth2)	Tableau REST API	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-6.3-01	Data integrity: Registration data transferred from CRM (Salesforce Education Cloud) to CRM (Salesforce Education Cloud) via RegisterPartnership must be complete and consistent	RegisterPartnership	Registration records in CRM (Salesforce Education Cloud) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.3-02	Data integrity: MOU data transferred from CRM (Salesforce Education Cloud) to Legal Document Mgmt via SubmitMOUForReview must be complete and consistent	SubmitMOUForReview	MOU records in Legal Document Mgmt match source; no data loss; reconciliation report confirms accuracy	High
AC-6.3-03	Data integrity: Employer data transferred from CRM (Salesforce Education Cloud) to SIS (Ellucian Banner) via CreateEmployerProfile must be complete and consistent	CreateEmployerProfile	Employer records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.3-04	Data integrity: Internship data transferred from Career Services Platform to SIS (Ellucian Banner) via SetupInternshipPipeline must be complete and consistent	SetupInternshipPipeline	Internship records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-6.3-05	Data integrity: Placement data transferred from Career Services Platform to Partner Portal via ForwardCandidate must be complete and consistent	ForwardCandidate	Placement records in Partner Portal match source; no data loss; reconciliation report confirms accuracy	High
AC-6.3-06	Data integrity: Metrics data transferred from CRM (Salesforce Education Cloud) to BI Platform (Tableau) via TrackPartnershipKPIs must be complete and consistent	TrackPartnershipKPIs	Metrics records in BI Platform (Tableau) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.3-07	Error handling: Failed transactions between CRM (Salesforce Education Cloud) and CRM (Salesforce Education Cloud) are logged with error context and trigger automatic retry or alert	RegisterPartnership	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-6.3-08	Security: All endpoints enforce authentication (Career API (REST), Salesforce REST API (OAuth2),	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	...) and encrypt data in transit via TLS 1.2+			

Library & Learning Support - BC133 BC135

Library resource management, digital content delivery, research support, and course resource lists integrated with LMS.

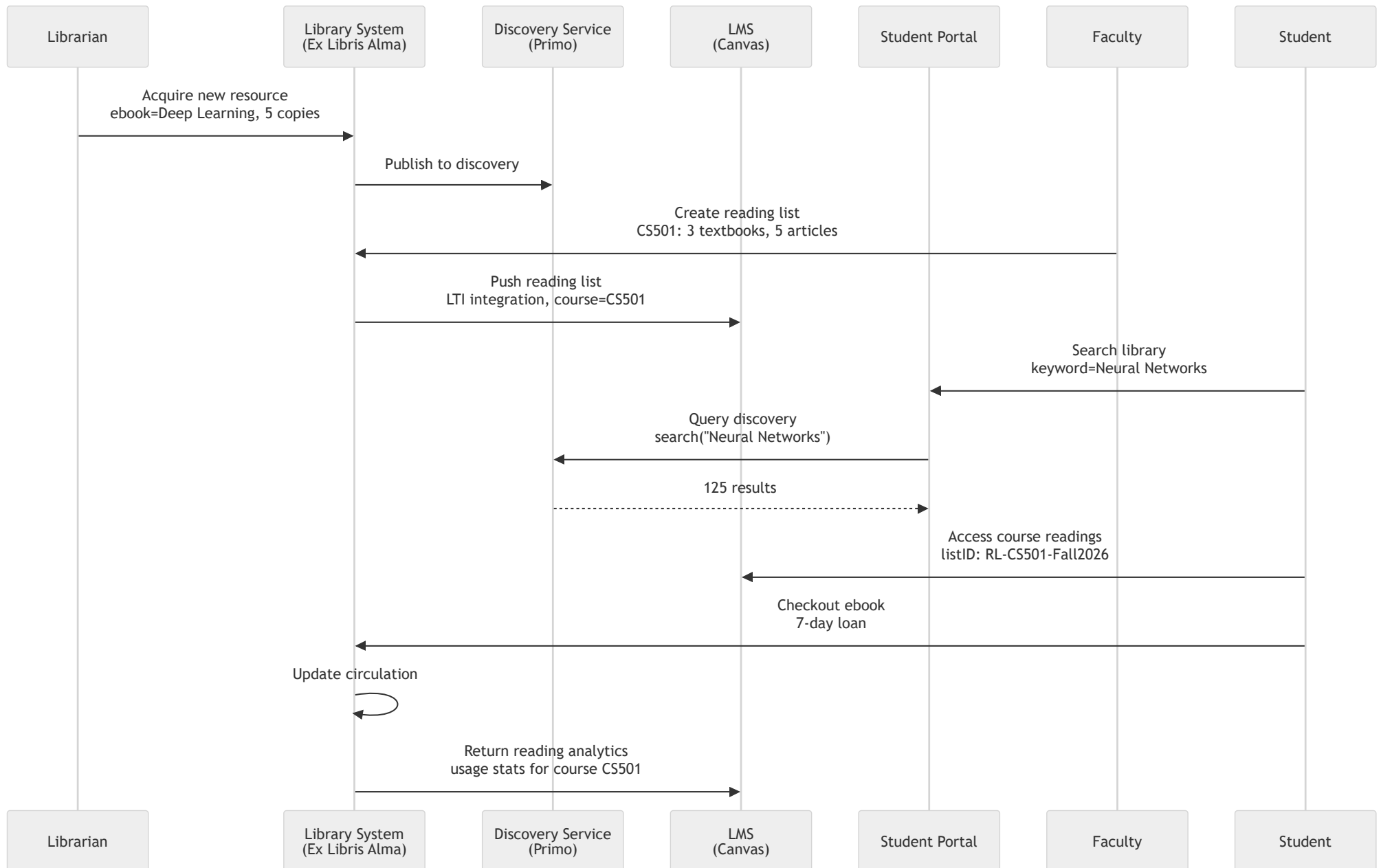
7.1 Library Resource Management

Description: Library acquires, catalogs, and circulates resources. Course reading lists linked to LMS. Digital resources accessed via portal.

Actors: Librarian, Student, Faculty

Systems: Library System, LMS, Student Portal, Discovery Service

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Library Resource Management	Reading List	PushReadingList	Library System (Ex Libris Alma)	LMS (Canvas)	Alma API (API Key)	Canvas LTI / REST API (OAuth2)	API-led (Real-time)	Medium
Library Resource Management	Search Query	SearchDiscovery	Student Portal	Discovery Service (Ex Libris Primo)	Portal REST API	Primo API (API Key)	API-led (Real-time)	Simple
Library Resource Management	Usage Analytics	SyncUsageAnalytics	Library System (Ex Libris Alma)	LMS (Canvas)	Alma API (API Key)	Canvas API (OAuth2)	Batch (Scheduled)	Simple
Library Resource Management	Acquisition	AcquireLibraryResource	Library System (Ex Libris Alma)	Vendor Portal	Alma API (API Key)	Vendor EDI / REST	API-led (Real-time)	Medium
Library Resource Management	Reserve	ManageCourseReserve	Library System (Ex Libris Alma)	LMS (Canvas)	Alma API (API Key)	Canvas LTI / REST API (OAuth2)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-7.1-01	Data integrity: Reading List data transferred from Library System (Ex Libris Alma) to LMS (Canvas) via PushReadingList must be complete and consistent	PushReadingList	Reading List records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	High
AC-7.1-02	Data integrity: Search Query data transferred from Student Portal to Discovery Service (Ex Libris Primo) via SearchDiscovery must be complete and consistent	SearchDiscovery	Search Query records in Discovery Service (Ex Libris Primo) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.1-03	Data integrity: Usage Analytics data transferred from Library System (Ex Libris Alma) to LMS (Canvas) via SyncUsageAnalytics must be complete and consistent	SyncUsageAnalytics	Usage Analytics records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.1-04	Data integrity: Acquisition data transferred from Library System (Ex Libris Alma) to Vendor Portal via AcquireLibraryResource must be complete and consistent	AcquireLibraryResource	Acquisition records in Vendor Portal match source; no data loss; reconciliation report confirms accuracy	High
AC-7.1-05	Data integrity: Reserve data transferred from Library System (Ex Libris Alma) to LMS (Canvas) via ManageCourseReserve must be complete and consistent	ManageCourseReserve	Reserve records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.1-06	Error handling: Failed transactions between Library System (Ex Libris Alma) and LMS (Canvas) are logged with error context and trigger automatic retry or alert	PushReadingList	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-7.1-07	Security: All endpoints enforce authentication (Alma API (API Key), Portal REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

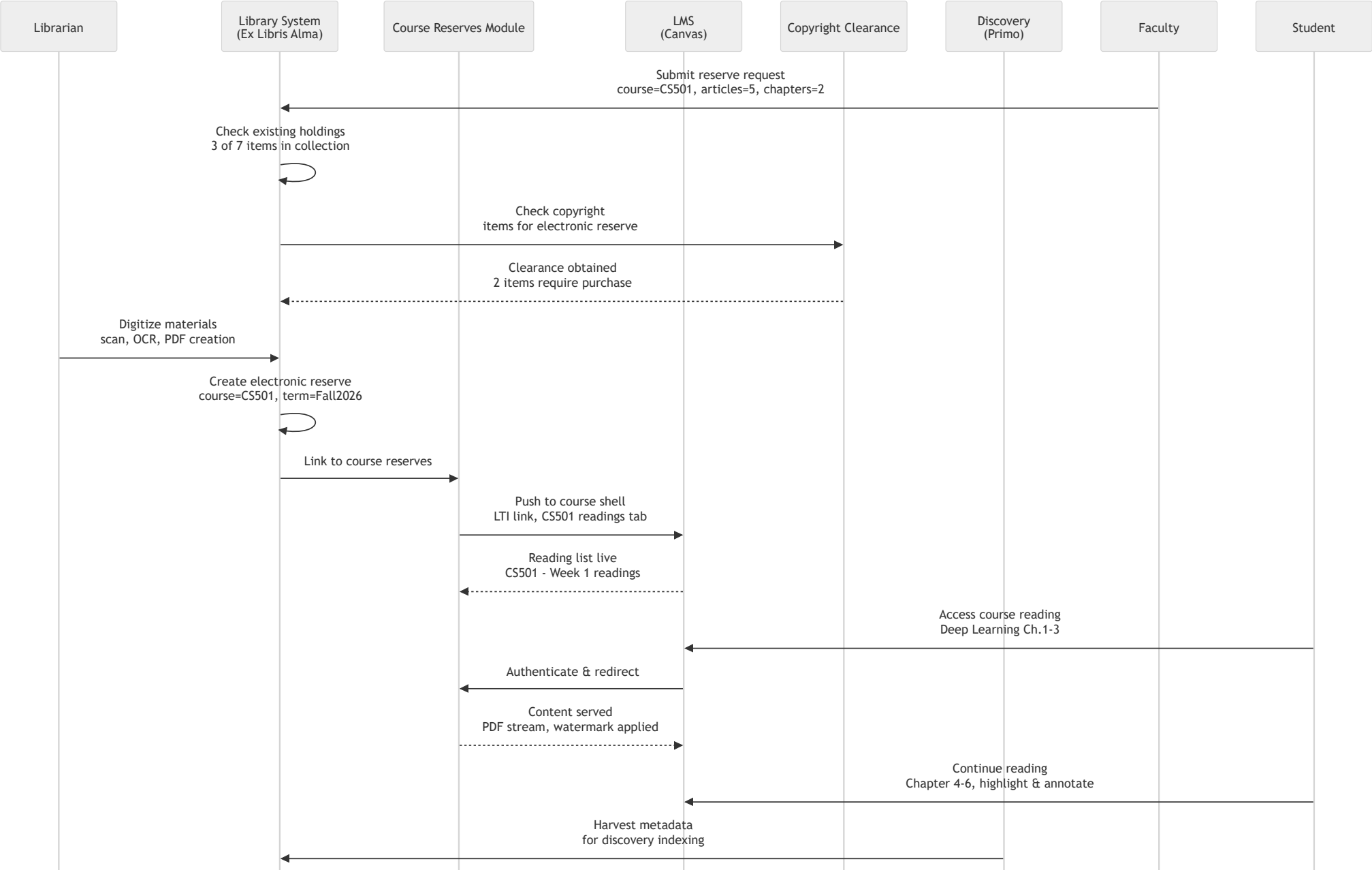
7.2 Digital Content Delivery & Course Reserves

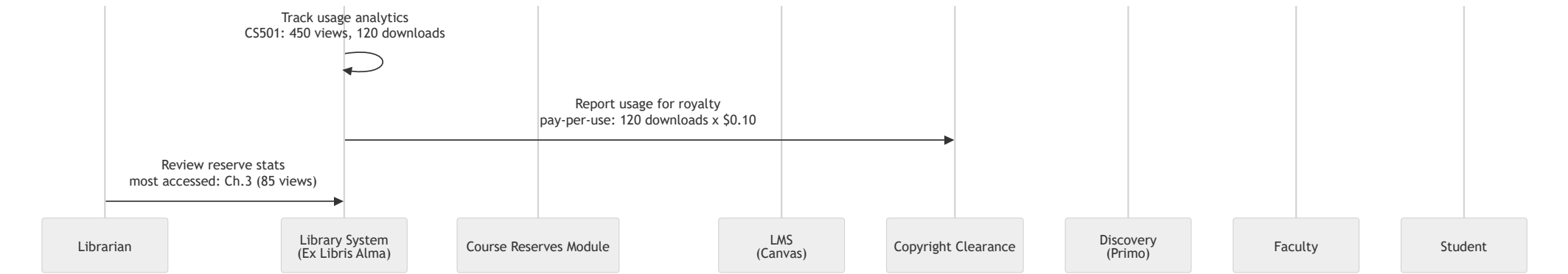
Description: Library digitizes course materials, manages electronic reserves, delivers content through LMS integration, tracks copyright compliance and usage analytics.

Actors: Librarian, Faculty, Student

Systems: Library System (Ex Libris Alma), Course Reserves Module, LMS (Canvas), Copyright Clearance Service, Discovery Service (Primo)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Digital Content	Reserve Request	SubmitReserveRequest	LMS (Canvas)	Library System (Ex Libris Alma)	Canvas LTI API (OAuth2)	Alma API (API Key)	API-led (Real-time)	Simple
Digital Content	Copyright	CheckCopyrightClearance	Library System (Ex Libris Alma)	Copyright Clearance Service	Alma API (API Key)	CCC API (REST)	API-led (Real-time)	Medium
Digital Content	Reserve	CreateElectronicReserve	Library System (Ex Libris Alma)	Course Reserves Module	Alma API (API Key)	Internal	API-led (Real-time)	Simple
Digital Content	Delivery	DeliverContentToLMS	Course Reserves Module	LMS (Canvas)	Reserves REST API	Canvas LTI / REST API (OAuth2)	API-led (Real-time)	Medium
Digital Content	Analytics	TrackUsageAnalytics	Library System (Ex Libris Alma)	Library System (Ex Libris Alma)	Internal	Internal	Event-driven	Simple
Digital Content	Royalty	ReportUsageForRoyalty	Library System (Ex Libris Alma)	Copyright Clearance Service	Alma API (API Key)	CCC API (REST)	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-7.2-01	Data integrity: Reserve Request data transferred from LMS (Canvas) to Library System (Ex Libris Alma) via SubmitReserveRequest must be complete and consistent	SubmitReserveRequest	Reserve Request records in Library System (Ex Libris Alma) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.2-02	Data integrity: Copyright data transferred from Library System (Ex Libris Alma) to Copyright Clearance Service via CheckCopyrightClearance must be complete and consistent	CheckCopyrightClearance	Copyright records in Copyright Clearance Service match source; no data loss; reconciliation report confirms accuracy	High
AC-7.2-03	Data integrity: Reserve data transferred from Library System (Ex Libris Alma) to Course Reserves Module via CreateElectronicReserve must be complete and consistent	CreateElectronicReserve	Reserve records in Course Reserves Module match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.2-04	Data integrity: Delivery data transferred from Course Reserves Module to LMS (Canvas) via DeliverContentToLMS must be complete and consistent	DeliverContentToLMS	Delivery records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	High
AC-7.2-05	Data integrity: Analytics data transferred from Library System (Ex Libris Alma) to Library System (Ex Libris Alma) via TrackUsageAnalytics must be complete and consistent	TrackUsageAnalytics	Analytics records in Library System (Ex Libris Alma) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.2-06	Data integrity: Royalty data transferred from Library System (Ex Libris Alma) to Copyright Clearance Service via ReportUsageForRoyalty must be complete and consistent	ReportUsageForRoyalty	Royalty records in Copyright Clearance Service match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.2-07	Error handling: Failed transactions between LMS (Canvas) and Library System (Ex Libris Alma) are logged with error context and trigger automatic retry or alert	SubmitReserveRequest	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-7.2-08	Security: All endpoints enforce authentication (Canvas	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	LTI API (OAuth2), Reserves REST API, ...) and encrypt data in transit via TLS 1.2+			

Enterprise Data & Analytics - BC201 BC135

Centralized data integration, data warehousing, business intelligence, and institutional reporting across all domains.

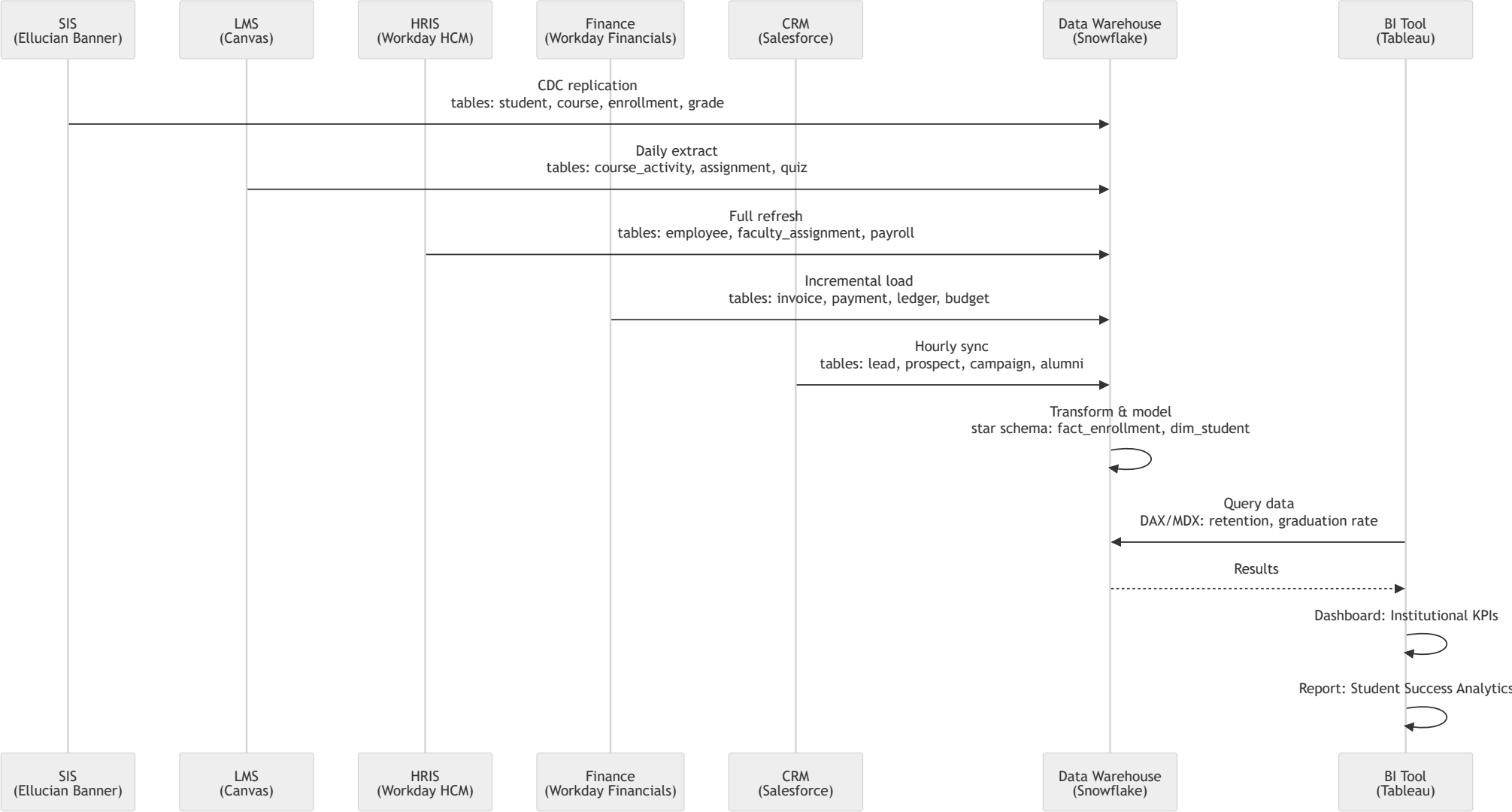
8.1 Data Integration & Warehousing

Description: Data from all source systems (SIS, LMS, HR, Finance, CRM) is ingested, transformed, and loaded into the enterprise data warehouse for reporting and analytics.

Actors: Data Engineer, Data Analyst

Systems: SIS, LMS, HRIS, Finance ERP, CRM, Data Warehouse, BI Tool

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Data Integration & Warehousing	Student Data	ReplicateStudent	SIS (Ellucian Banner)	Data Warehouse (Snowflake)	Banner CDC (HVR/Attunity)	Snowflake JDBC / Snowpipe	Database Replication (CDC)	High
Data Integration & Warehousing	Course Activity	ExtractCourseActivity	LMS (Canvas)	Data Warehouse (Snowflake)	Canvas API (OAuth2)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium
Data Integration & Warehousing	Employee Data	SyncEmployee	HRIS (Workday HCM)	Data Warehouse (Snowflake)	Workday API (WS-Security)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium
Data Integration & Warehousing	Financial Data	LoadFinancialData	Finance ERP (Workday Financials)	Data Warehouse (Snowflake)	Workday API (WS-Security)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium
Data Integration & Warehousing	CRM Data	SyncCRMDData	CRM (Salesforce Education Cloud)	Data Warehouse (Snowflake)	Salesforce REST API (OAuth2)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium
Data Integration & Warehousing	Metadata	UpdateDataCatalog	Data Warehouse (Snowflake)	Data Catalog (Collibra)	Snowflake JDBC	Collibra API (REST)	Event-driven	Medium
Data Integration & Warehousing	Quality	RunDataQualityCheck	Data Warehouse (Snowflake)	Data Quality Tool (Informatica)	Snowflake JDBC	Informatica API (REST)	Batch (ETL)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-8.1-01	Data integrity: Student Data data transferred from SIS (Ellucian Banner) to Data Warehouse (Snowflake) via ReplicateStudent must be complete and consistent	ReplicateStudent	Student Data records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-8.1-02	Data integrity: Course Activity data transferred from LMS (Canvas) to Data Warehouse (Snowflake) via ExtractCourseActivity must be complete and consistent	ExtractCourseActivity	Course Activity records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-03	Data integrity: Employee Data data transferred from HRIS (Workday HCM) to Data Warehouse (Snowflake) via SyncEmployee must be complete and consistent	SyncEmployee	Employee Data records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-04	Data integrity: Financial Data data transferred from Finance ERP (Workday Financials) to Data Warehouse (Snowflake) via LoadFinancialData must be complete and consistent	LoadFinancialData	Financial Data records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-05	Data integrity: CRM Data data transferred from CRM (Salesforce Education Cloud) to Data Warehouse (Snowflake) via SyncCRMDData must be complete and consistent	SyncCRMDData	CRM Data records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-06	Data integrity: Metadata data transferred from Data Warehouse (Snowflake) to Data Catalog (Collibra) via UpdateDataCatalog must be complete and consistent	UpdateDataCatalog	Metadata records in Data Catalog (Collibra) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-07	Data integrity: Quality data transferred from Data Warehouse (Snowflake) to Data Quality Tool (Informatica) via RunDataQualityCheck must be complete and consistent	RunDataQualityCheck	Quality records in Data Quality Tool (Informatica) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-08	Error handling: Failed transactions between SIS (Ellucian Banner) and Data Warehouse (Snowflake) are logged with error context and trigger automatic retry or alert	ReplicateStudent	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-8.1-09	Security: All endpoints enforce authentication (Salesforce REST API (OAuth2), Snowflake JDBC, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

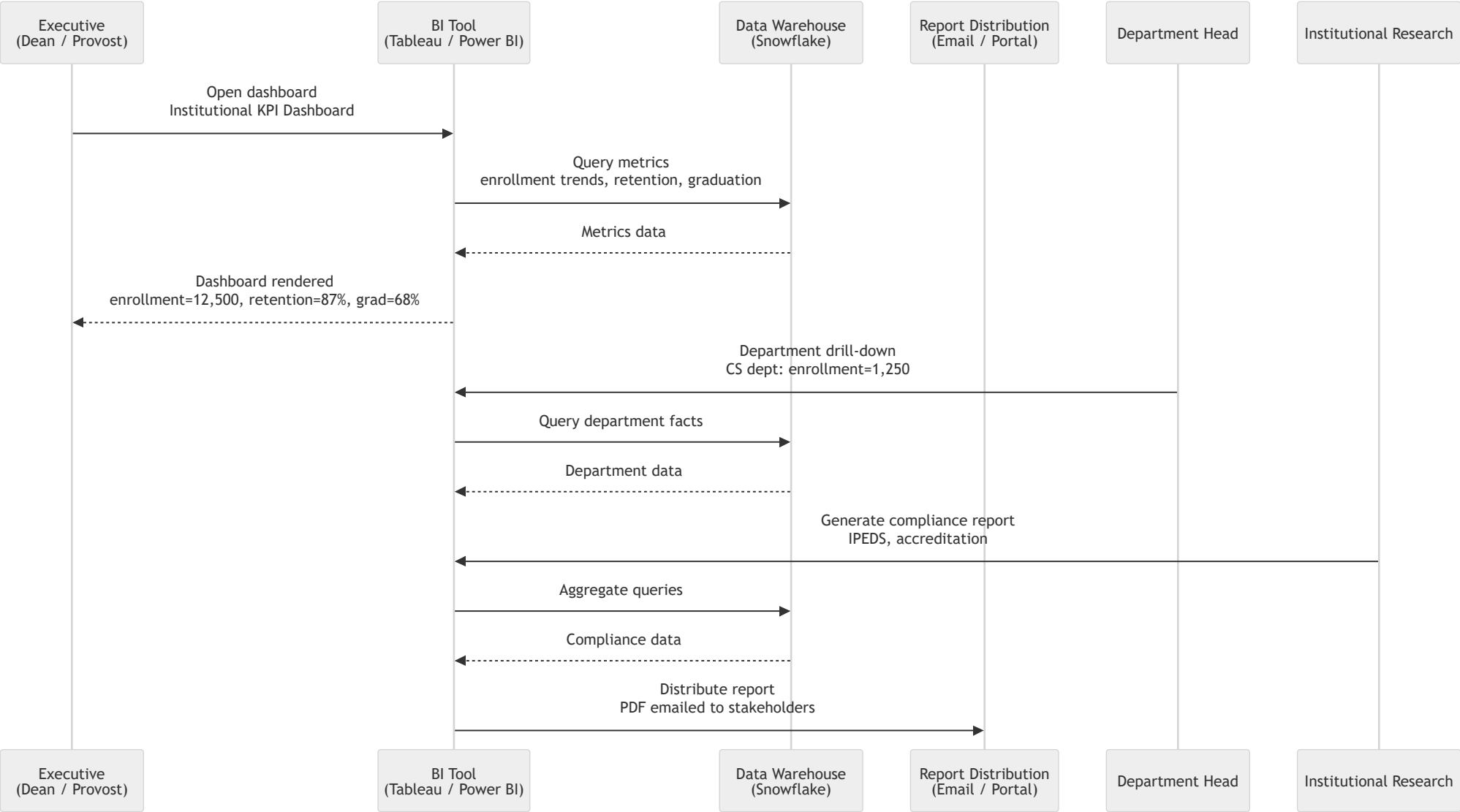
8.2 Institutional Reporting & BI

Description: Dashboards and reports for institutional KPIs: enrollment trends, retention rates, graduation rates, financial health, faculty metrics.

Actors: Executive, Department Head, Institutional Research

Systems: Data Warehouse, BI Tool, Report Distribution

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Institutional Reporting & BI	Dashboard Metrics	QueryDashboard	BI Tool (Tableau / Power BI)	Data Warehouse (Snowflake)	Tableau Server / Power BI Gateway	Snowflake ODBC / JDBC	API-led (Real-time)	Simple
Institutional Reporting & BI	Compliance Report	GenerateComplianceReport	BI Tool (Tableau / Power BI)	Report Distribution (Email / Portal)	Tableau REST API / Power BI API	SMTP / Portal API	Batch (Scheduled)	Medium
Institutional Reporting & BI	Data	RefreshDataset	BI Tool (Tableau / Power BI)	Data Warehouse (Snowflake)	Tableau Server API / Power BI API	Snowflake ODBC / JDBC	Batch (Scheduled)	Simple
Institutional Reporting & BI	Schedule	ScheduleReportRefresh	BI Tool (Tableau / Power BI)	BI Tool (Tableau / Power BI)	Internal	Internal	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-8.2-01	Data integrity: Dashboard Metrics data transferred from BI Tool (Tableau / Power BI) to Data Warehouse (Snowflake) via QueryDashboard must be complete and consistent	QueryDashboard	Dashboard Metrics records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-8.2-02	Data integrity: Compliance Report data transferred from BI Tool (Tableau / Power BI) to Report Distribution (Email / Portal) via GenerateComplianceReport must be complete and consistent	GenerateComplianceReport	Compliance Report records in Report Distribution (Email / Portal) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.2-03	Data integrity: Data data transferred from BI Tool (Tableau / Power BI) to Data Warehouse (Snowflake) via RefreshDataset must be complete and consistent	RefreshDataset	Data records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-8.2-04	Data integrity: Schedule data transferred from BI Tool (Tableau / Power BI) to BI Tool (Tableau / Power BI) via ScheduleReportRefresh must be complete and consistent	ScheduleReportRefresh	Schedule records in BI Tool (Tableau / Power BI) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-8.2-05	Error handling: Failed transactions between BI Tool (Tableau / Power BI) and Data Warehouse (Snowflake) are logged with error context and trigger automatic retry or alert	QueryDashboard	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-8.2-06	Security: All endpoints enforce authentication (Tableau Server / Power BI Gateway, Tableau Server API / Power BI API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

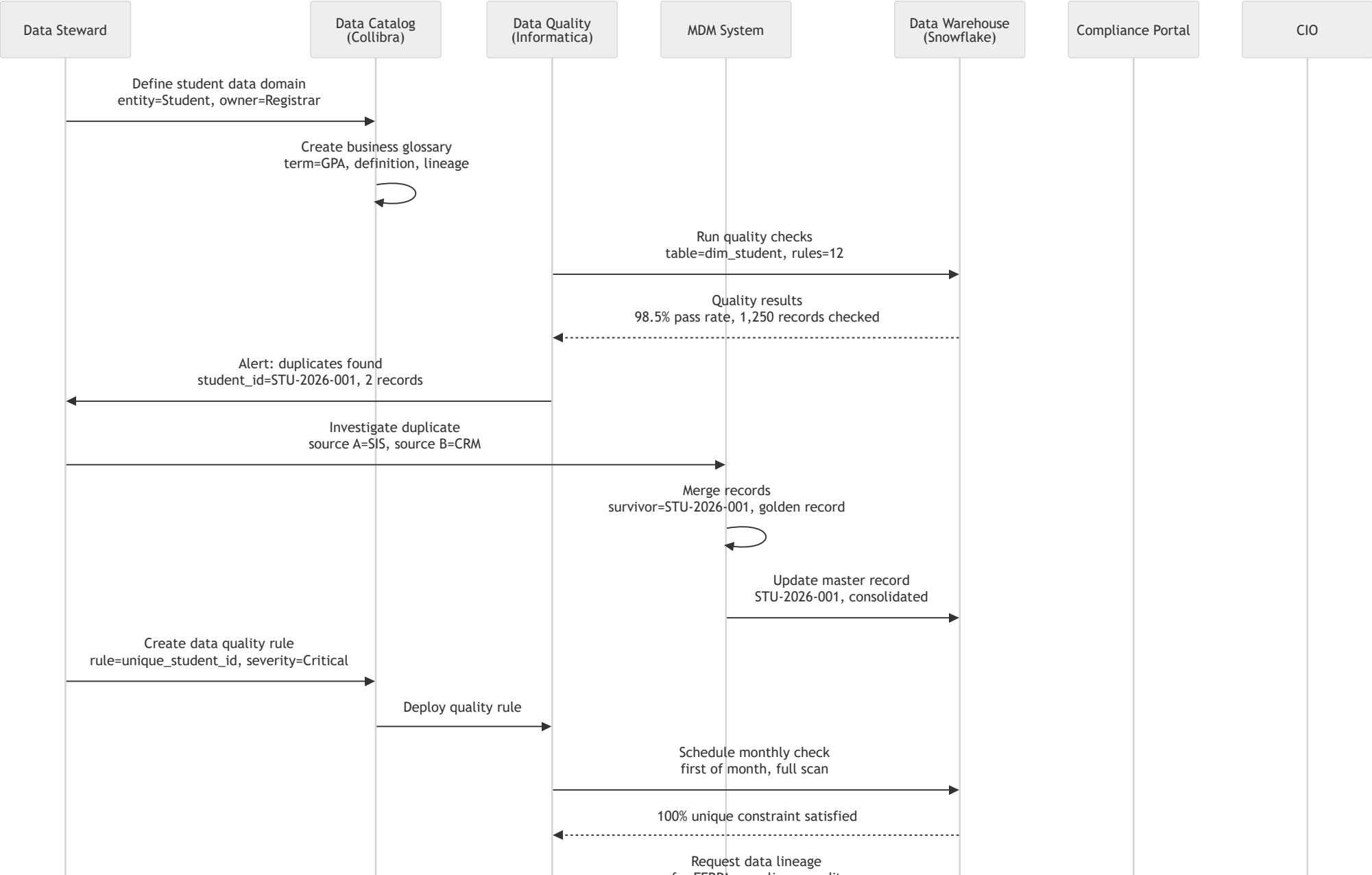
8.3 Data Governance & Quality Management

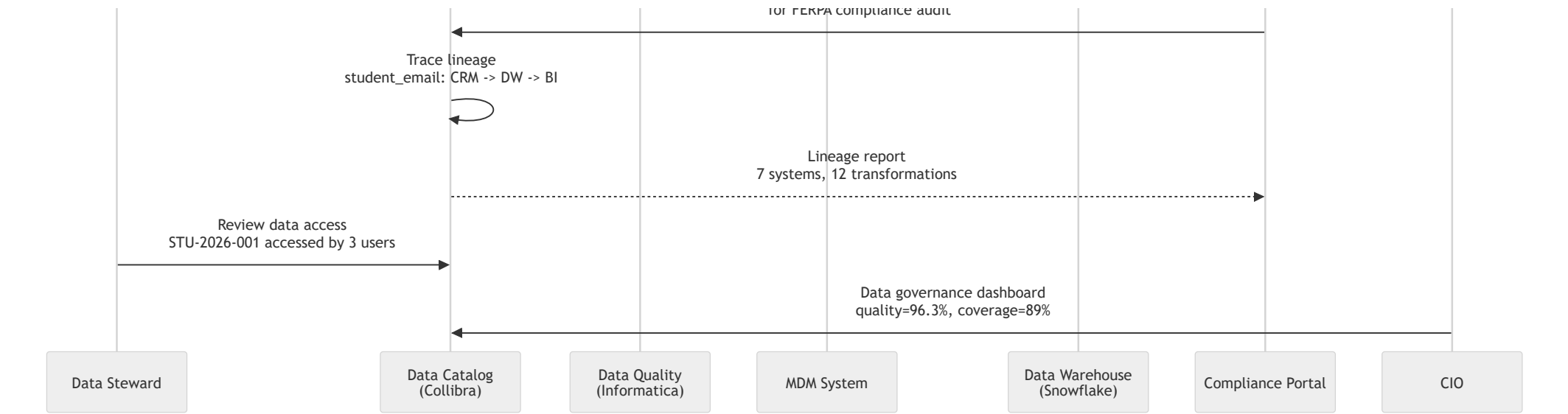
Description: Data governance team manages data catalog, quality rules, stewardship, and master data management. Ensures data across all systems is accurate, consistent, and compliant.

Actors: Data Steward, Data Quality Analyst, Data Engineer, CIO

Systems: Data Warehouse (Snowflake), Data Catalog (Collibra), Data Quality Tool (Informatica), MDM System, SIS, LMS, HRIS, Compliance Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Data Governance	Glossary	CreateBusinessGlossary	Data Catalog (Collibra)	Data Catalog (Collibra)	Internal	Internal	API-led (Real-time)	Simple
Data Governance	Quality	RunDataQualityCheck	Data Quality (Informatica)	Data Warehouse (Snowflake)	Informatica API (REST)	Snowflake JDBC	Batch (Scheduled)	Medium
Data Governance	Duplicate	AlertDuplicateRecord	Data Quality (Informatica)	Data Steward Portal	Informatica API (REST)	Workflow API	Event-driven	Simple
Data Governance	MDM	MergeDuplicateRecord	MDM System	Data Warehouse (Snowflake)	MDM API (SOAP)	Snowflake JDBC	Batch (ETL)	Medium
Data Governance	Lineage	TraceDataLineage	Data Catalog (Collibra)	Compliance Portal	Collibra API (REST)	Compliance API (REST)	API-led (Real-time)	Medium
Data Governance	Dashboard	GenerateGovernanceDashboard	Data Catalog (Collibra)	BI Platform (Tableau)	Collibra API (REST)	Tableau REST API	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-8.3-01	Data integrity: Glossary data transferred from Data Catalog (Collibra) to Data Catalog (Collibra) via CreateBusinessGlossary must be complete and consistent	CreateBusinessGlossary	Glossary records in Data Catalog (Collibra) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-8.3-02	Data integrity: Quality data transferred from Data Quality (Informatica) to Data Warehouse (Snowflake) via RunDataQualityCheck must be complete and consistent	RunDataQualityCheck	Quality records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.3-03	Data integrity: Duplicate data transferred from Data Quality (Informatica) to Data Steward Portal via AlertDuplicateRecord must be complete and consistent	AlertDuplicateRecord	Duplicate records in Data Steward Portal match source; no data loss; reconciliation report confirms accuracy	Medium
AC-8.3-04	Data integrity: MDM data transferred from MDM System to Data Warehouse (Snowflake) via MergeDuplicateRecord must be complete and consistent	MergeDuplicateRecord	MDM records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	High
AC-8.3-05	Data integrity: Lineage data transferred from Data Catalog (Collibra) to Compliance Portal via TraceDataLineage must be complete and consistent	TraceDataLineage	Lineage records in Compliance Portal match source; no data loss; reconciliation report confirms accuracy	High
AC-8.3-06	Data integrity: Dashboard data transferred from Data Catalog (Collibra) to BI Platform (Tableau) via GenerateGovernanceDashboard must be complete and consistent	GenerateGovernanceDashboard	Dashboard records in BI Platform (Tableau) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-8.3-07	Error handling: Failed transactions between Data Catalog (Collibra) and Data Catalog (Collibra) are logged with error context and trigger automatic retry or alert	CreateBusinessGlossary	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-8.3-08	Security: A	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	II endpoints enforce authentication (Colibra API (REST), MDM API (SOAP), ...) and encrypt data in transit via TLS 1.2+			

Student Services & Wellbeing - BC052 BC019

Student support services: academic counselling, health and wellbeing, career services, financial guidance, and disability support. These services ensure student retention and success beyond academics.

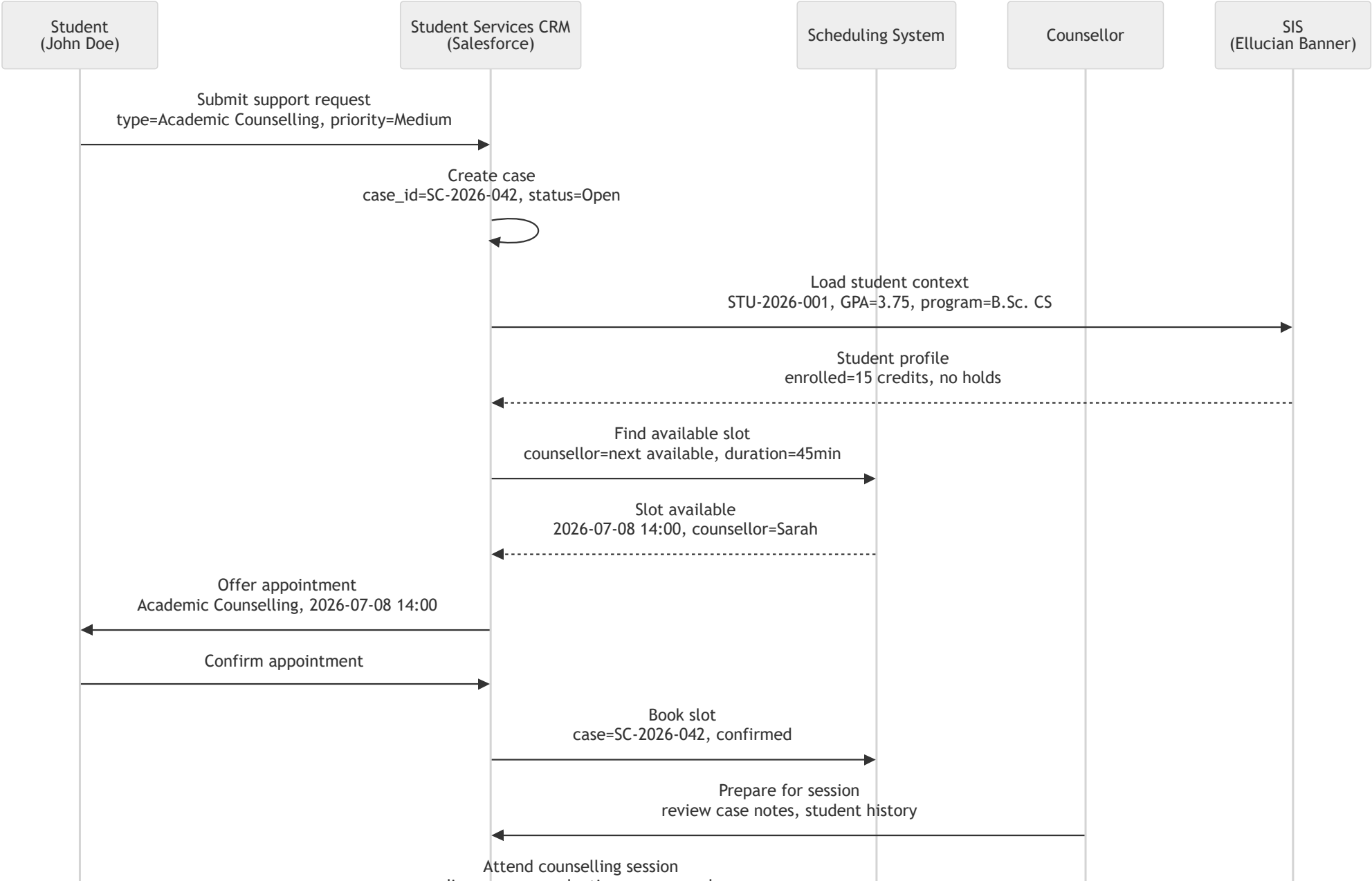
9.1 Student Support & Counselling Services

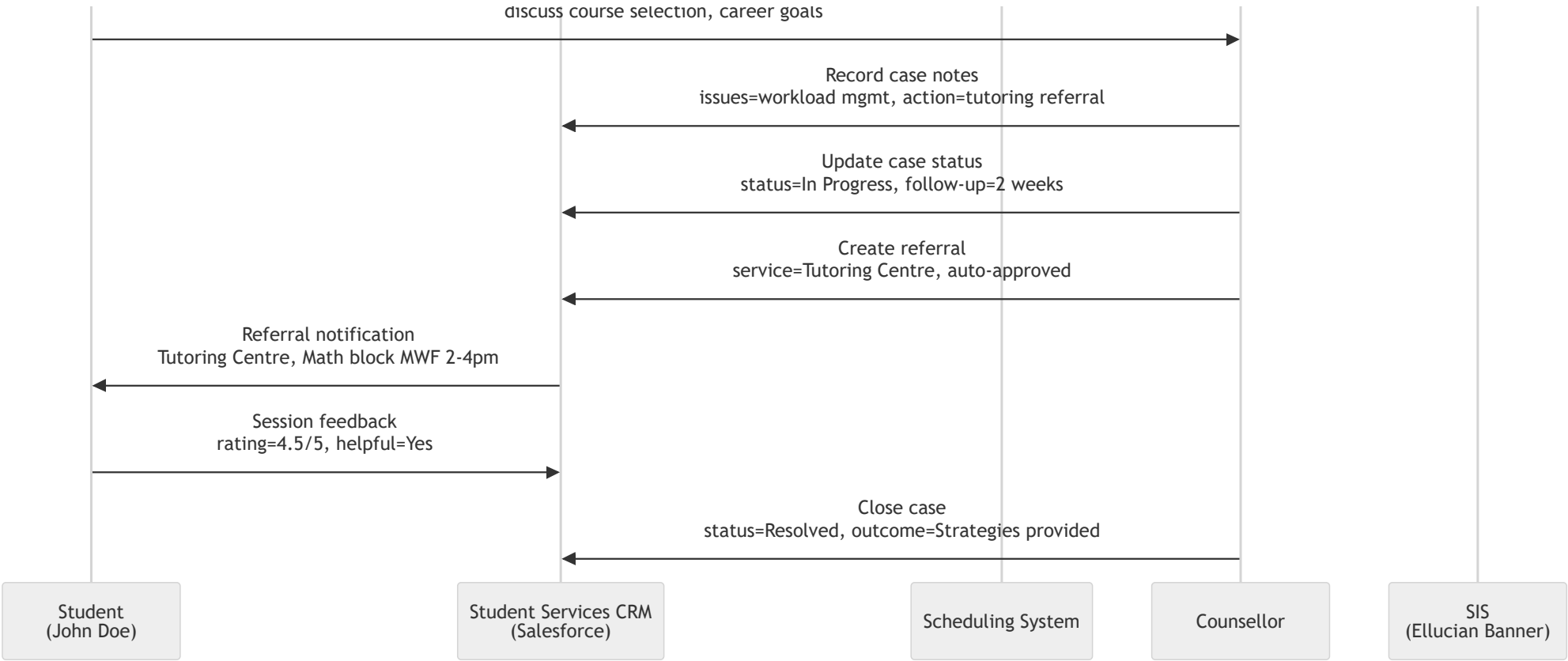
Description: Students submit support requests. Counselling team triages, schedules appointments, manages case notes, tracks outcomes, and escalates as needed.

Actors: Student, Counsellor, Student Support Manager

Systems: Student Services CRM (Salesforce Education Cloud), Scheduling System, SIS (Ellucian Banner), Case Management System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Student Support	Case	CreateSupportCase	Student Portal	Student Services CRM (Salesforce)	Portal REST API	Salesforce REST API (OAuth2)	API-led (Real-time)	Simple
Student Support	Profile	LoadStudentContext	Student Services CRM (Salesforce)	SIS (Ellucian Banner)	Salesforce REST API (OAuth2)	Banner API (API Key)	API-led (Real-time)	Simple
Student Support	Appointment	BookCounsellingSlot	Student Services CRM (Salesforce)	Scheduling System	Salesforce REST API (OAuth2)	Scheduling API (REST)	API-led (Real-time)	Medium
Student Support	Notes	RecordCaseNotes	Counsellor Portal	Student Services CRM (Salesforce)	Portal REST API	Salesforce REST API (OAuth2)	API-led (Real-time)	Simple
Student Support	Referral	CreateReferral	Student Services CRM (Salesforce)	Student Services CRM (Salesforce)	Internal	Internal	Event-driven	Simple
Student Support	Feedback	CaptureSessionFeedback	Student Portal	Student Services CRM (Salesforce)	Portal REST API	Salesforce REST API (OAuth2)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-9.1-01	Data integrity: Case data transferred from Student Portal to Student Services CRM (Salesforce) via CreateSupportCase must be complete and consistent	CreateSupportCase	Case records in Student Services CRM (Salesforce) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.1-02	Data integrity: Profile data transferred from Student Services CRM (Salesforce) to SIS (Ellucian Banner) via LoadStudentContext must be complete and consistent	LoadStudentContext	Profile records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.1-03	Data integrity: Appointment data transferred from Student Services CRM (Salesforce) to Scheduling System via BookCounsellingSlot must be complete and consistent	BookCounsellingSlot	Appointment records in Scheduling System match source; no data loss; reconciliation report confirms accuracy	High
AC-9.1-04	Data integrity: Notes data transferred from Counsellor Portal to Student Services CRM (Salesforce) via RecordCaseNotes must be complete and consistent	RecordCaseNotes	Notes records in Student Services CRM (Salesforce) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.1-05	Data integrity: Referral data transferred from Student Services CRM (Salesforce) to Student Services CRM (Salesforce) via CreateReferral must be complete and consistent	CreateReferral	Referral records in Student Services CRM (Salesforce) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.1-06	Data integrity: Feedback data transferred from Student Portal to Student Services CRM (Salesforce) via CaptureSessionFeedback must be complete and consistent	CaptureSessionFeedback	Feedback records in Student Services CRM (Salesforce) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.1-07	Error handling: Failed transactions between Student Portal and Student Services CRM (Salesforce) are logged with error context and trigger automatic retry or alert	CreateSupportCase	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-9.1-08	Security: All endpoints enforce authentication (Salesforce REST API (OAuth2), Internal, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

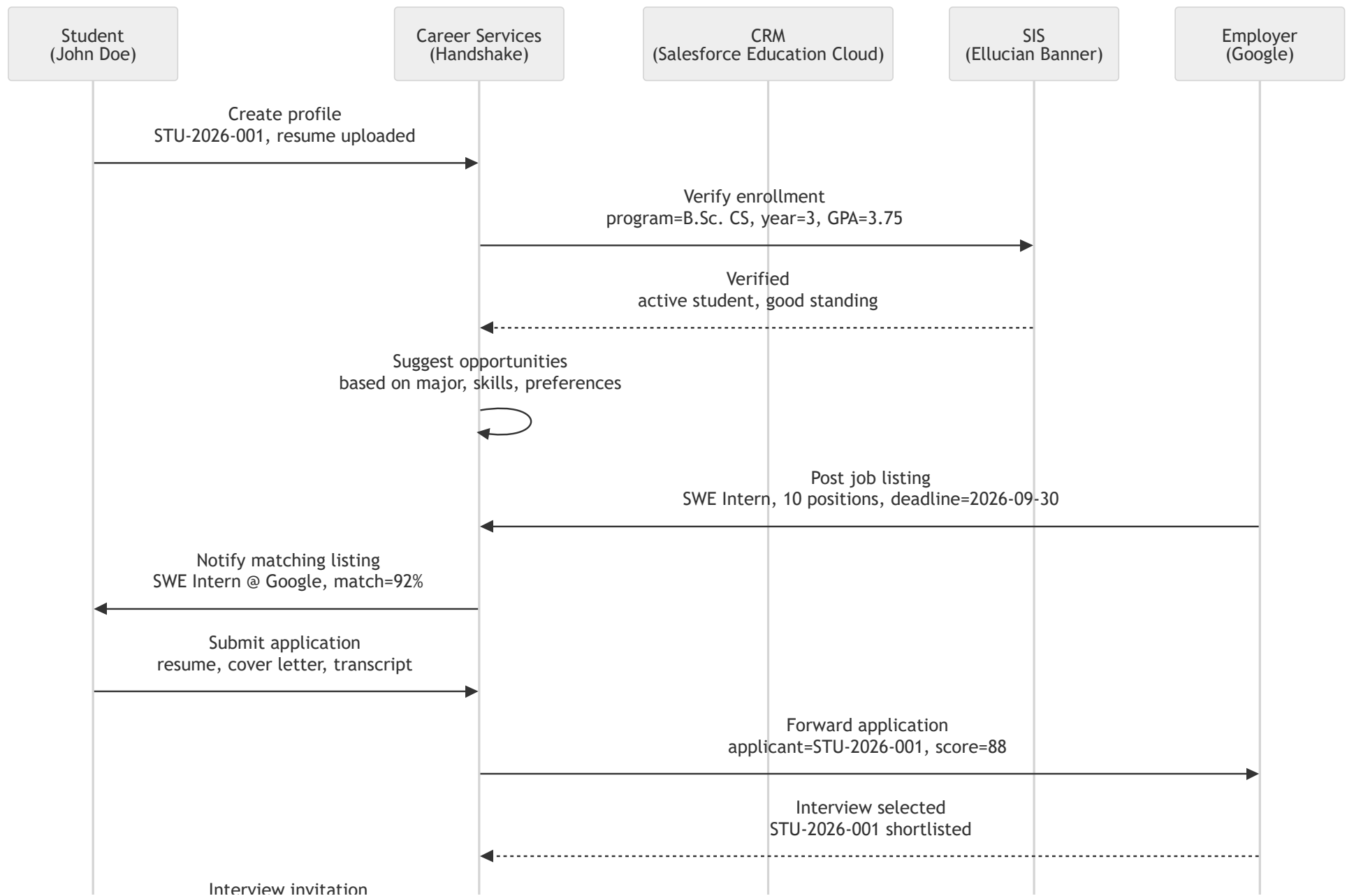
9.2 Career Services & Placement

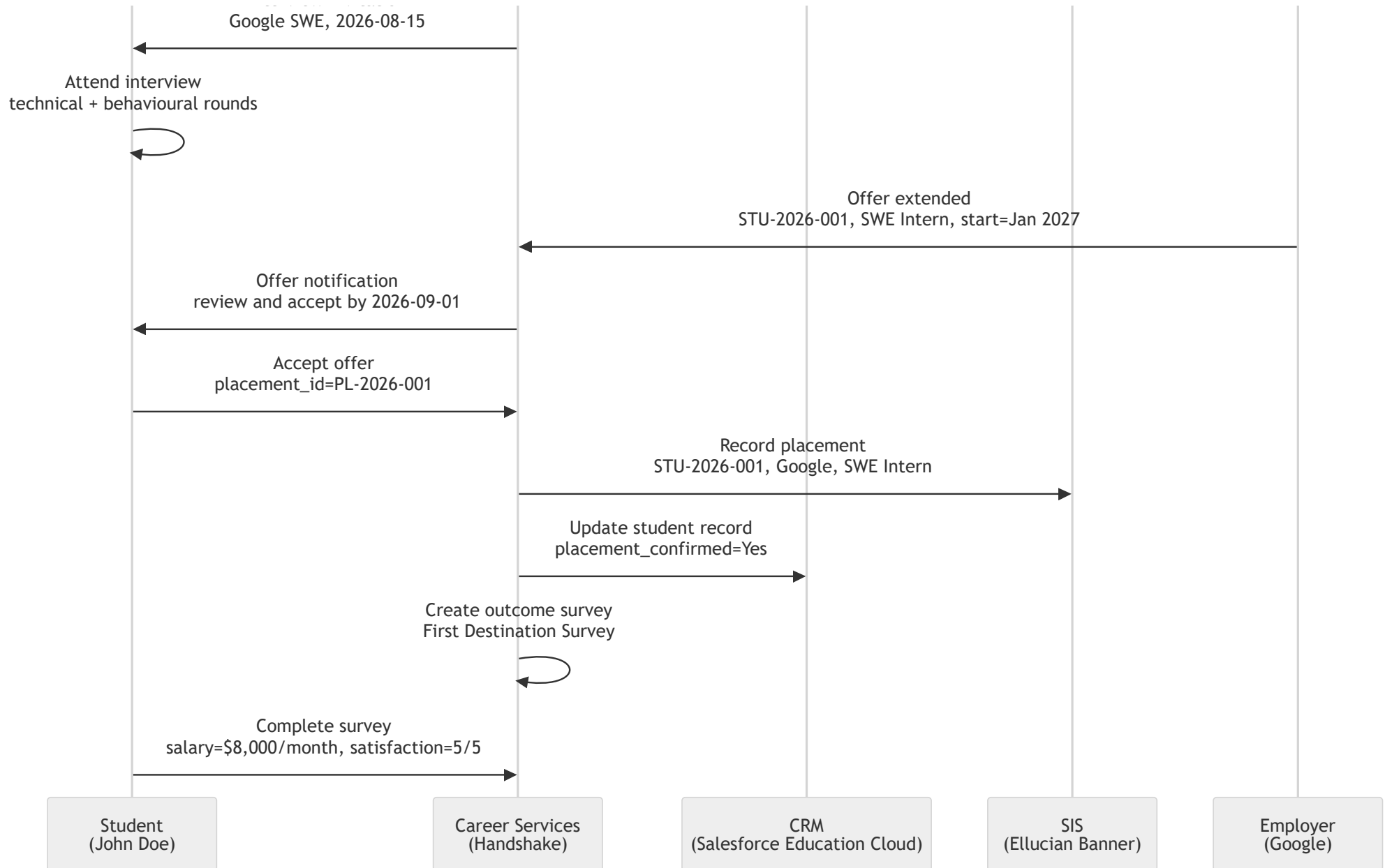
Description: Career services team manages job listings, resume workshops, interview preparation, employer networking events, and tracks graduate employment outcomes.

Actors: Student, Career Advisor, Employer, Alumni

Systems: Career Services Platform (Handshake), CRM (Salesforce Education Cloud), SIS (Ellucian Banner), Alumni CRM (Blackbaud), LMS (Canvas)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Career Services	Profile	CreateStudentProfile	Career Services (Handshake)	SIS (Ellucian Banner)	Handshake API (OAuth2)	Banner API (API Key)	API-led (Real-time)	Simple
Career Services	Listing	PostJobListing	Employer Portal	Career Services (Handshake)	Employer API (OAuth2)	Handshake API (OAuth2)	API-led (Real-time)	Simple
Career Services	Application	SubmitApplication	Student Portal	Career Services (Handshake)	Portal REST API	Handshake API (OAuth2)	API-led (Real-time)	Simple
Career Services	Placement	RecordPlacement	Career Services (Handshake)	SIS (Ellucian Banner)	Handshake API (OAuth2)	Banner API (API Key)	Event-driven	Medium
Career Services	Survey	CompleteOutcomeSurvey	Student Portal	Career Services (Handshake)	Portal REST API	Handshake API (OAuth2)	API-led (Real-time)	Simple
Career Services	Analytics	TrackPlacementRate	Career Services (Handshake)	BI Platform (Tableau)	Handshake API (OAuth2)	Tableau REST API	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-9.2-01	Data integrity: Profile data transferred from Career Services (Handshake) to SIS (Ellucian Banner) via CreateStudentProfile must be complete and consistent	CreateStudentProfile	Profile records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.2-02	Data integrity: Listing data transferred from Employer Portal to Career Services (Handshake) via PostJobListing must be complete and consistent	PostJobListing	Listing records in Career Services (Handshake) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.2-03	Data integrity: Application data transferred from Student Portal to Career Services (Handshake) via SubmitApplication must be complete and consistent	SubmitApplication	Application records in Career Services (Handshake) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.2-04	Data integrity: Placement data transferred from Career Services (Handshake) to SIS (Ellucian Banner) via RecordPlacement must be complete and consistent	RecordPlacement	Placement records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-9.2-05	Data integrity: Survey data transferred from Student Portal to Career Services (Handshake) via CompleteOutcomeSurvey must be complete and consistent	CompleteOutcomeSurvey	Survey records in Career Services (Handshake) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.2-06	Data integrity: Analytics data transferred from Career Services (Handshake) to BI Platform (Tableau) via TrackPlacementRate must be complete and consistent	TrackPlacementRate	Analytics records in BI Platform (Tableau) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-9.2-07	Error handling: Failed transactions between Career Services (Handshake) and SIS (Ellucian Banner) are logged with error context and trigger automatic retry or alert	CreateStudentProfile	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-	Security: All endpoints enforce authentication (Employer API (OAuth2), Portal REST API, ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
9.2-08				

Facilities & Campus Operations - BC125 BC114

Management of physical campus infrastructure: building maintenance, room scheduling and space allocation, security and access control, and ancillary services including catering, printing, and transportation.

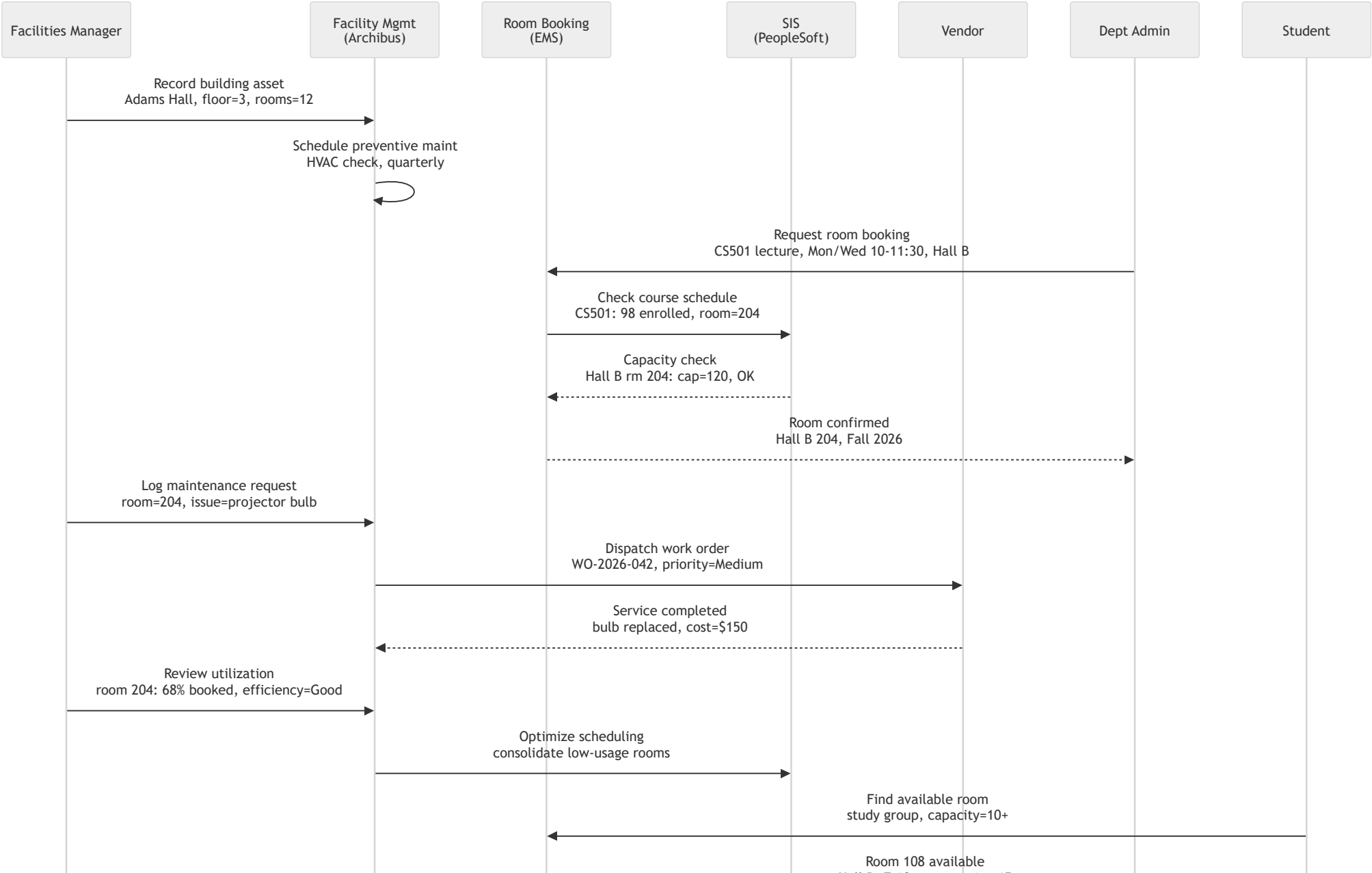
10.1 Facility Management & Space Allocation

Description: Facilities team manages building maintenance requests, room booking and allocation, space utilization tracking, and capital project planning.

Actors: Facilities Manager, Department Admin, Vendor, Student

Systems: Facility Management System (Archibus), Room Booking (EMS), SIS (PeopleSoft), Procurement (Coupa), Vendor Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Facility Management	Asset	RecordBuildingAsset	Facility Mgmt (Archibus)	Facility Mgmt (Archibus)	Internal	Internal	API-led (Real-time)	Simple
Facility Management	Room	BookRoom	Room Booking (EMS)	SIS (PeopleSoft Campus Solutions)	EMS Web API (Basic Auth)	PeopleSoft REST API (JWT)	API-led (Real-time)	Medium
Facility Management	Maintenance	LogMaintenanceRequest	Facility Mgmt (Archibus)	Vendor Portal	Archibus API (REST)	Vendor API (EDI)	Event-driven	Medium
Facility Management	Utilization	ReviewSpaceUtilization	Facility Mgmt (Archibus)	BI Platform (Tableau)	Archibus API (REST)	Tableau REST API	Batch (Scheduled)	Simple
Facility Management	Optimize	OptimizeRoomSchedule	Facility Mgmt (Archibus)	SIS (PeopleSoft Campus Solutions)	Archibus API (REST)	PeopleSoft REST API (JWT)	Batch (Scheduled)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-10.1-01	Data integrity: Asset data transferred from Facility Mgmt (Archibus) to Facility Mgmt (Archibus) via RecordBuildingAsset must be complete and consistent	RecordBuildingAsset	Asset records in Facility Mgmt (Archibus) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.1-02	Data integrity: Room data transferred from Room Booking (EMS) to SIS (PeopleSoft Campus Solutions) via BookRoom must be complete and consistent	BookRoom	Room records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	High
AC-10.1-03	Data integrity: Maintenance data transferred from Facility Mgmt (Archibus) to Vendor Portal via LogMaintenanceRequest must be complete and consistent	LogMaintenanceRequest	Maintenance records in Vendor Portal match source; no data loss; reconciliation report confirms accuracy	High
AC-10.1-04	Data integrity: Utilization data transferred from Facility Mgmt (Archibus) to BI Platform (Tableau) via ReviewSpaceUtilization must be complete and consistent	ReviewSpaceUtilization	Utilization records in BI Platform (Tableau) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.1-05	Data integrity: Optimize data transferred from Facility Mgmt (Archibus) to SIS (PeopleSoft Campus Solutions) via OptimizeRoomSchedule must be complete and consistent	OptimizeRoomSchedule	Optimize records in SIS (PeopleSoft Campus Solutions) match source; no data loss; reconciliation report confirms accuracy	High
AC-10.1-06	Error handling: Failed transactions between Facility Mgmt (Archibus) and Facility Mgmt (Archibus) are logged with error context and trigger automatic retry or alert	RecordBuildingAsset	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-10.1-07	Security: All endpoints enforce authentication (Archibus API (REST), EMS Web API (Basic Auth), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

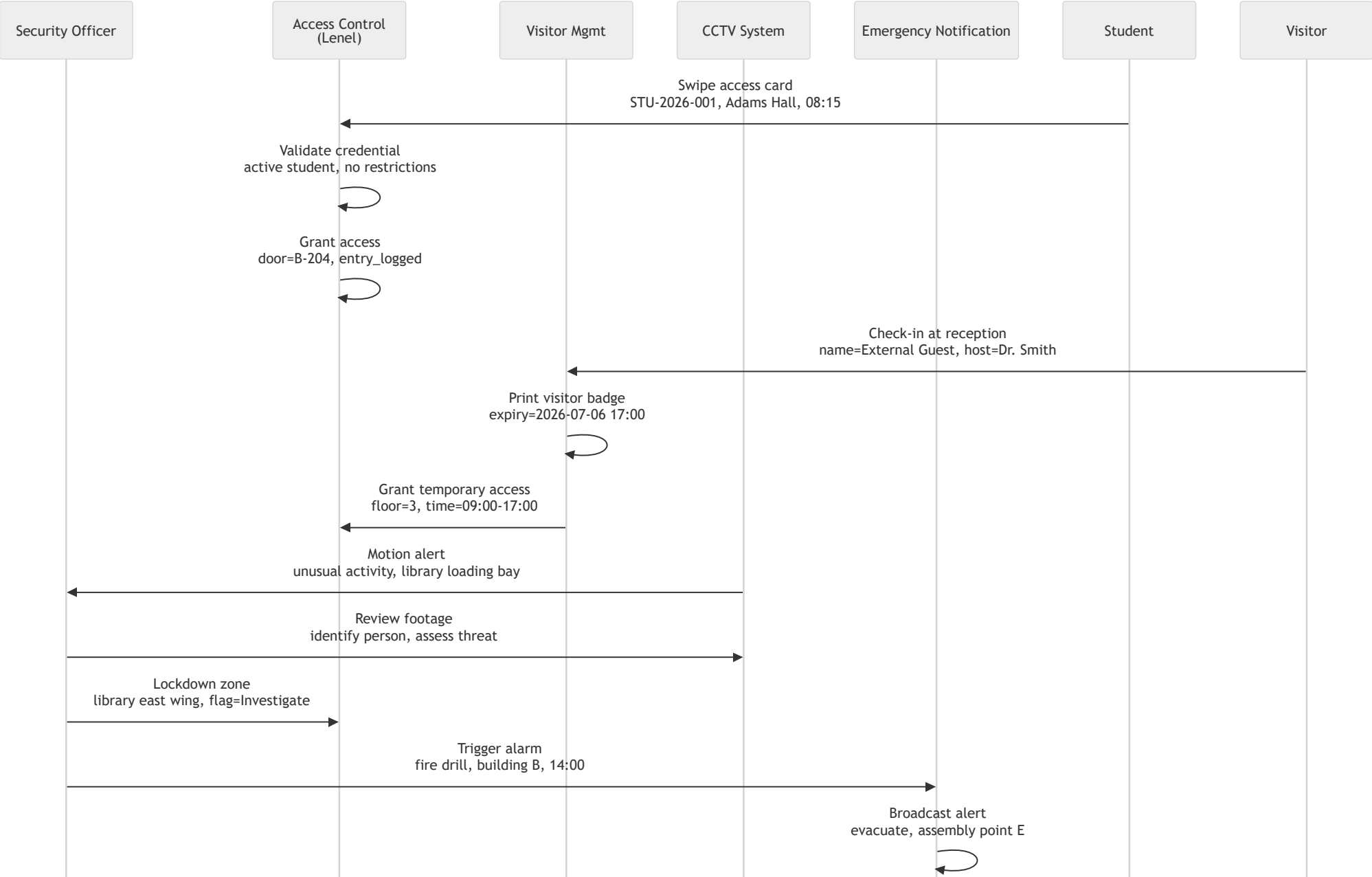
10.2 Security & Access Control

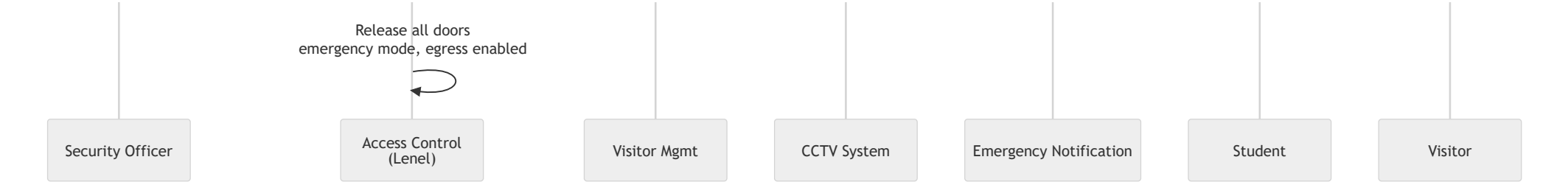
Description: Security team manages campus access control, visitor management, CCTV monitoring, incident response, and emergency notification systems.

Actors: Security Officer, Student, Visitor, Emergency Coordinator

Systems: Access Control System (Lenel), Visitor Mgmt System, CCTV System, Emergency Notification, SIS

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Security & Access	Credential	ValidateAccessCard	Access Control (Lenel)	SIS (Ellucian Banner)	Lenel API (SOAP)	Banner API (API Key)	API-led (Real-time)	Medium
Security & Access	Visitor	RegisterVisitor	Visitor Mgmt System	Access Control (Lenel)	Visitor API (REST)	Lenel API (SOAP)	API-led (Real-time)	Simple
Security & Access	CCTV	ReviewCCTVFootage	CCTV System	Security Console	CCTV API (REST)	Internal	API-led (Real-time)	Simple
Security & Access	Lockdown	InitiateLockdown	Security Console	Access Control (Lenel)	Internal	Lenel API (SOAP)	Event-driven	High
Security & Access	Emergency	BroadcastEmergencyAlert	Emergency Notification System	Notification Service	ENS API (REST)	Push/SMS/Email	Event-driven	High

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-10.2-01	Data integrity: Credential data transferred from Access Control (Lenel) to SIS (Ellucian Banner) via ValidateAccessCard must be complete and consistent	ValidateAccessCard	Credential records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-10.2-02	Data integrity: Visitor data transferred from Visitor Mgmt System to Access Control (Lenel) via RegisterVisitor must be complete and consistent	RegisterVisitor	Visitor records in Access Control (Lenel) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.2-03	Data integrity: CCTV data transferred from CCTV System to Security Console via ReviewCCTVFootage must be complete and consistent	ReviewCCTVFootage	CCTV records in Security Console match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.2-04	Data integrity: Lockdown data transferred from Security Console to Access Control (Lenel) via InitiateLockdown must be complete and consistent	InitiateLockdown	Lockdown records in Access Control (Lenel) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-10.2-05	Data integrity: Emergency data transferred from Emergency Notification System to Notification Service via BroadcastEmergencyAlert must be complete and consistent	BroadcastEmergencyAlert	Emergency records in Notification Service match source; no data loss; reconciliation report confirms accuracy	Critical
AC-10.2-06	Error handling: Failed transactions between Access Control (Lenel) and SIS (Ellucian Banner) are logged with error context and trigger automatic retry or alert	ValidateAccessCard	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-10.2-07	Security: All endpoints enforce authentication (Internal, Visitor API (REST), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

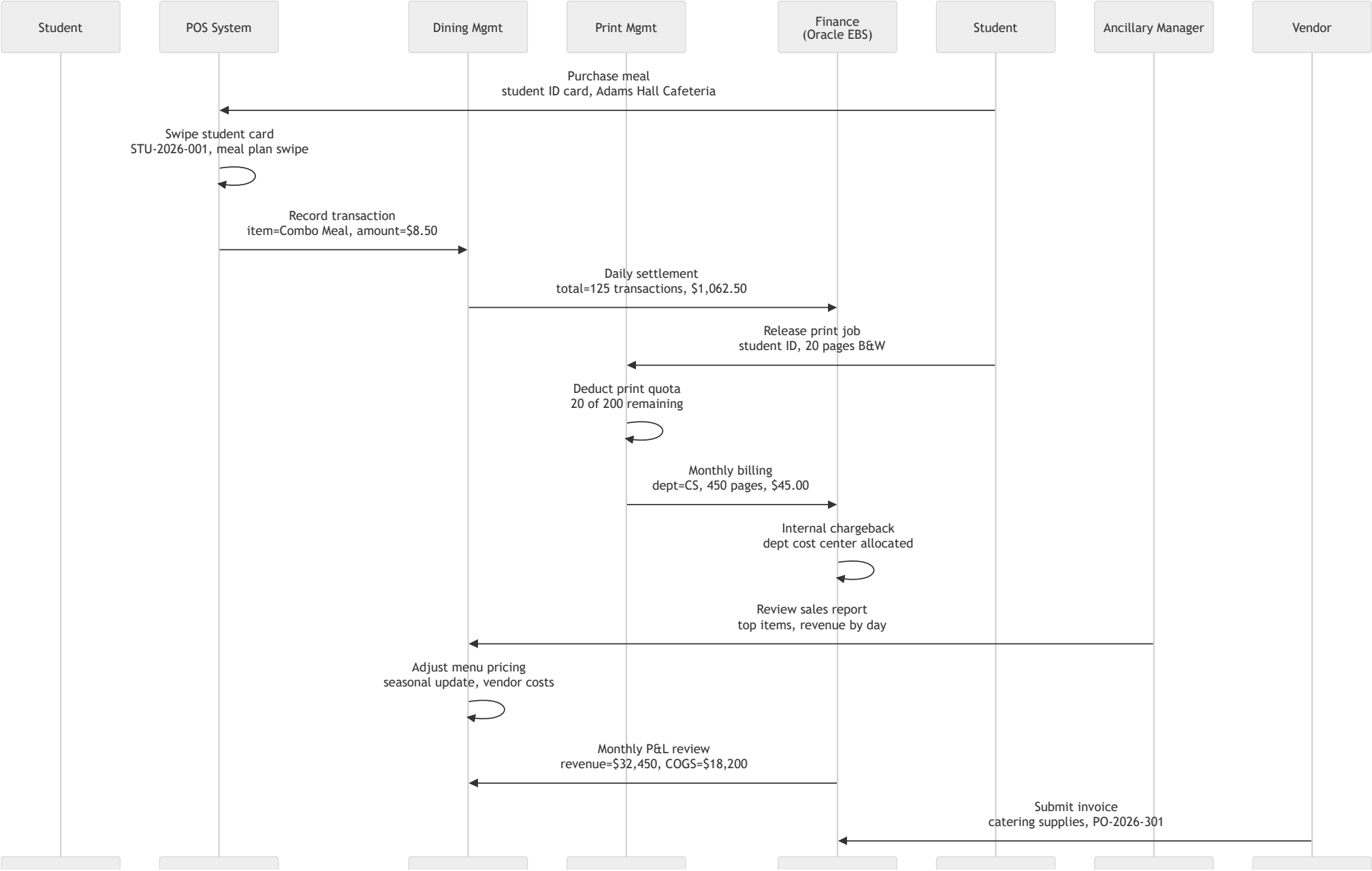
10.3 Ancillary Services Management

Description: Management of auxiliary services: campus dining, printing/copy centres, bookstores, transportation, and vending services with POS integration and financial reconciliation.

Actors: Student, Ancillary Manager, Vendor, Finance

Systems: POS System, Dining Management, Print Management, Finance ERP (Oracle EBS), Student ID/Card System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Ancillary Services	Meal	ProcessMealTransaction	POS System	Dining Management System	POS API (REST)	Dining API (REST)	API-led (Real-time)	Simple
Ancillary Services	Settlement	DailySettlement	Dining Management System	Finance ERP (Oracle EBS)	Dining API (REST)	Oracle EBS REST API (SOAP)	Batch (Scheduled)	Medium
Ancillary Services	Print	ProcessPrintJob	Print Management System	Student ID System	Print API (REST)	Card API	API-led (Real-time)	Simple
Ancillary Services	Chargeback	InternalDeptChargeback	Finance ERP (Oracle EBS)	Finance ERP (Oracle EBS)	Internal	Internal	Batch (Scheduled)	Medium
Ancillary Services	P&L	MonthlyPnLReview	Dining Management System	Finance ERP (Oracle EBS)	Dining API (REST)	Oracle EBS REST API (SOAP)	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-10.3-01	Data integrity: Meal data transferred from POS System to Dining Management System via ProcessMealTransaction must be complete and consistent	ProcessMealTransaction	Meal records in Dining Management System match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.3-02	Data integrity: Settlement data transferred from Dining Management System to Finance ERP (Oracle EBS) via DailySettlement must be complete and consistent	DailySettlement	Settlement records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	High
AC-10.3-03	Data integrity: Print data transferred from Print Management System to Student ID System via ProcessPrintJob must be complete and consistent	ProcessPrintJob	Print records in Student ID System match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.3-04	Data integrity: Chargeback data transferred from Finance ERP (Oracle EBS) to Finance ERP (Oracle EBS) via InternalDeptChargeback must be complete and consistent	InternalDeptChargeback	Chargeback records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	High
AC-10.3-05	Data integrity: P&L data transferred from Dining Management System to Finance ERP (Oracle EBS) via MonthlyPnLReview must be complete and consistent	MonthlyPnLReview	P&L records in Finance ERP (Oracle EBS) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-10.3-06	Error handling: Failed transactions between POS System and Dining Management System are logged with error context and trigger automatic retry or alert	ProcessMealTransaction	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures<	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			/td>	
AC-10.3-07	Security: All endpoints enforce authentication (POS API (REST), Dining API (REST), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

Governance, Risk & Compliance - BC160 BC155

Institutional governance framework: policy management, accreditation compliance, risk assessment, internal audit, legal services and contract management.

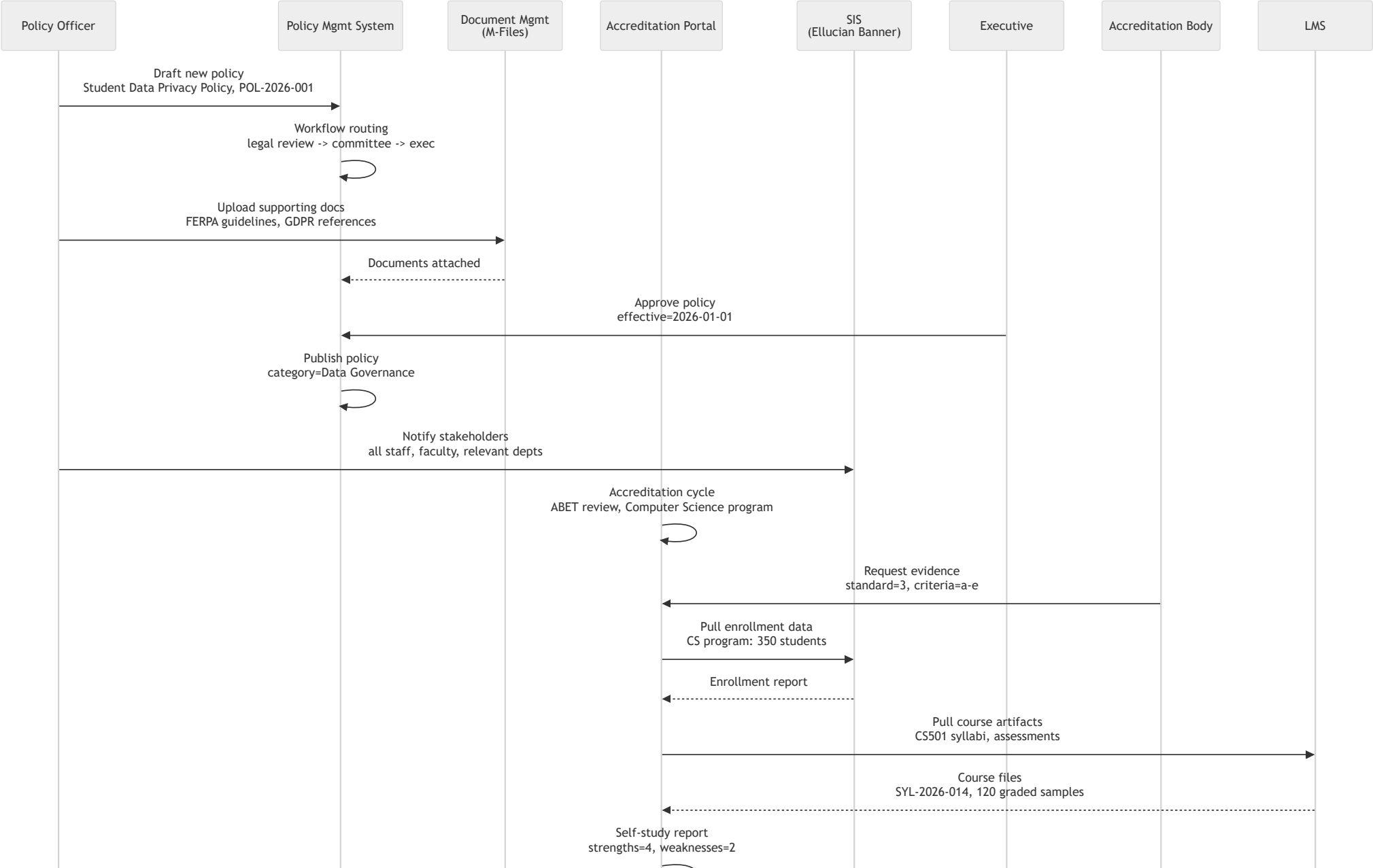
11.1 Policy Management & Accreditation

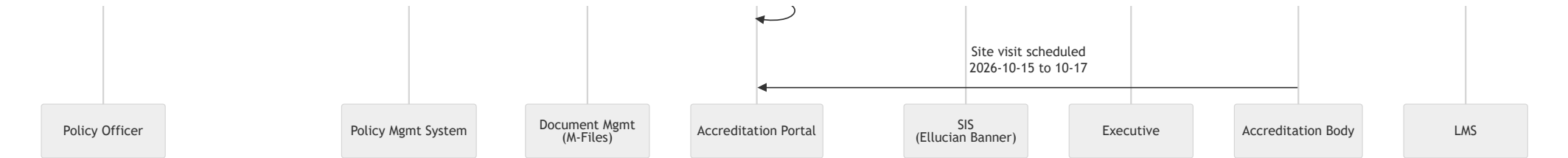
Description: Institutional policies created, reviewed, approved and published. Accreditation bodies evaluated. Self-study reports, evidence collection, and site visit coordination managed.

Actors: Policy Officer, Compliance Officer, Accreditation Body, Executive

Systems: Policy Management System, Document Mgmt (M-Files), Accreditation Portal, SIS, LMS

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Policy Management	Draft	CreatePolicyDraft	Policy Mgmt System	Document Mgmt (M-Files)	Policy API (REST)	M-Files REST API (API Key)	API-led (Real-time)	Simple
Policy Management	Approval	ApprovePolicyWorkflow	Policy Mgmt System	Policy Mgmt System	Internal	Internal	API-led (Real-time)	Medium
Policy Management	Notification	NotifyPolicyStakeholders	Policy Mgmt System	SIS (Ellucian Banner)	Policy API (REST)	Banner API (API Key)	Event-driven	Simple
Accreditation	Evidence	RequestAccreditationEvidence	Accreditation Portal	SIS (Ellucian Banner)	Accreditation API (REST)	Banner API (API Key)	API-led (Real-time)	Medium
Accreditation	Artifacts	PullCourseArtifacts	Accreditation Portal	LMS (Canvas)	Accreditation API (REST)	Canvas API (OAuth2)	Batch (Scheduled)	Medium
Accreditation	Report	GenerateSelfStudyReport	Accreditation Portal	Document Mgmt (M-Files)	Accreditation API (REST)	M-Files REST API (API Key)	API-led (Real-time)	High

Acceptance Criteria

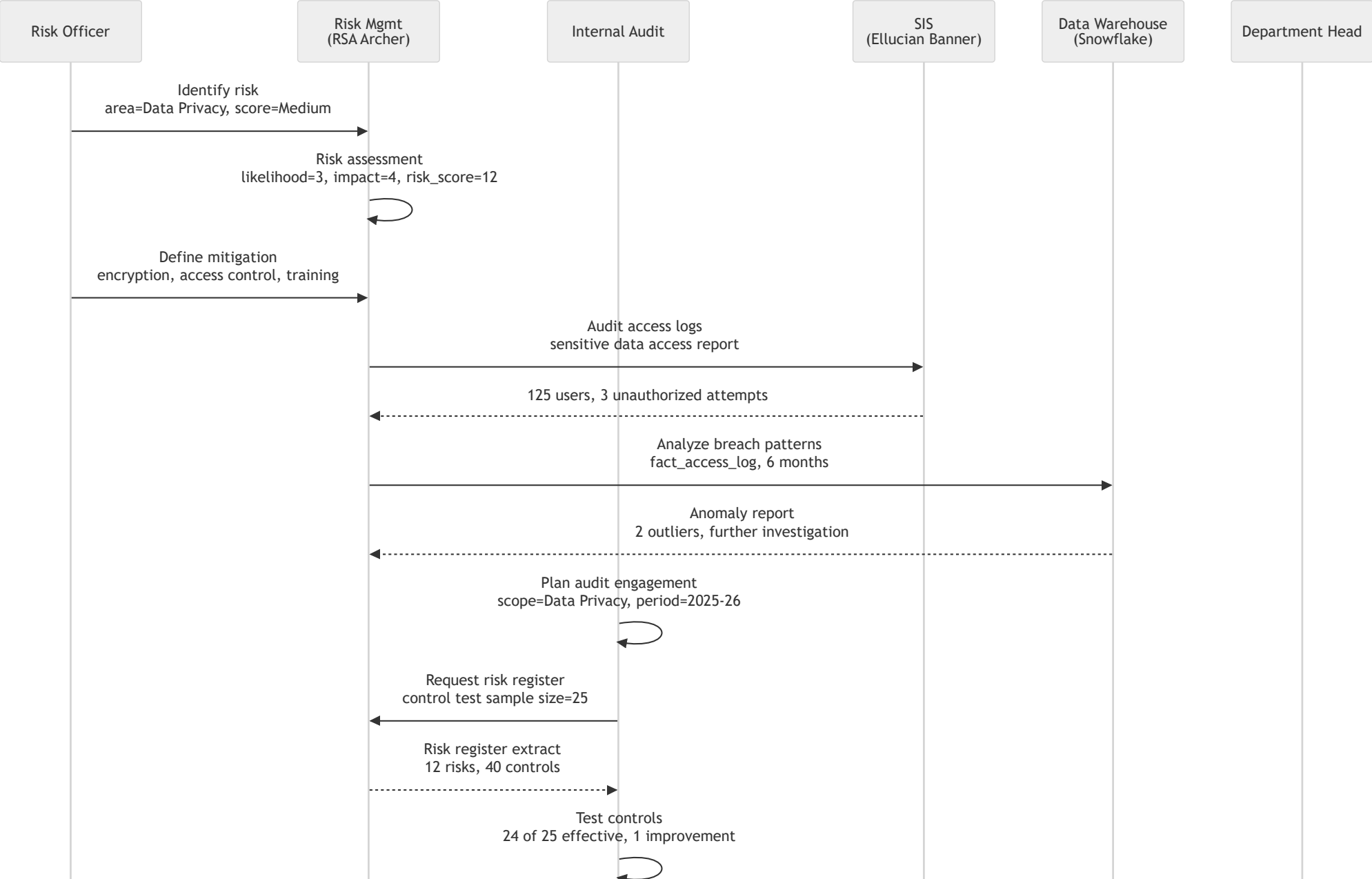
AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-11.1-01	Data integrity: Draft data transferred from Policy Mgmt System to Document Mgmt (M-Files) via CreatePolicyDraft must be complete and consistent	CreatePolicyDraft	Draft records in Document Mgmt (M-Files) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-11.1-02	Data integrity: Approval data transferred from Policy Mgmt System to Policy Mgmt System via ApprovePolicyWorkflow must be complete and consistent	ApprovePolicyWorkflow	Approval records in Policy Mgmt System match source; no data loss; reconciliation report confirms accuracy	High
AC-11.1-03	Data integrity: Notification data transferred from Policy Mgmt System to SIS (Ellucian Banner) via NotifyPolicyStakeholders must be complete and consistent	NotifyPolicyStakeholders	Notification records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-11.1-04	Data integrity: Evidence data transferred from Accreditation Portal to SIS (Ellucian Banner) via RequestAccreditationEvidence must be complete and consistent	RequestAccreditationEvidence	Evidence records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-11.1-05	Data integrity: Artifacts data transferred from Accreditation Portal to LMS (Canvas) via PullCourseArtifacts must be complete and consistent	PullCourseArtifacts	Artifacts records in LMS (Canvas) match source; no data loss; reconciliation report confirms accuracy	High
AC-11.1-06	Data integrity: Report data transferred from Accreditation Portal to Document Mgmt (M-Files) via GenerateSelfStudyReport must be complete and consistent	GenerateSelfStudyReport	Report records in Document Mgmt (M-Files) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-11.1-07	Error handling: Failed transactions between Policy Mgmt System and Document Mgmt (M-Files) are logged with error context and trigger automatic retry or alert	CreatePolicyDraft	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-11.1-08	Security: All endpoints enforce authentication (Policy API (REST), Accreditation API (REST), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

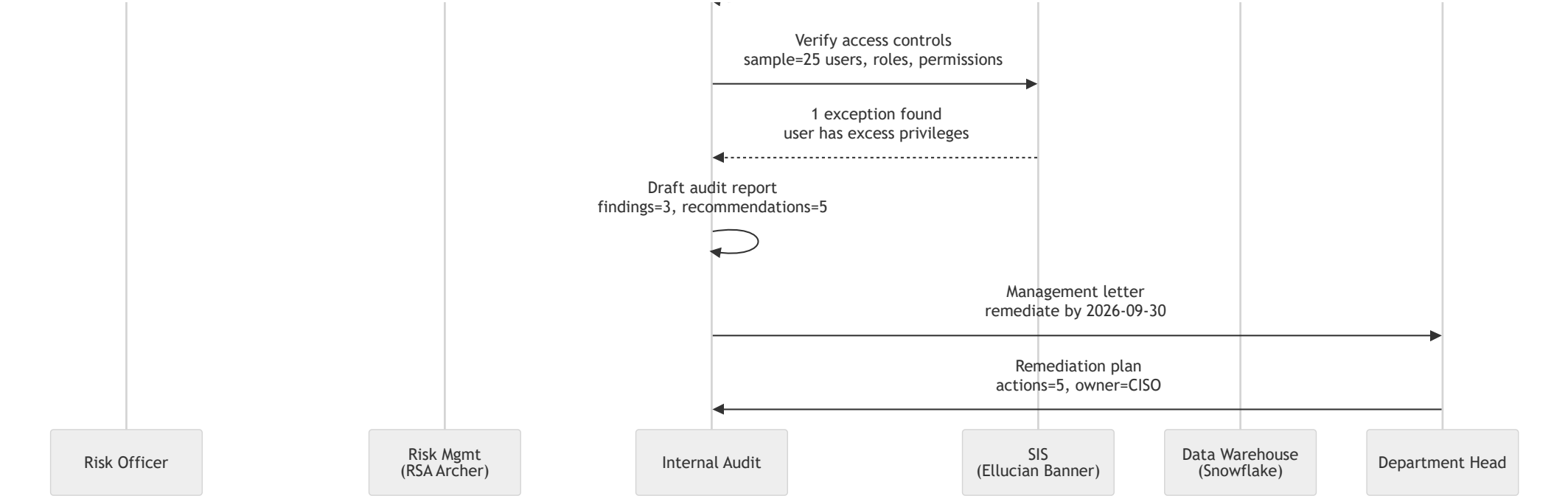
11.2 Risk Management & Audit

Description: Enterprise risk management: risk register, control testing, internal audit planning, issue tracking, and management reporting to audit committee.

Actors: Risk Officer, Internal Auditor, Department Head, Audit Committee
Systems: Risk Management System (RSA Archer), Audit Management System, SIS, Data Warehouse (Snowflake), BI Platform (Tableau)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Risk Management	Assessment	PerformRiskAssessment	Risk Mgmt (RSA Archer)	Risk Mgmt (RSA Archer)	Internal	Internal	API-led (Real-time)	Medium
Risk Management	Log	AuditAccessLogs	Risk Mgmt (RSA Archer)	SIS (Ellucian Banner)	Archer API (SOAP)	Banner API (API Key)	Batch (Scheduled)	Medium
Risk Management	Anomaly	AnalyzeBreachPatterns	Risk Mgmt (RSA Archer)	Data Warehouse (Snowflake)	Archer API (SOAP)	Snowflake JDBC	Batch (ETL)	High
Audit	Engagement	PlanAuditEngagement	Internal Audit System	Risk Mgmt (RSA Archer)	Audit API (REST)	Archer API (SOAP)	API-led (Real-time)	Simple
Audit	Testing	TestControls	Internal Audit System	SIS (Ellucian Banner)	Audit API (REST)	Banner API (API Key)	API-led (Real-time)	Medium
Audit	Report	IssueAuditReport	Internal Audit System	BI Platform (Tableau)	Audit API (REST)	Tableau REST API	Batch (Scheduled)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-11.2-01	Data integrity: Assessment data transferred from Risk Mgmt (RSA Archer) to Risk Mgmt (RSA Archer) via PerformRiskAssessment must be complete and consistent	PerformRiskAssessment	Assessment records in Risk Mgmt (RSA Archer) match source; no data loss; reconciliation report confirms accuracy	High
AC-11.2-02	Data integrity: Log data transferred from Risk Mgmt (RSA Archer) to SIS (Ellucian Banner) via AuditAccessLogs must be complete and consistent	AuditAccessLogs	Log records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-11.2-03	Data integrity: Anomaly data transferred from Risk Mgmt (RSA Archer) to Data Warehouse (Snowflake) via AnalyzeBreachPatterns must be complete and consistent	AnalyzeBreachPatterns	Anomaly records in Data Warehouse (Snowflake) match source; no data loss; reconciliation report confirms accuracy	Critical
AC-11.2-04	Data integrity: Engagement data transferred from Internal Audit System to Risk Mgmt (RSA Archer) via PlanAuditEngagement must be complete and consistent	PlanAuditEngagement	Engagement records in Risk Mgmt (RSA Archer) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-11.2-05	Data integrity: Testing data transferred from Internal Audit System to SIS (Ellucian Banner) via TestControls must be complete and consistent	TestControls	Testing records in SIS (Ellucian Banner) match source; no data loss; reconciliation report confirms accuracy	High
AC-11.2-06	Data integrity: Report data transferred from Internal Audit System to BI Platform (Tableau) via IssueAuditReport must be complete and consistent	IssueAuditReport	Report records in BI Platform (Tableau) match source; no data loss; reconciliation report confirms accuracy	Medium
AC-11.2-07	Error handling: Failed transactions between Risk Mgmt (RSA Archer) and Risk Mgmt (RSA Archer) are logged with error context and trigger automatic retry or alert	PerformRiskAssessment	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-11.2-08	Security: All endpoints enforce authentication (Archer API (SOAP), Audit API (REST), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical